Detecting Green Roofs in Baden-Württemberg by Integrating LoD2 CityGML and PlanetScope Imagery in a Supervised Machine Learning Framework

by

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A dissertation presented in partial fulfillment of the requirements for the degree of Master of Science in the Department of Geomatics, Computer Science and Mathematics, Stuttgart University of Applied Sciences

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Master Course Photogrammetry and Geoinformatics

Detecting Green Roofs in Baden-Württemberg by Integrating LoD2 CityGML and PlanetScope Imagery in a Supervised Machine Learning Framework

This thesis develops a 2025 roof-plane–level green roof inventory for Baden-Württemberg by combining LoD2 building geometry, semantic attributes and PlanetScope winter/summer imagery in a supervised machine learning workflow. Roof planes in four cities (Stuttgart, Karlsruhe, Freiburg im Breisgau, Tübingen; n=18,216) were manually labelled from high-resolution imagery and used to train and compare Random Forest (RF) and Multilayer Perceptron (MLP) models under a Leave-One-City-Out (LOCO) protocol. RF consistently outperformed MLP and achieved moderate spatial generalization (PR-AUC ≈ 0.69, best F1 ≈ 0.65) and strong within-sample performance under random splits (PR-AUC ≈ 0.84, best F1 ≈ 0.77). Permutation importance and ablation experiments highlight NDVI, roof slope, area, height and semantic building function as key predictors. Probability-based thresholding and ranking strategies translate RF outputs into different inventory "slices" that trade precision against completeness, providing a flexible, state-wide green roof layer for planning, monitoring and further research.

Keywords: Green roofs, Roof plane classification, LoD2, PlanetScope, Supervised machine learning, Random Forest

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Abbreviations

AOI	Area of Interest
AP	Average Precision
API	Application Programming Interface
BKG	Bundesamt für Kartographie und Geodäsie
BW	Baden-Württemberg
CLI	Command Line Interface
CRS	Coordinate Reference System
CNN	Convolutional Neural Network
EPSG	European Petroleum Survey Group (code)
GDAL	Geospatial Data Abstraction Library
GIS	Geographic Information System
GML	Geography Markup Language
LoD	Level of Detail
LOCO	Leave-One-City-Out
ML	Machine Learning
MLP	Multilayer Perceptron
NDVI	Normalized Difference Vegetation Index
NIR	Near-Infrared
PR	Precision–Recall
PR-AUC	Precision–Recall Area Under the Curve
RF	Random Forest
ROC	Receiver Operating Characteristic
RSKF	Repeated Stratified K-Fold
SR	Surface Reflectance
UTM	Universal Transverse Mercator
VG2500 KRS	Verwaltungsgebiete 1:2 500 000, Kreise
WMS	Web Map Service
XML	eXtensible Markup Language

1 Introduction

1.1 Background

Urban green roofs provide critical ecosystem services, including stormwater retention, urban cooling, biodiversity support, and improved air quality (Mentens, et al., 2006). Their integration into building infrastructure has gained attention as cities seek to combat the environmental effects of rapid urbanization and climate change. Despite their benefits and growing implementation, there is no comprehensive spatial inventory of green roofs, especially at regional or state-wide scales. Traditional mapping methods such as field surveys or visual inspection of orthophotos are labor-intensive, subjective, and not scalable (Carter & Fowler, 2008). Remote sensing technologies now offer viable alternatives for large-area monitoring.

Figure 1: Inference of the Vegetated Roof from Marmy et al. (2025)

A similar effort has been carried out in the PROJ-VEGROOFS project in Switzerland (Marmy, et al., 2025), which focused on identifying vegetated rooftops in the city of Lausanne using slope and NDVI thresholds derived from LoD2 data and aerial orthophotos. While effective, their approach was rule-based and limited to a single urban context. This thesis aims to build upon that methodology by expanding the spatial extent to an entire German state (Baden-Württemberg), and by focusing on mapping existing vegetated roofs rather than inferring greening potential, in order to keep the workflow transferable and as automated as possible.

Satellite imagery from the PlanetScope constellation provides frequent, high-resolution (3–5 m) imagery in four spectral bands (R, G, B, NIR), enabling vegetation monitoring through indices like the Normalized Difference Vegetation Index (NDVI). Combined with detailed 3D building data from LoD2 models available via the Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL-BW), an integrated method of rooftop vegetation detection becomes technically feasible.

This study proposes a scalable approach to detect vegetated rooftops using high-resolution satellite imagery and 3D building data, applying a reliable geospatial pipeline to automate classification. The outcome will support environmental planning, urban sustainability strategies, and spatial policymaking in the region.

1.2 Problem Statement

While some city-level inventories exist, there is currently no systematic and automated method for identifying vegetated rooftops across Baden-Württemberg. Manual digitization is not feasible at the state scale. Moreover, the full potential of combining high-resolution spectral data with 3D geometric information from LoD2 models remains underexplored in urban vegetation mapping.

Urban planners and policymakers lack consistent spatial data to evaluate the distribution and extent of existing green roofs or identify opportunities for further implementation. This research seeks to address this gap by developing a reliable pipeline.

1.3 Motivation

This thesis is motivated by the feasibility of combining:

- LoD2 building models (roof-plane geometry suitable for roof-surface analysis), provided as CityGML from the Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL), and
- PlanetScope multispectral imagery which supports vegetation detection via NDVI and related spectral statistics.

A roof plane representing greenery is attractive because it allows predictions to be produced at the roof surface level (rather than at coarse building blocks or pixel grids), making outputs easier to interpret and aggregate for planning and reporting.

1.4 Contribution

This thesis makes the following contributions:

- A roof plane feature dataset that combines LoD2 roof geometry (e.g., slope, area, aspect, height) with PlanetScope-derived spectral summaries (R, G, B, NIR, NDVI and band means/std).
- A state-scale geospatial ML pipeline that automates roof-plane classification (vegetated vs non-vegetated) and produces an inventory-ready output layer.
- A cross-city generalization evaluation using city hold-out validation (LOCO)
- A 2025 Baden-Württemberg Green Roof Inventory, as a spatial geospatial layer containing probability of vegetated roof with different slicing thresholds to produce binary classification per roof polygon across BW.

2 Theoretical Background

2.1 Green roofs

Green roofs are commonly defined as roofs that include a vegetated surface and a substrate, i.e., systems that support plant growth rather than merely green-colored roofing materials (Oberndorfer, et al., 2007). Green roofs are discussed as a form of urban ecosystem that can provide ecosystem services such as improved stormwater management, better regulation of building temperatures, reduced urban heat-island effects, and increased habitat potential (Oberndorfer, et al., 2007).

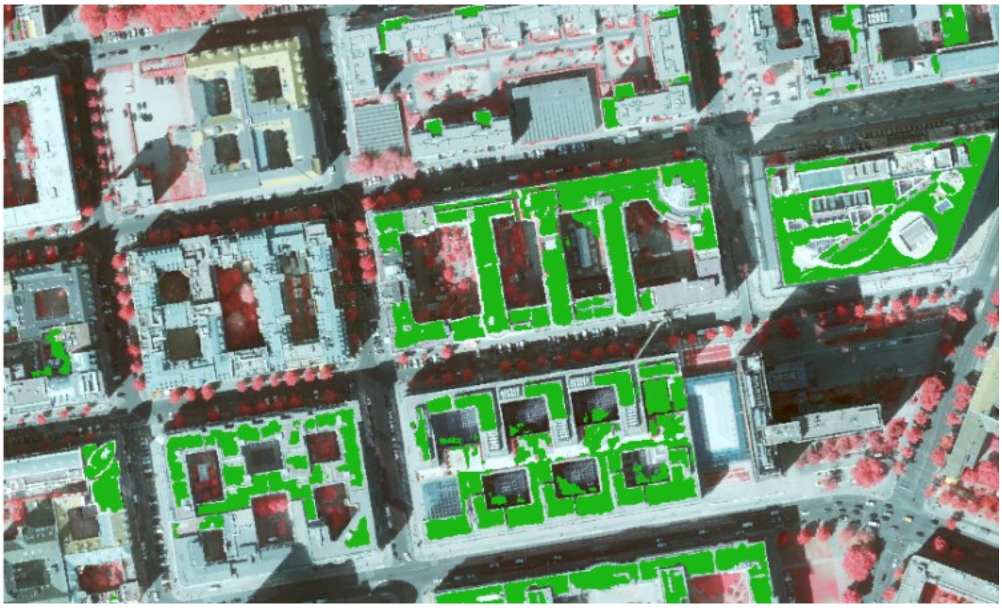


Figure 2: Green Roofs from Pauligk et al. (2022)

A widely used typology distinguishes extensive and intensive green roofs. Extensive roofs typically have thinner substrates, lower weight loads, and low-maintenance vegetation (often sedum-type plantings), while intensive roofs generally involve deeper substrates that support more diverse vegetation but require higher structural capacity and more maintenance. This differentiation matters for remote sensing because vegetation structure, moisture conditions, and phenology can vary between roof types, influencing spectral response and the stability of vegetation indices (Oberndorfer, et al., 2007).

In this thesis, a roof plane is classified as vegetated (green roof) if its spectral response indicates photosynthetically active vegetation on the roof surface, supported by NDVI and multispectral reflectance statistics. Non-vegetated surfaces that may appear green (e.g., coatings or artificial turf) are not intended to be part of the vegetated class. This definition is intentionally operational: it aligns the target concept with what can be measured from multispectral imagery according to Rouse et al. (1974) and Tucker (1979).

2.2 Remote sensing for roof vegetation detection

Vegetation detection in remote sensing relies on characteristic spectral behavior: healthy vegetation typically exhibits relatively low reflectance in the red band (chlorophyll absorption) and higher reflectance in the near-infrared band (leaf internal structure), enabling discrimination from many non-vegetated materials. A classic and widely used index that exploits this contrast is the Normalized Difference Vegetation Index (NDVI) according to Rouse et al. (1974) and Tucker (1979).

NDVI is commonly defined as:

$$NDVI = (NIR - Red) / (NIR + Red)$$

According to Rouse et al. (1974) and Tucker (1979), because NDVI is physically linked to vegetation reflectance patterns, it is widely applied as an interpretable indicator of vegetation presence and relative greenness at scale including green roofs.

Detecting vegetation on rooftops is more difficult than mapping vegetation in open landscapes because roofs introduce confounding factors: roof materials and coatings vary widely, roof structures create shadowing and adjacency effects, and non-vegetated materials can sometimes produce misleading spectral signals in visible bands (Oberndorfer, et al., 2007). As a result, purely threshold-based NDVI methods may suffer from false positives and false negatives when applied to heterogeneous urban roofscapes.

To improve robustness, many applied urban remote sensing workflows combine NDVI with additional spectral summaries (e.g., band means and within-roof variability) and contextual information, rather than relying on a single threshold rule (Oberndorfer, et al., 2007).

2.3 LoD2 as 3D building data with roof planes

In semantic 3D city models, the Level of Detail (LoD) concept describes the geometric and semantic richness of represented objects such as buildings. The LoD concept is formalized in CityGML and has been widely discussed as a multi-scale representation scheme for urban data (Biljecki, et al., 2016). In LoD2, buildings commonly include differentiated roof structures, enabling explicit representation of roof surfaces as geometric objects rather than a single extruded block. Current OGC CityGML specifications explicitly discuss building geometries including roof surface representations as part of the standard's modeling concept (Open Geospatial Consortium, 2023)

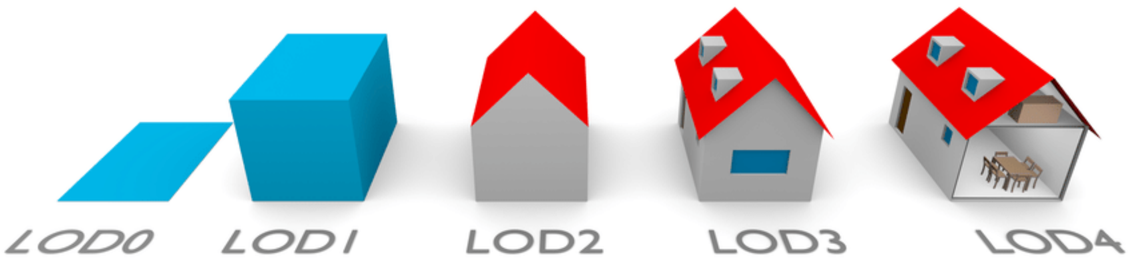


Figure 3: Level of Details (LoD) Illustration from Biljecki et al. (2016)

A roof-plane representation provides an analysis unit that is directly interpretable and inventory-ready: outputs can be summarized by roof count, roof area, or administrative aggregation without requiring complex post-processing of pixel maps. In addition, roof-plane geometries support extraction of geometric attributes (e.g., slope, aspect, and surface area) that are not available from imagery alone, enabling the integration of physical roof context with spectral indicators for classification.

In addition to geometric representations, the LoD2 building models for Baden-Württemberg include semantic attributes that describe roof shape and building use. The roof type is encoded as a categorical code indicating general roof geometry (e.g. flat, gabled, hipped, mixed forms), while the building function attribute characterizes the primary use of the building (e.g. residential, mixed residential–commercial, industrial, public or agricultural). Both attributes are derived from the official ALKIS/LGL code lists and provide contextual information that is not available from imagery alone. In this thesis, roof type and building function are used as semantic predictors alongside geometric and spectral features in the supervised learning framework, under the assumption that structural suitability and land-use context influence where green roofs are most likely to occur.

2.4 Supervised learning for roof-plane classification

Machine learning (ML) techniques are increasingly used to classify land cover and urban features from remote sensing data. Deep learning (DL) methods, including Convolutional Neural Networks (CNNs) and Multilayer Perceptrons (MLPs), have also been applied to urban mapping tasks. While CNNs excel at pixel-level image classification, MLPs are effective for tabular feature data (Zhu, et al., 2017) where geometric, spectral, and temporal attributes are combined. Hybrid approaches such as integrating handcrafted features (slope, NDVI, area) with deep learning, have shown improved performance over rule-based or single-model techniques (Maxwell, et al., 2018).

2.4.1 Random Forest (RF)

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions. It introduces randomness via bootstrapped sampling and random feature selection at splits, which reduces variance and improves generalization (Breiman, 2001). RF is frequently used for remote sensing classification tasks involving heterogeneous feature sets because it can model non-linear relationships and interactions without requiring strict parametric assumptions.

2.4.2 Multilayer Perceptron (MLP)

A Multilayer Perceptron (MLP) is a feedforward neural network trained via backpropagation, commonly used for supervised learning with tabular feature vectors. In practice, MLP models depend on training choices such as optimization method and regularization; for example, the Adam optimizer is widely used in neural network training and is referenced directly within standard MLP implementations (Kingma & Ba, 2015).

2.4.3 Feature Importance

For model explanation, it is common to report feature importance (e.g., tree-based importance measures for RF) and to use model-agnostic interpretability approaches such as SHAP, which provides a unified framework for additive feature attributions based on Shapley-value principles (Lundberg & Lee, 2017). This supports the analysis of which geometric and spectral features contribute most to a green-roof prediction, beyond reporting performance metrics alone.

3 Related research

3.1 Existing green roofs mapping effort

Related work on green roof mapping sits at the intersection of (i) urban remote sensing, (ii) building/roof modeling, and (iii) applied inventory production for planning. A recurring theme is that many cities want a green roof inventory, but the methods used to produce one vary widely in automation level, data requirements, and transferability across locations (Wu & Biljecki, 2021; Pauligk, et al., 2022)

Broadly, the literature and applied projects can be grouped into three families:

- Manual or semi-manual mapping, often based on orthophoto interpretation, sometimes supported by preliminary automated hints.
- Rule-based detection, usually using vegetation indices (e.g. NDVI) and thresholds, sometimes constrained by building footprints or roof geometry. In practice, many operational workflows combine such thresholds with supervised classifiers (Pauligk, et al., 2022).
- Machine learning / deep learning approaches, which learn decision boundaries from labeled data and aim to improve robustness and generalization (Wu & Biljecki, 2021; Marmy, et al., 2025; Li, 2025).

This thesis builds directly on the third family while borrowing practical lessons from the first two (especially around post-processing and quality control for inventories).

3.2 Rule-based and Spectral Thresholding

A common baseline idea is simple: vegetated roofs should show a vegetation-like spectral signature; therefore, an NDVI threshold can flag candidate green roofs. In practice, many applied inventory workflows combine this with additional constraints and manual review steps because roofs are spectrally messy and urban shadowing is brutal.

A concrete example is the workflow by Pauligk, et al. (2022). Their method begins with an automated pre-mapping step inside building outlines, where NDVI thresholds (e.g. average NDVI > 0.1) and segmentation are used to flag candidate green-roof areas. These candidates then feed into a supervised CNN step and are followed by extensive rule-based post-processing and manual checking. The NDVI thresholds therefore act as a pre-filter and as part of later categorization (e.g. extensive vs intensive), rather than as a standalone classifier.

These hybrid workflows are important prior art because they show what is often required for "inventory-grade" outputs: a pipeline that acknowledges uncertainty, includes post-checks, and uses explicit mapping rules to clean results for final reporting. At the same time, reliance on NDVI thresholds and local post-processing rules can make transferability difficult when roof materials, illumination conditions, or image products differ between regions.

3.3 Supervised learning approaches

3.3.1 Traditional ML using engineered roof descriptors

A practical middle-ground between thresholding and end-to-end deep learning is supervised learning using engineered features. This typically means computing statistics such as mean/variance of spectral bands or NDVI within roof objects, sometimes combined with geometry or height features, and then training a classifier.

The Swiss Territorial Data Lab project PROJ-VEGROOFS describes a workflow that derives statistical descriptors from airborne imagery and uses supervised learning to predict whether a roof is green. The project reports that a combination of models (random forest and logistic regression) trained on pixel statistics from candidate vegetated areas defined by NDVI and luminosity thresholds achieved high performance on their test set (Marmy, et al., 2025). This is highly relevant to this thesis approach because it explicitly shows (a) roof-level statistics as inputs and (b) the value of combining multiple feature cues rather than relying on a single index.

3.3.2 Deep learning for roof typology

Deep learning approaches have recently been used to map roof typologies at scale, including green roofs, often using convolutional neural networks (CNNs) applied to imagery. The Roofpedia project is a prominent example: it presents a deep learning + GIS pipeline to automatically map and quantify green and solar roofs and reports an open roof registry covering more than one million buildings across 17 cities (Wu & Biljecki, 2021). Roofpedia is especially relevant as related work because it frames roof inventories as a data scarcity problem and demonstrates a workflow explicitly designed to generalize across multiple cities.

Similarly, Pauligk, et al. (2022) methodology explicitly documents a CNN-based supervised classification step and cites U-Net as the architecture used in that project's approach, followed by substantial post-processing due to data limitations and the need for inventory-quality mapping. This CNN step is embedded in a broader pipeline that uses NDVI-based pre-segmentation and extensive post-processing, illustrating a pragmatic hybrid between deep learning and rule-based inventory cleaning. This combination of deep learning for preliminary mapping plus human or rule-based correction shows a real-world production strategy rather than a purely academic benchmark.

A branch of related work focuses not on detecting existing green roofs, but on identifying roofs suitable for greening. For example, (Li, 2025) propose a screening approach integrating remote sensing and deep learning to identify roof-greening potential. While methodologically adjacent, suitability mapping is conceptually different from an inventory of existing green roofs, because it targets "where could greening happen" rather than "where is greening already present". This distinction matters for evaluation: suitability studies may prioritize different ground truth definitions and error tolerances than inventory studies.

3.4 Analysis Units

A key axis of difference in related work is the analysis unit. Pixel-based approaches classify individual image pixels, while object-based approaches aggregate image information within meaningful objects (e.g., buildings, roof planes, segments), then classify those objects. In practice, many inventory workflows constrain the search to building outlines or roof objects to reduce false positives from ground vegetation and to make outputs directly inventory-ready (Pauligk, et al., 2022; Wu & Biljecki, 2021).

Pauligk, et al. (2022) methodology explicitly states that the analysis is performed within building outlines and combines segmentation/cluster-based steps with supervised classification, reinforcing that roof-focused inventories often benefit from object constraints and post-processing rules. Roofpedia also emphasizes a geospatial pipeline that produces a registry at building scale, again aligning the mapping unit with reporting needs (Wu & Biljecki, 2021).

This thesis follows the same "inventory logic," but focuses on roof-plane objects rather than whole buildings, enabling integration of roof-plane geometry (e.g., slope) with spectral summaries.

3.5 Gaps and limitations in existing work

Across the literature and applied inventories, several gaps emerge that directly motivate this research design:

- Transferability across cities/regions

Most existing inventories rely on very high-resolution orthophotos and city-specific data products (e.g. Berlin's CIR orthophotos and local height models), which makes direct transfer to other regions difficult. This thesis explicitly tests cross-city generalization using medium-resolution satellite imagery (PlanetScope) and state-wide LoD2 data

- Object granularity

  Some inventories operate at building scale, but do not explicitly model roof planes; this can hide within-building heterogeneity (e.g., partial green roofs) and limits integration of geometric roof-plane features.

- Inventory-grade post-processing

  Real inventories frequently depend on rule-based filtering and/or manual review to meet quality expectations, especially under limited training data.

- Problem framing differences

  A non-trivial portion of related work targets roof-greening potential rather than current green roof existence, which changes ground truth definition, evaluation, and interpretation.

In summary, prior work demonstrates that automated mapping of green roofs is feasible, and recent projects show promising results using both engineered-feature classifiers and deep learning pipelines (Wu & Biljecki, 2021; Marmy, et al., 2025; Pauligk, et al., 2022). However, consistent challenges remain around transferability, object granularity, and producing inventory-grade outputs with explicit quality controls. This thesis addresses these gaps by implementing a roof-plane-level workflow that integrates multispectral vegetation cues with 3D roof geometry and evaluates cross-city generalization before producing a state-scale 2025 inventory.

4 Objectives

4.1 Main objectives

The primary objective of this thesis is to produce a 2025 roof-plane-level green roof inventory for Baden-Württemberg using a reproducible workflow that integrates (i) LoD2 roof-plane geometry and (ii) multispectral vegetation indicators derived from PlanetScope imagery, combined within a supervised machine learning framework. The inventory is designed to be inventory-ready, meaning outputs can be summarized and mapped directly at roof-plane level for spatial reporting and planning applications.

4.2 Specific objectives

To achieve the main objective, the thesis pursues the following specific objectives:

O1: Roof-plane dataset construction.

Construct a roof-plane dataset where each roof plane is represented as a feature vector combining geometric descriptors (e.g., slope, aspect, roof area, height proxies) and multispectral summaries (e.g., NDVI and band statistics) aggregated over the roof polygon.

O2: Feature design for robust vegetation discrimination.

Design and test feature families that reduce common urban remote sensing confusions (e.g., shadowing, heterogeneous roofing materials) by combining spectral indices with geometric context.

O3: Model training and comparison.

Train and compare at least two supervised models—Random Forest and Multilayer Perceptron—on the same roof-plane feature set to evaluate relative performance on this classification task.

O4: Cross-city generalization evaluation.

Evaluate generalization under a city hold-out design (leave-one-city-out across the training cities) to test whether patterns learned in some cities transfer to an unseen city, which is critical for scaling an inventory beyond a single case study.

O5: State-scale inference and inventory production.

Apply the best-performing validated model to generate a roof-plane classification layer for Baden-Württemberg for the year 2025, including quality control logic suitable for inventory outputs.

4.3 Research questions

To operationalize the objectives, this thesis addresses the following research questions:

RQ1: Feasibility and accuracy.

How accurately can green roofs be detected at roof-plane level in Baden-Württemberg by integrating LoD2 roof geometry and PlanetScope multispectral features within a supervised machine learning framework?

RQ2: Model comparison.

How do Random Forest and Multilayer Perceptron compare in classification performance under identical features and evaluation strategy?

RQ3: Feature contribution and interpretability.

Which geometric and spectral features contribute most to discriminating vegetated vs non-vegetated roof planes, and are these contributions consistent across different city hold-outs?

RQ4: Inventory-grade output behavior.

To what extent do post-processing and probability-based decision rules influence the number and spatial distribution of predicted green roofs in the final inventory, compared to using a single fixed probability cut-off?

4.4 Scope and delimitation

This thesis produces a single-year inventory for 2025 and is explicitly not a multi-year change detection study.

In scope:

- Binary classification of roof planes into vegetated vs non-vegetated based on roof-plane objects derived from LoD2.
- Use of multispectral indicators (including NDVI) and roof-plane geometry as features.
- City hold-out evaluation to assess cross-city generalization, followed by state-scale inference for Baden-Württemberg.

Out of scope (delimitations):

- Multi-year temporal trend analysis (e.g., 2020–2025 change); seasonal summaries may be used as features for robustness, but the output product is a 2025 inventory rather than a time series.
- Roof suitability/potential mapping (where green roofs *could* be built), which is conceptually different from mapping existing green roofs.

4.5 Expected outputs

- A roof-plane-level geospatial dataset for Baden-Württemberg (2025) containing predicted class labels and model confidence/probability values per roof plane.
- A validated workflow suitable for scaling inventory production.
- Evaluation results demonstrating cross-city generalization performance and model comparison.

5 Methodology

This thesis implements a roof-plane-level workflow to detect vegetated roofs and produce a 2025 green roof inventory for Baden-Württemberg. The workflow integrates roof-plane geometry from LoD2 building models with multispectral vegetation cues from PlanetScope Basemaps Monthly mosaics. The methodology is structured into five phases below

Phase 1: Geometry and Spectral Feature Engineering

Roof planes are extracted from LoD2 CityGML and cleaned to form the object units of analysis. PlanetScope Basemaps Monthly mosaics are used to compute roof-plane spectral summaries (e.g., band means and NDVI statistics) through polygon-based aggregation. The resulting dataset consists of roof-plane objects with geometric and spectral features.

Phase 2: Manual Labeling

A binary reference dataset is created via manual labeling in four cities (Stuttgart, Karlsruhe, Freiburg im Breisgau, and Tübingen). Each labeled roof plane is assigned to one of two classes: vegetated (green roof) or non-vegetated. This labeled dataset provides the supervised learning target required for model training.

Phase 3: Model Selection

Candidate classifier models (Random Forest and Multilayer Perceptron) are trained on manually labeled roof planes from four cities (Stuttgart, Karlsruhe, Freiburg im Breisgau, Tübingen). Model performance is evaluated using leave-one-city-out (LOCO) validation to test transferability to unseen cities.

Phase 4: Refine Features

Model refinement is conducted to improve performance. This study tried to tune in/out several input parameters based on intuitive reason, feature importance, and group ablation test (e.g., geometric vs spectral groups, seasonal subsets). This phase reduces reliance on incidental correlations or redundancies.

Phase 5: State-scale inference and inventory post-processing.

The best-performing model is applied to all roof planes in Baden-Württemberg to generate probabilities and predicted labels. Post-processing rules transform probabilistic outputs into a stable inventory product, using several threshold slicing approaches.

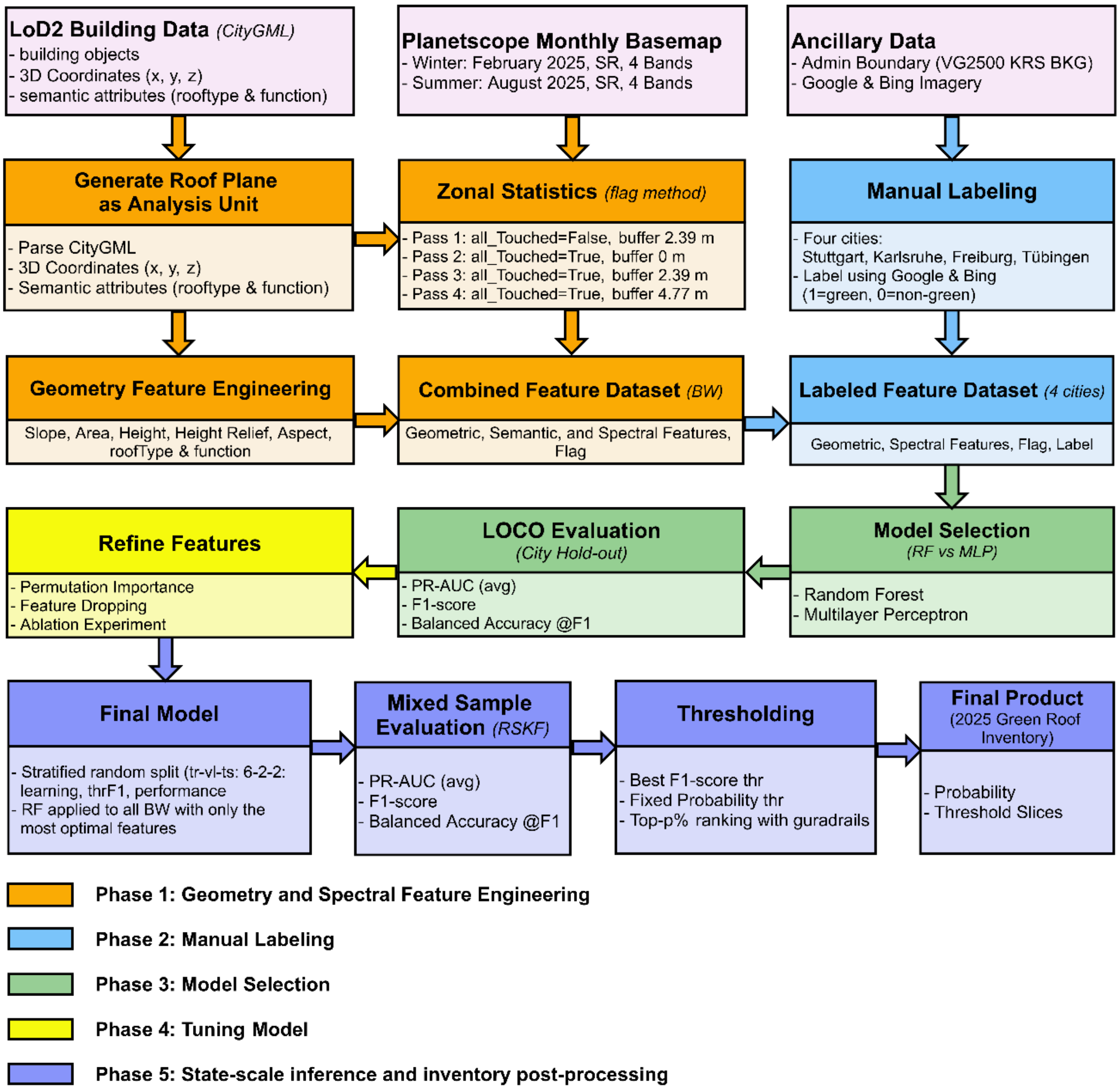


Figure 4: Workflow Diagram

5.1 Data

5.1.1 Study area

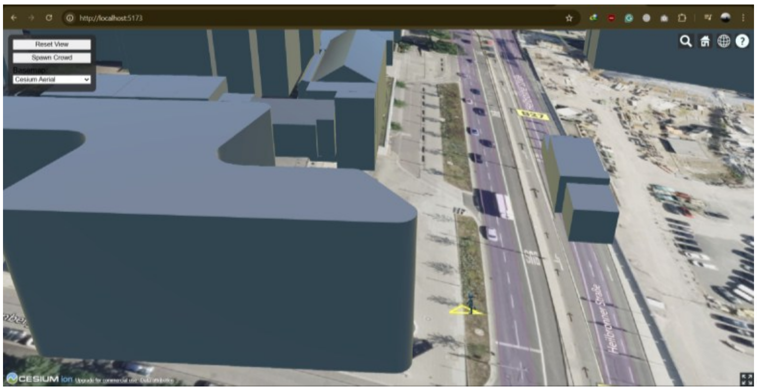
The study is conducted in the federal state of Baden-Württemberg, Germany, which spans approximately 35,751 km² and encompasses a diverse range of urban, suburban, and rural environments. Key cities such as Stuttgart, Karlsruhe, Freiburg im Breisgau, and Tübingen are included in the training phase due to their architectural and vegetation contexts. This geographic diversity provides a solid foundation for testing the spatial generalization capabilities of the model. The methodological goal is a state-scale inventory, which requires a workflow that remains stable under varying building typologies, roof materials, and local illumination/seasonal conditions across multiple cities. Cross-city evaluation is therefore treated as a methodological requirement rather than an optional experiment.

5.1.2 Data Sources

Table 1: Datasets used within the study

Dataset*	Description	Source
LoD2 Building Data	3D building models with flat or inclined roof surfaces encoded as CityGML	LGL BW Open Data Portal
PlanetScope Imagery	Fine resolution (~5 m) 4-band mosaics (R, G, B, NIR); monthly frequency	Planet Labs via LGL BW License & API
Google & Bing Imagery	High resolution of open data image to help labeling process	Google & Bing Image WMS
Administrative Boundaries	District and municipal boundaries for masking and statistics	Bundesamt für Kartographie und Geodäsie (BKG)

*All spatial datasets are projected to EPSG:25832 (ETRS89 / UTM zone 32N) for consistency.



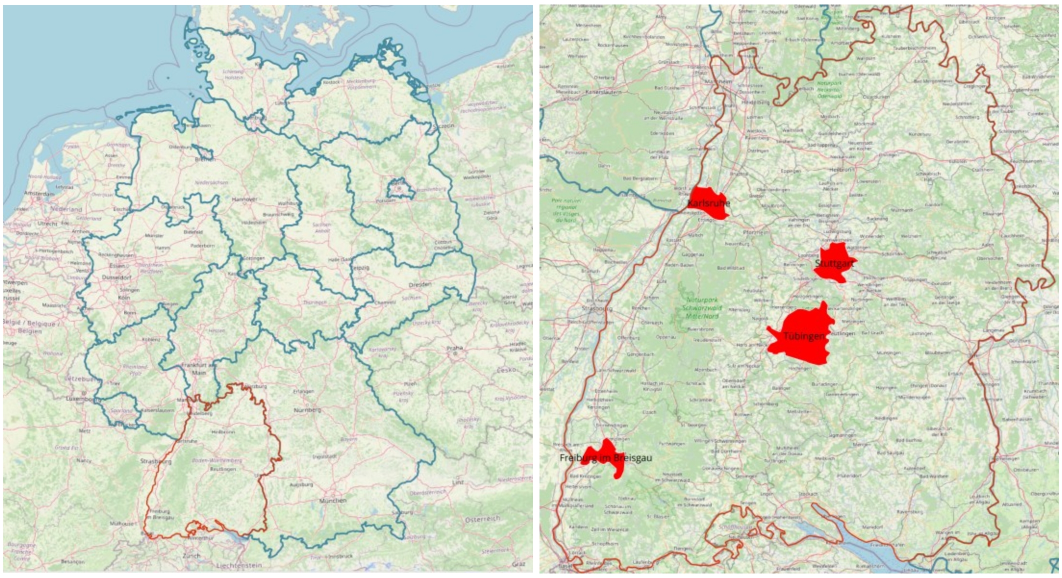


Figure 5: Dataset used within study

LoD2 Building Data

The LoD2 dataset provides detailed 3D building models. These models enable geometric analysis of rooftops, such as slope, aspect, and area, which are critical for identifying suitable surfaces for green roof detection. The main limitation is that the dataset may not capture recent construction or demolition, as source and update varies.

PlanetScope Imagery

PlanetScope provides daily, high-resolution imagery at 3–5 m spatial resolution in four spectral bands (Red, Green, Blue, and Near-Infrared). This enables vegetation monitoring through indices such as NDVI. The imagery's high temporal frequency supports seasonal and inter-annual monitoring. The Planetscope Images used within this study are the Monthly Basemap at 4.77 m resolution recorded in February 2025 to represent the Winter and August 2025 to represent the Summer.

Administrative Boundaries

District and municipal boundary data (VG2500 KRS) are used to mask analysis to Baden-Württemberg and to aggregate results at meaningful administrative levels. This allows the production of city-level statistics and comparisons across municipalities. While these boundaries do not affect classification directly, they provide essential spatial context for policy-relevant reporting and chunking at state-level production phase.

Google & Bing Imagery

High-resolution imagery from Google and Bing was used exclusively to support the manual validation process during labeling. These basemap images provided visual confirmation for the presence or absence of vegetation on rooftops. Importantly, these datasets were not used as direct input into the machine learning models but rather served as external reference material to improve label accuracy.

5.2 Geometry Feature Engineering

The roof-plane geometry used in this thesis is derived from LoD2 (Level of Detail 2) building models provided in CityGML/GML format directly from the Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL-BW). LoD2 building data represents semantically structured 3D city models in which buildings are modeled as volumetric objects with explicitly defined roof, wall, and ground surfaces, each represented as 3D polygon geometries with XYZ coordinates. The extraction workflow converts 3D roof surface geometries into 2D polygon objects and computes per-roof geometric attributes (slope, aspect, area, and height proxies) that are later used as features for classification.

gml_id	DEBW_0010009MHGN
creationDate	2024-10-17
informationSystem	http://repository.gdi-de.org/schemas/adv/citygml/fdv/art.htm#_9100
name	DEBWL0010009MHGN
Gemeindeschluessel	8336014
Geometrietyp2DReferenz	3000
DatenquelleDachhoehe	1000
Grundrissaktualitaet	2024-04-01
DatenquelleBodenhoehe	1100
DatenquelleLage	1000
function	31001_2140
roofType	3100
measuredHeight	5.869
measuredHeight_uom	urn:adv:uom:m
name_	DEBWL0010009MHGN
CountryName	Germany
LocalityName	Efringen-Kirchen
ThoroughfareName	Neue Straße
ThoroughfareNumber	NULL
consistsOfBuildingPart|BuildingPart|creationDate	NULL
consistsOfBuildingPart|BuildingPart|roofType	NULL
consistsOfBuildingPart|BuildingPart|measuredHeight	NULL
consistsOfBuildingPart|BuildingPart|measuredHeight_uom	NULL

Figure 6: Attribute of LGL Baden-Württemberg LoD2 Data

5.2.1 Parsing strategy and namespace handling

CityGML datasets encode geometry and semantics using XML namespaces. Therefore, the parser explicitly defines namespaces for GML and CityGML building/core modules and queries elements using namespace-qualified XPath expressions. To handle minor file truncations or non-critical XML inconsistencies in large-scale processing, a forgiving XML parser configuration is used ("recover" mode) so that partially corrupted files can still be processed where possible.

```
# For parsing GML namespaces
ns = {
```

```
    'gml': 'http://www.opengis.net/gml',
    'bldg': 'http://www.opengis.net/citygml/building/1.0',
    'core': 'http://www.opengis.net/citygml/1.0'
}

# Forgiving XML parser: recovers from minor file truncations
XML_PARSER = etree.XMLParser(recover=True, huge_tree=True)

LOD_dir = Path(r"D:\LGL\01 LoD2 All")

# find all .gml files
gml_files = list(LOD_dir.glob("**/*.gml"))

print(f"Found {len(gml_files)} GML files:")
for f in gml_files:
    print(f)
```

5.2.2 Roof surface polygon extraction and vertex decoding

For each `bldg:Building` object, the workflow searches for `bldg:RoofSurface` elements and extracts their geometry from embedded gml:Polygon elements. Vertex coordinates are read from either: `gml:posList` (a single flattened coordinate list), or multiple `gml:pos` elements (coordinate triplets stored as repeated nodes). This logic is consistent with common GML geometry encodings, where polygon vertex positions can be provided either as a list or as separate coordinate nodes (Open Geospatial Consortium, 2023).

```
# finding RoofSurface
# inside the process_gml()
for roof in b.findall(".//bldg:RoofSurface", namespaces=ns):
    ...
    for poly in roof.findall(".//gml:Polygon", namespaces=ns):
        pts = _xyz_from_polygon(poly)
        if not pts:
            continue
```

The vertex decoding step first searches each `<gml:Polygon>` for a `<gml:posList>` element. If present, the coordinate text is parsed into floating-point numbers and then grouped into (x,y,z) triplets to reconstruct the polygon's 3D vertex sequence. If the polygon ring is explicitly closed (the last vertex is identical to the first) the duplicate closing vertex is removed to prevent double counting in later computations (e.g., slope, area). The resulting vertex list is only accepted if it contains at least three valid vertices; otherwise, the geometry is treated as invalid for roof-plane processing and is skipped.

```
# decodes vertices from <gml:posList>
# Inside _xyz_from_polygon(poly_elem):
pos = poly_elem.find(".//gml:posList", namespaces=ns)
if pos is not None and pos.text and pos.text.strip():
    nums = list(map(float, pos.text.strip().split()))
    ...
    pts = [tuple(nums[i:i+3]) for i in range(0, len(nums), 3)]
    if len(pts) > 1 and pts[0] == pts[-1]:
        pts = pts[:-1]
    return pts if len(pts) >= 3 else []
```

When a roof polygon does not contain a <gml:posList>, the workflow falls back to parsing multiple <gml:pos> elements inside the same <gml:Polygon>. In this fallback, all coordinate strings stored in the repeated gml:pos nodes are read and merged into a single numeric sequence, which is then re-grouped into (x,y,z) triplets to reconstruct the polygon's vertex list. If the polygon ring is explicitly closed (i.e., the last vertex repeats the first), the duplicate closing vertex is removed to avoid double-counting in subsequent geometric computations, and the cleaned list is returned only when at least three valid vertices are available.

```
# fallback that decodes vertices from multiple <gml:pos>
# Still inside _xyz_from_polygon(poly_elem):
poss = poly_elem.findall(".//gml:pos", namespaces=ns)
...
for pn in poss:
    vals = list(map(float, pn.text.strip().split()))
    if len(vals) >= 3 and (len(vals) % 3) == 0:
        nums.extend(vals)
...
pts = [tuple(nums[i:i+3]) for i in range(0, len(nums), 3)]
if len(pts) > 1 and pts[0] == pts[-1]:
    pts = pts[:-1]
return pts if len(pts) >= 3 else []
```

After extracting vertices with _xyz_from_polygon(poly), the code skips any polygon that does not yield a valid 3D vertex list. This happens in process_gml() via the guard clause if not pts: continue, which filters out polygons with missing/invalid coordinate encodings or fewer than three usable vertices.

```
# Guardrail
pts = _xyz_from_polygon(poly)
if not pts:
    continue
```

Each roof polygon is then stored as a 2D polygon geometry using only (x,y) coordinates for spatial operations such as overlay and zonal statistics, while the original z values are preserved for computing slope and height-related features.

5.2.3 Slope and aspect computation using a plane normal

For each roof polygon, a plane normal vector is computed using the cross product of two edge vectors derived from the first three vertices. Specifically, given three 3D points p_1, p_2, p_3, the vectors $v_1 = p_2 - p_1$ and $v_2 = p_3 - p_1$ are formed and their cross-product $n = v_1 \times v_2$ yields a normal vector orthogonal to the polygon plane. The vector is normalized to unit length to ensure stable slope/aspect calculations.

```
# Compute Normal Vector
def compute_normal(points):
    """Unit normal from first 3 points of a 3D polygon."""
    p1, p2, p3 = np.array(points[0]), np.array(points[1]), np.array(points[2])
    v1 = p2 - p1
    v2 = p3 - p1
    n = np.cross(v1, v2)
```

```
    norm = np.linalg.norm(n)
    if norm == 0:
        return np.array([0, 0, 1], dtype=float)
    return n / norm
```

Slope and aspect are then derived from the plane orientation. The slope is computed as the inclination of the plane relative to the horizontal, where 0^0 corresponds to a flat surface and 90^0 corresponds to a vertical surface. Aspect is reported as a compass direction clockwise from North, consistent with common GIS conventions. For near-flat surfaces, aspect is undefined and is therefore stored as `NaN` to avoid introducing arbitrary directions. Roof planes with near-zero are treated as vertical and receive a slope near 90^0 because a stable aspect is not meaningful in that configuration.

```
# Compute Slope and Ascpect
def slope_aspect_from_normal(n):
    """
    Slope/Aspect from plane z = a x + b y + d derived from normal n.
    - slope_deg: 0 flat → 90 vertical
    - aspect_deg: 0=N, 90=E, clockwise; NaN if nearly flat
    """
    nx, ny, nz = float(n[0]), float(n[1]), float(n[2])
    if abs(nz) < 1e-12:
        return 90.0, np.nan

    a = -nx / nz
    b = -ny / nz

    slope_rad = np.arctan(np.hypot(a, b))
    slope_deg = np.degrees(slope_rad)

    # math angle from +x (east), CCW
    ang_math = np.degrees(np.arctan2(b, a))
    # convert to compass: 0=N, 90=E, clockwise
    aspect_deg = (450.0 - ang_math) % 360.0

    if slope_deg < 1e-6:
        aspect_deg = np.nan
    return slope_deg, aspect_deg
```

5.2.4 Roof-plane area computation in 3D

Roof-plane area is computed from the 3D polygon geometry by projecting polygon vertices into a local 2D coordinate system aligned with the polygon plane and then applying the polygon area formula (shoelace formula) in that local 2D space. The shoelace formula (also known as Gauss's area formula) computes polygon area from ordered planar vertices using cross-multiplication of coordinates (O'Rourke, 1998). This projection-based approach allows the area to represent the true roof surface area rather than only the horizontal footprint area. In degenerate cases where a stable polygon plane cannot be derived (e.g., collinear points), the workflow falls back to computing the area in the (x,y) plane using the shoelace formula.

```
# Compute Area
def compute_area(points):
    """3D polygon area via projection to the polygon plane + shoelace."""
    pts = np.asarray(points, dtype=float)
```

```
    if pts.shape[0] < 3:
        return 0.0
    p1, p2, p3 = pts[0], pts[1], pts[2]
    v1 = p2 - p1
    v2 = p3 - p1
    n = np.cross(v1, v2)
    n_norm = np.linalg.norm(n)
    if n_norm == 0:
        x, y = pts[:, 0], pts[:, 1]
        return 0.5 * abs(np.dot(x, np.roll(y, 1)) - np.dot(y, np.roll(x, 1)))
    n = n / n_norm
    u = v1 / (np.linalg.norm(v1) + 1e-12)
    v = np.cross(n, u)
    rel = pts - p1
    xu = rel @ u
    yv = rel @ v
    return 0.5 * abs(np.dot(xu, np.roll(yv, 1)) - np.dot(yv, np.roll(xu, 1)))
```

5.2.5 Height estimation using GroundSurface median z (and fallback relief height)

A height proxy is computed for each roof plane to represent elevation relative to the building ground reference. For each building, the workflow estimates a ground elevation z_{ground} as the median z value across all vertices found in `bldg:GroundSurface` polygons. Using the median would mitigate against outliers and local irregularities. The roof height for a roof plane is then computed as:

$$h = \max(0, z_{max} - z_{ground})$$

where z_{max} is the maximum vertex elevation of the roof plane. If a GroundSurface is not available, a fallback height descriptor is computed as relief height, defined as the roof plane's internal z-range:

$$h_{relief} = \max(z) - \min(z)$$

This fallback does not represent true building height above terrain but still provides a geometric descriptor of vertical variation within the roof plane.

```
# Compute Height
def relief_height(points):
    """No-DTM relief per facet: max(Z) - min(Z)."""
    zs = [p[2] for p in points if len(p) == 3]
    if not zs:
        return None
    return max(0.0, float(max(zs) - min(zs)))

def _xyz_from_polygon(poly_elem):
    """
    Extract XYZ triplets from <gml:Polygon> via <gml:posList> or multiple <gml:pos>.
    Returns list[(x,y,z)] or [] if not 3D.
    """
    pos = poly_elem.find(".//gml:posList", namespaces=ns)
    if pos is not None and pos.text and pos.text.strip():
        nums = list(map(float, pos.text.strip().split()))
        if len(nums) >= 9 and (len(nums) % 3) == 0:
            pts = [tuple(nums[i:i+3]) for i in range(0, len(nums), 3)]
```

```
            if len(pts) > 1 and pts[0] == pts[-1]:
                pts = pts[:-1]
            return pts if len(pts) >= 3 else []
        return []

    # fallback: multiple <gml:pos>
    poss = poly_elem.findall(".//gml:pos", namespaces=ns)
    if not poss:
        return []
    nums = []
    for pn in poss:
        vals = list(map(float, pn.text.strip().split()))
        if len(vals) >= 3 and (len(vals) % 3) == 0:
            nums.extend(vals)
    if len(nums) >= 9 and (len(nums) % 3) == 0:
        pts = [tuple(nums[i:i+3]) for i in range(0, len(nums), 3)]
        if len(pts) > 1 and pts[0] == pts[-1]:
            pts = pts[:-1]
        return pts if len(pts) >= 3 else []
    return []

def _median_ground_z_for_building(bldg_elem):
    """
    Estimate ground elevation for a Building as median Z
    across all vertices in its bldg:GroundSurface surfaces.
    """
    z_vals = []
    for gs in bldg_elem.findall(".//bldg:GroundSurface", namespaces=ns):
        # collect z from any polygons under GroundSurface
        for poly in gs.findall(".//gml:Polygon", namespaces=ns):
            pts = _xyz_from_polygon(poly)
            if pts:
                z_vals.extend([p[2] for p in pts])
    if z_vals:
        return float(np.median(z_vals))
    return None
```

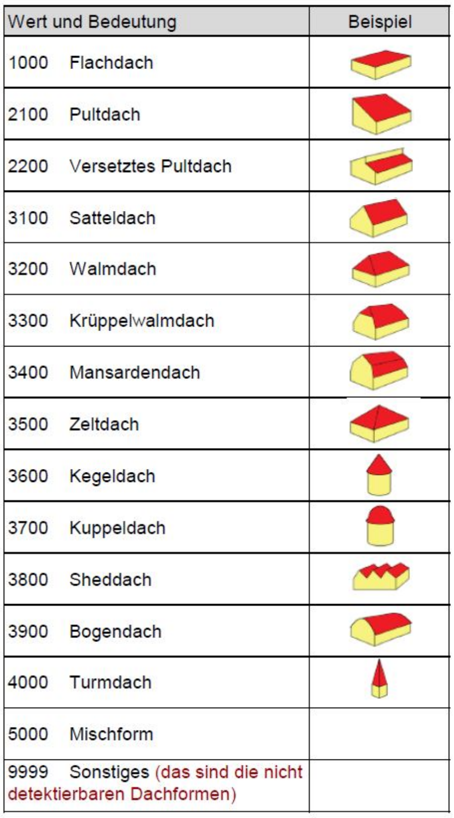
5.2.6 Attribute extraction and output storage

In addition to geometric features, the workflow extracts semantic attributes where available, such as `roofType` and `function`, using CityGML element lookups at building level and roof-surface level (with roof-surface values taking precedence when present). Each roof plane is written as one record containing:

- 2D polygon geometry,
- slope (degrees), aspect (degrees), area (m^2),
- height estimate (m) and relief height (m),
- building identifier and source file metadata,
- optional semantic attributes (`roofType`, `function`).

All outputs are stored incrementally into a GeoPackage layer. Batch writing is used to improve performance and to prevent memory growth when processing very large LoD2 datasets.

It it important to note that the LoD2 building data for Baden-Württemberg provide a semantic `roofType` classification that encodes the dominant roof shape per building or roof surface (Figure 4). Meanwhile, `function` describes the primary use of the building at a relatively fine level of detail, distinguishing for example purely residential buildings (Wohngebäude, Wohnheim), mixed-use buildings with housing and services, commercial and industrial buildings (Bürogebäude, Fabrik, Lagerhalle), accommodation and food-service buildings (Hotel, Jugendherberge, Gaststätte), public and institutional buildings (Schule, Krankenhaus, Rathaus), agricultural buildings (Scheune, Stall, Gewächshaus), and various technical and special-purpose structures (e.g. energy, water or transport infrastructure).



Wert und Bedeutung	Beispiel
1000 Flachdach	
2100 Pultdach	
2200 Versetztes Pultdach	
3100 Satteldach	
3200 Walmdach	
3300 Krüppelwalmdach	
3400 Mansardendach	
3500 Zeltdach	
3600 Kegeldach	
3700 Kuppeldach	
3800 Sheddach	
3900 Bogendach	
4000 Turmdach	
5000 Mischform	
9999 Sonstiges (das sind die nicht detektierbaren Dachformen)	

Figure 7: Roof Type Dictionary Provided by LGL-BW

During parsing, the corresponding code is extracted for each roof plane where available and stored as a categorical variable `roofType` and `function`. No attempt is made to reclassify or simplify the original scheme at the geometric level; instead, the raw codes are later handled via imputation and one-hot encoding in the machine-learning pipeline (Section 5.5.3). Conceptually, roof type and building function provides a coarse prior on structural suitability for greening, because certain roof type and building function are more likely to host vegetation systems than the other.

5.3 Spectral Feature Engineering

5.3.1 Accessing Planetscope Images

PlanetScope Basemaps Monthly mosaics were accessed through the Planet platform using the Planet command-line interface (CLI). Authentication was performed using the CLI login workflow, after which available mosaics were queried and inspected using the mosaics listing and metadata inspection commands. This step provides the mosaic identifier, product type, band configuration, declared coordinate system, and grid properties needed to confirm that the selected mosaics match the intended inputs. Planet mosaics are distributed as a grid of GeoTIFF "quads" (tiles), commonly on a Web Mercator tiling grid, and can be downloaded for a specified area of interest (AOI) using a GeoJSON geometry.

```
# Planet Login
import subprocess, textwrap

def planet(cmd: str):
    """
    Run Planet CLI command from inside the notebook and print output nicely.
    Example: planet("mosaics list | Select-String 2025")
    """
    full_cmd = f"planet {cmd}"
    print(">", full_cmd)
    p = subprocess.run(full_cmd, shell=True, capture_output=True, text=True)
    print(p.stdout)
    if p.stderr:
        print(p.stderr)
    return p.stdout

planet("--version")
planet("auth login")
```

For this thesis, two monthly basemap mosaics (February and August) from 2025 were selected as winter and summer representations, and the AOI was defined as the Baden-Württemberg boundary polygon provided as a GeoJSON file converted from VG2500 KRS by the Bundesamt für Kartographie und Geodäsie (BKG). The mosaics were downloaded using the CLI's geometry-based download option, which retrieves all quads intersecting the AOI and stores them to a structured output directory This produced a set of monthly basemap GeoTIFF tiles for Baden-Württemberg for subsequent feature extraction.

```
BW_GEOJSON = r"D:\LGL\bw.geojson"

OUT_FEB_8B = r"...basemap_norm8b_202502"
OUT_AUG_8B = r"...basemap_norm8b_202508"

ID_FEB_8B = "fb9529df-c3ed-4122-bbd7-f740fb7850f8"
ID_AUG_8B = "31303f6f-b5c7-4659-bd47-ad234510ce4c"
```

5.3.2 Reprojecting basemap tiles

Planet basemap quads are commonly delivered in Web Mercator (EPSG:3857) or geographic coordinates depending on configuration. The basemap product specification explicitly lists EPSG:3857 as a supported coordinate system for quad GeoTIFFs. In this workflow, the downloaded quads were treated as EPSG:3857 and reprojected to ETRS89 / UTM Zone 32N (EPSG:25832) to match the coordinate reference system used for the LoD2 roof planes and state geodata processing in Baden-Württemberg. Using a common projected CRS reduces CRS mismatch during overlay operations and avoids repeated on-the-fly reprojection during zonal statistics. Reprojection was performed using GDAL's gdalwarp utility, which supports raster reprojection and resampling to a target coordinate system.

```
SRC_EPSG = "EPSG:3857"
DST_EPSG = "EPSG:25832"
RESAMPLING = "bilinear"
CREATE_OPTS = ["COMPRESS=DEFLATE", "TILED=YES"]
NUM_THREADS = "ALL_CPUS"
```

To scale to many tiles reliably, warping was executed in parallel using a thread pool and a safe-write strategy. Each warped output was first written to a temporary file and moved one by one into its final location only after successful completion. Existing outputs were skipped unless overwriting was explicitly enabled, preventing unnecessary recomputation.

Overviews are optionally generated using gdaladdo to improve visualization performance in GIS and to accelerate multi-scale reads, using factors 2, 4, 8, and 16 with average resampling.

```
cmd = ["gdaladdo", "-r", "average", str(tif), "2", "4", "8", "16"]
proc = subprocess.run(cmd, capture_output=True, text=True)
```

5.3.3 Preparing GeoPackage schema for zonal statistics

Zonal statistics requires a destination schema to store per-roof spectral summaries for each month/season. Because the roof-plane dataset is stored as a GeoPackage (SQLite-based), additional columns are added directly using `SQL ALTER TABLE`. In SQLite, adding a column does not rewrite the full table; existing rows receive `NULL` values until populated. This approach is computationally efficient for large roof-plane datasets and allows zonal statistics to be computed incrementally.

```
TARGET_COLUMNS = [
    "R_avg_Winter","G_avg_Winter","B_avg_Winter","NIR_avg_Winter",
    "NDVI_avg_Winter","NDVI_std_Winter","Pixel_Count_Winter",
    "R_avg_Summer","G_avg_Summer","B_avg_Summer","NIR_avg_Summer",
    "NDVI_avg_Summer","NDVI_std_Summer","Pixel_Count_Summer",
    "method_winter","method_summer"
]

with sqlite3.connect(GPKG) as conn:
    layer = _get_layer_name(conn)
    for col in TARGET_COLUMNS:
```

```
        _add_column_if_missing(conn, layer, col, "REAL")
```

To create a stable explicit ID usable for joins and debugging, the workflow adds a `roof_id` column and copies values from the existing `fid`. This is a pragmatic "ID mirror" so downstream steps do not depend on implicit GeoPackage row identifiers.

```
if new_field.lower() not in existing_cols:
    conn.execute(f'ALTER TABLE "{layer}" ADD COLUMN "{new_field}" INTEGER')

conn.execute(f'UPDATE "{layer}" SET "{new_field}" = "{source_field}"')
conn.commit()
```

5.3.4 General configuration and band selection

Roof-plane spectral features are computed using polygon-based zonal statistics on PlanetScope monthly basemap quads. For each Planet quad (GeoTIFF), roof-plane polygons intersecting the tile extent are rasterized onto the tile grid and used as labels, allowing per-roof aggregation of pixel values. This approach avoids repeated clipping operations per polygon and enables efficient, tile-wise processing at state scale. Rasterization is performed with `rasterio.features.rasterize`, where the `all_touched` flag determines whether only pixels whose centers fall inside a polygon are counted (`False`, default) or whether any pixel touched by polygon boundaries is included (`True`).

The workflow computes features separately for Winter and Summer using Planet basemap folders already reprojected to EPSG:25832. PlanetScope normalized SR basemaps are multi-band. For this workflow selects Blue, Green, Red, and NIR bands and uses the alpha band as a nodata indicator.

```
RASTER_FOLDERS = {
    "Winter": r"...\basemap_norm8b_202502_25832",
    "Summer": r"...\basemap_norm8b_202508_25832",
}

BAND_B   = 2
BAND_G   = 4
BAND_R   = 6
BAND_NIR = 8
BAND_A   = 9  # Alpha
```

5.3.5 Tile-wise roof selection and buffering

For each raster tile, only roofs intersecting the tile bounding box are loaded from the GeoPackage. To reduce edge effects where roof polygons align poorly with the raster grid, a half-pixel buffer is applied in strict mode. The buffer is set to half of the Planet pixel size (≈2.39 m), which increases the chance that roofs intersect at least one pixel center in all_touched=False mode.

```
PLANET_PIXEL_M = 4.7773143
BUFFER_M = PLANET_PIXEL_M / 2.0   # ≈ 2.39 m

```

```
with fiona.open(gpkg_path, layer=layer) as src:
    for feat in src.filter(bbox=bbox):
        rid = int(feat["properties"].get("roof_id"))
        g = shape(feat["geometry"]).buffer(buffer_m)
        yield (mapping(g), rid)
```

5.3.6 Rasterizing roof planes into a label grid

Roof polygons (optionally buffered) are rasterized into an integer label array matching the raster tile shape. Each pixel is assigned the corresponding `roof_id` if it belongs to a roof; pixels not covered by roofs receive 0. The `all_touched` parameter is used to switch between strict centroid-based inclusion and a more permissive boundary-touch inclusion.

```
labels = np.zeros((ds.height, ds.width), dtype=np.int64)

rasterize(
    shapes=shapes,            # (geom, roof_id)
    out=labels,
    transform=ds.transform,
    fill=0,
    all_touched=all_touched_flag,
    dtype=np.int64,
)
```

5.3.7 Valid pixel masking using band masks and Planet alpha band

Pixels are included in the zonal statistics only if (i) they belong to a roof (labels > 0), (ii) they are valid in the rasterio per-band masks (mask > 0 indicates valid pixels), and (iii) the alpha band mask is valid. Rasterio's mask semantics follow GDAL's valid data mask convention: non-zero values indicate valid data, while zero indicates invalid/nodata. The Planet basemaps alpha band indicates regions with no imagery data and is therefore used to further exclude nodata regions.

```
mB = ds.read_masks(BAND_B)   > 0
mG = ds.read_masks(BAND_G)   > 0
mR = ds.read_masks(BAND_R)   > 0
mN = ds.read_masks(BAND_NIR) > 0
mA = ds.read_masks(BAND_A)   > 0

valid = mB & mG & mR & mN & mA & (labels > 0)
if not np.any(valid):
    return ""
```

5.3.8 NDVI computation and per-roof aggregation (mean, standard deviation, pixel count)

NDVI is computed per pixel as (NIR−Red)/(NIR+Red), a standard vegetation index that leverages high vegetation reflectance in NIR and absorption in red wavelengths (Tucker, 1979). Pixels with zero denominator are safely excluded from division. To efficiently compute roof-level statistics, the workflow aggregates per roof using sum and sum-of-squares for NDVI and bands, enabling reconstruction of means and variances after tile reduction.

```
# NDVI Calculation
denom = (NIR + R)
ndvi = np.zeros_like(NIR, dtype=np.float32)
nz = denom != 0
ndvi[nz] = (NIR[nz] - R[nz]) / denom[nz]
```

```
# sum/sumsq aggregation per roof_id
arrays = {"ndvi": ndvi, "r": R, "g": G, "b": B, "nir": NIR}
sums_df = compute_group_sums(labels, valid, arrays)
```

After merging tile-level partial outputs, final per-roof means and NDVI standard deviation are computed using:

$$\mu = \frac{\sum x}{n}, \sigma = \sqrt{\max\left(0, \frac{\sum x^2}{n} - \mu^2\right)}$$

This variance-from-moments formulation is widely used in streaming and tile-based aggregation because it is decomposable across partitions.

```
mean = np.divide(s, c, out=np.zeros_like(s), where=c>0)
ex2  = np.divide(ss, c, out=np.zeros_like(ss), where=c>0)
var  = np.clip(ex2 - mean*mean, 0.0, None)
out[std_name] = np.sqrt(var)
```

5.3.9 Alpha median per roof plane

In addition to mean reflectance/NDVI, the workflow stores an alpha median per roof for debugging purpose. The debugging serve as an indicator of whether the related roof plane has proper extraction of zonal statistics later on. In a normal case, a NULL spectral value on certain roof plane might be caused by the NULL or 0 value in alpha median. The median is computed exactly per tile using a label-wise sorted grouping strategy. During reduction, each roof can appear in multiple tiles; to avoid mixing partial medians across tiles, the workflow selects the tile-level median with the largest alpha_count (i.e., the tile contributing the most valid roof pixels), treating it as the most representative alpha statistic.

```
# exact median per tile + best selection
alpha_df = segmented_median(labels, valid, A.astype(np.float64))
alpha_best = (
    big.sort_values(["roof_id","alpha_count"], ascending=[True, False])
       .drop_duplicates(subset=["roof_id"])[["roof_id","alpha_median","alpha_count"]]
)
```

5.3.10 Zonal statistics strategies

A strict centroid-based rasterization (`all_touched=False`) can fail to assign any pixels to very small or narrow roof planes due to grid alignment. To maximize coverage while controlling false inclusions, the workflow uses a two-pass hybrid strategy:

- Pass 1 (strict, method = 1): buffer = 2.39 m (half pixel), `all_touched=False`
- Pass 2 (fallback/touch, method = 2): buffer = 0.0 m, `all_touched=True` applied only to roofs still missing alpha after strict pass

The use of `all_touched=True` is explicitly recognized as more permissive (includes any touched pixel), whereas `all_touched=False` is centroid-based and more conservative. The workflow stores the applied method as method_winter / method_summer so downstream modeling can trace which roofs relied on fallback extraction.

```
# Pass 1
p = process_tile(tif, season, "strict", buffer_m=2.39, all_touched_flag=False)
...
update_gpkg_for_season(conn, season, df, method_label=METHOD_STRICT)

# detect fallback candidates
sql = f"SELECT roof_id FROM {LAYER_NAME} WHERE {alpha_field} IS NULL"
fallback_roof_ids[season] = {int(r[0]) for r in cur.execute(sql).fetchall()}

# Pass 2
p = process_tile(tif, season, "touch", buffer_m=0.0, all_touched_flag=True, roof_id_filter=ids)
...
update_gpkg_for_season(conn, season, df, method_label=METHOD_TOUCH, only_if_alpha_null=True)
```

Additional permissive variants were tested (methods 3–4), combining all_touched=True with buffering of 2.39 m and 4.77 m.

- Pass 3 (fallback/touch, method = 3): buffer = 2.39 m (half pixel), `all_touched=True` applied only to roofs still missing alpha after Pass 2
- Pass 4 (fallback/touch, method = 4): buffer = 4.77 m(one pixel), `all_touched=True` applied only to roofs still missing alpha after Pass 3

These variants reduced the number of roofs with NULL alpha values (i.e., increased coverage) but introduced visually implausible associations where pixels were assigned to roof-plane geometries that do not represent valid roof surfaces for vegetation detection (e.g., very small artifacts such as poles/towers or non-roof structures encoded as roof planes). Because the objective of this thesis is an inventory-grade roof-plane classification, the pipeline prioritizes geometric plausibility and conservative pixel assignment over maximal coverage. Therefore, the final workflow uses method 1 and method 2 only, and remaining NULL-alpha roofs are retained as missing rather than forcing unreliable statistics.

Table 2: Zonal statistics method variants (0–4) and their trade-offs

Method	all_touched	buffer_m	Intended behavior	Typical risk	Final decision
0	False	0.00	Conservative centroid-based pixel inclusion.	Many small/narrow roofs get 0 pixels (missing stats)	Tested (baseline), not used

1	False	2.39	Conservative + half-pixel buffer to reduce edge misses.	Still misses extreme small artifacts; may slightly expand roofs.	Used (primary)
2	True	0.00	Permissive inclusion for remaining NULL-alpha roofs only.	Boundary-touch pixels may introduce noise at edges.	Used (fallback only)
3	True	2.39	More permissive + buffer to maximize coverage.	Higher chance of implausible pixel assignment to odd roof planes.	Rejected (QA issues)
4	True	4.77	Most permissive + ~1 pixel buffer to maximize coverage.	Strongly increases spurious assignments (e.g., tiny/odd structures).	Rejected (QA issues)

5.4 Labeling dataset

A reference dataset was created through manual labeling of roof planes in QGIS. Labels were required to train and evaluate supervised classifiers at roof-plane level, consistent with the object unit used throughout the workflow. Labeling was performed for four cities (i.e. Stuttgart, Karlsruhe, Freiburg im Breisgau, and Tübingen) as they represent diverse landscape of urban, rural, and vegetation aspect.

5.4.1 Labeling environment and imagery sources

Roof-plane geometries were loaded in QGIS and visually interpreted using high-resolution basemap layers from Google and Bing as contextual reference. PlanetScope basemaps were not used for manual interpretation because their spatial detail was insufficient (4.77m resolution) for reliably distinguishing vegetated roof structures at roof-plane scale in many cases. The basemap from Google and Bing Imagery served strictly as a visual aid for interpretation during labelling and not to be treated as input for learning model.

Figure 8: Comparison of Planetscope, Google, and Bing Imagery (left to right)

5.4.2 Operational definition of the target class

Labels were assigned as a binary target variable, where 0 indicates a non-vegetated roof plane and 1 indicates a vegetated roof plane (“green roof”). A roof plane was labeled as vegetated if approximately 60% or more of the roof-plane area was covered by visible vegetation. Vegetation included typical green roof surfaces such as sedum mats, rooftop gardens, or grass-like cover. Roof surfaces that appeared green due to materials, coatings, or imaging artifacts were not intended to be labeled as vegetated. Roof planes that were strongly shadowed or ambiguous were handled using a conservative decision rule

Figure 9: Various sample of Green Roof (green), Non-green Roof (Red), and Unlabeled (blue)

5.4.3 Sampling strategy

To reduce spatial bias, roofs were labeled using a spatially distributed sampling strategy within each city. The goal was to include roof planes across different neighborhoods and urban contexts, and to capture variability in roof geometry (e.g., flat and pitched roofs), roof size, and surrounding land use. Both positive (vegetated) and negative (non-vegetated) examples were actively included to avoid selection based only on visually salient green roofs.

5.4.4 Handling ambiguity and label consistency

Not all roof planes can be interpreted with high confidence due to factors such as cast shadows, rooftop installations, mixed surface materials, or partial occlusion. To reduce label noise, the labeling process was designed to avoid ambiguous cases whenever possible, focusing primarily on roof planes whose vegetation status could be interpreted clearly from high-resolution reference imagery. If an ambiguous roof plane was nevertheless included (e.g., because vegetation cover was still visually dominant), it was labeled using the same operational threshold applied throughout the dataset. Roof planes with approximately ≥ 60% visible vegetation cover were labeled as 1 (vegetated), whereas roof planes below this threshold were labeled as 0 (non-vegetated). This conservative strategy reduces the probability of systematic mislabeling at the cost of leaving some borderline roofs unlabeled. Ambiguous cases were under-sampled by design; therefore, the training dataset emphasizes clear examples and may under-represent borderline roofs.

To support consistency over time, the same decision rule was applied across all four cities, and a subset of labeled roofs was revisited during labeling to verify that the threshold interpretation remained stable.

5.4.5 Output of the labeling phase

The output of this phase is a roof-plane training dataset stored as a filtered GeoPackage for the four training cities. Each roof plane contains a unique identifier (roof_id), city identifier, and a binary label field indicating vegetated (1) or non-vegetated (0). These labels are used in the subsequent model training.

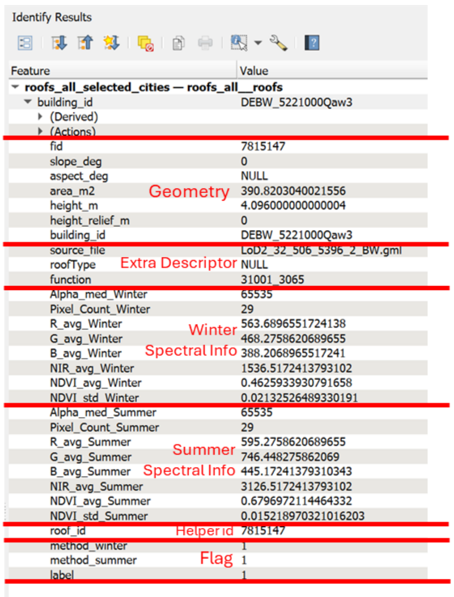


Figure 10: Labeling Dataset Output

5.4.6 Label dataset summary and quality control

After manual labeling, the resulting training dataset was filtered to include only roof planes from the four training cities and to ensure each roof plane appears only once. A primary quality control step was to verify the uniqueness of the identifier used throughout the workflow. The check "Duplicate `roof_id` in filtered training: 0" confirms that each roof plane in the training dataset has a unique `roof_id` and that no roof plane was accidentally duplicated during filtering, exporting, or merging. This is critical because duplicated objects can bias model fitting and inflate validation performance if the same object appears multiple times in training or testing splits.

The overall class distribution shows a clear imbalance, with 83.789% non-vegetated roofs (label = 0) and 16.211% vegetated roofs (label = 1). Such imbalance is expected because green roofs represent a minority of the overall roof stock in most urban contexts. Class imbalance is relevant for model training and evaluation because naive accuracy can become misleading under strong imbalance; therefore, later evaluation emphasizes imbalance-aware metrics (e.g., precision–recall behavior) and appropriate training strategies (Saito & Rehmsmeier, 2015).

City-wise counts indicate that the labeled dataset is distributed across all four cities, but the positive class frequency varies by city

Table 3: Labeled Dataset Summary

City	Labeled roofs (n)	Vegetated (label=1)	Non-vegetated (label=0)	Share vegetated (%)
Stuttgart	5475	1234	4241	22.5
Karlsruhe	4583	752	3831	16.4
Freiburg im Breisgau	4304	550	3754	12.8
Tübingen	3854	417	3437	10.8

These differences reflect that the labeled sample was not forced into equal class proportions across cities, but instead aimed to capture representative urban variability while still ensuring sufficient positive examples for supervised learning. The city-wise variation in class frequency is also relevant for leave-one-city-out validation, because each fold's training set inherits the class composition of the included cities.

5.5 Model selection

Model selection aims to identify a classifier that generalizes across cities and is therefore suitable for statewide inference. Candidate models were trained on the manually labeled roof-plane dataset and evaluated under a `leave-one-city-out` (LOCO) protocol, where each fold holds out one complete city as the test domain. This group-based validation is implemented with `LeaveOneGroupOut`, which ensures that samples from the same group (city) never appear in both training and testing within a fold.

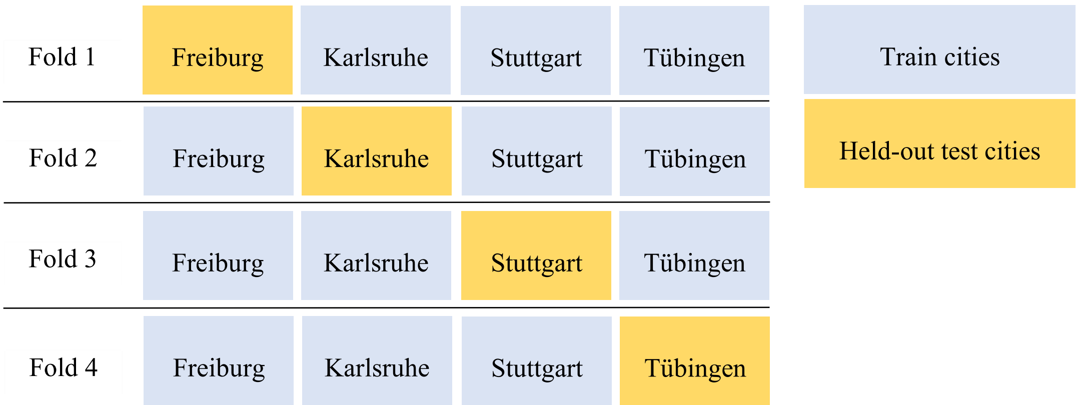


Figure 11: Leave One City Out (LOCO) Illustration

5.5.1 Training dataset loading and quality filtering

The labeled roof-plane dataset is stored as GeoParquet and loaded with GeoPandas. GeoParquet defines how geospatial vector data are stored in Parquet, enabling efficient columnar access and consistent CRS/geometry metadata storage.

```
gdf_train = gpd.read_parquet(TRAIN_PATH)

mask_good = (
    gdf_train["method_winter"].isin(ALLOWED_METHODS) &
    gdf_train["method_summer"].isin(ALLOWED_METHODS)
)
gdf_train_f = gdf_train.loc[mask_good].copy()
```

Before model training, roof planes are filtered using flagged quality (`method_winter`, `method_summer`). Only roofs whose zonal-statistics extraction succeeded under Pass1 and Pass 2 are retained. This avoids training on samples with missing or unreliable spectral features. This filtering reduces the initial training set from 19,758 to 18,216 roof planes (dropped: 1,542), keeping the model to be trained on samples with consistent feature availability.

5.5.2 LOCO-ready feature matrix and grouping variable

For machine learning, only the feature columns and essential identifiers are retained. Geometry is excluded from the feature matrix because model fitting operates on tabular features; the `roof_id` is kept separately for traceability. The target vector y is binary (0/1) and the grouping variable groups stores the city name, which drives LOCO splitting.

```
df_ml = pd.DataFrame(gdf_train_f[FEATURE_COLS + [TARGET_COL, GROUP_COL, ID_COL] + TRACE_COLS])
y = df_ml[TARGET_COL].astype(int).values
groups = df_ml[GROUP_COL].astype(str).values
X = df_ml[FEATURE_COLS].copy()
```

5.5.3 Preprocessing strategy

Fixed feature set was used consisting of roof-plane geometric attributes (slope, aspect, area, and height proxies) and seasonal spectral summaries (band means and NDVI statistics for Winter and Summer). Two auxiliary extraction diagnostics which are `Pixel_Count_*` and `Alpha_med_*` were excluded from the baseline numeric feature set because they only used initially for debugging within zonal statistics (i.e., how many pixels were available and whether the tile contained valid imagery), rather than roof biophysical characteristics.

```
# Numeric baseline features (pixel_count + alpha_med excluded)
NUM_COLS = [
    "slope_deg", "aspect_deg", "area_m2", "height_m", "height_relief_m",
    "R_avg_Winter", "G_avg_Winter", "B_avg_Winter", "NIR_avg_Winter",
    "NDVI_avg_Winter", "NDVI_std_Winter",
    "R_avg_Summer", "G_avg_Summer", "B_avg_Summer", "NIR_avg_Summer",
    "NDVI_avg_Summer", "NDVI_std_Summer",
]
FEATURE_COLS = NUM_COLS + CAT_COLS
```

Preprocessing is implemented with a ColumnTransformer so that numerical and categorical variables can be handled appropriately without leakage. Numerical features are median-imputed and missingness indicators are added. Categorical features are imputed using the most frequent category and then one-hot encoded. One-hot encoding uses `handle_unknown="ignore"` to ensure that categories appearing only in a held-out test city do not crash inference

```
numeric_preprocess = SimpleImputer(strategy="median", add_indicator=True)

categorical_preprocess = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("ohe", OneHotEncoder(handle_unknown="ignore", sparse_output=True)),
])

preprocess_common = ColumnTransformer(
    transformers=[
        ("num", numeric_preprocess, NUM_COLS),
        ("cat", categorical_preprocess, CAT_COLS),
    ],
    remainder="drop",
    sparse_threshold=0.3,
)
```

5.5.4 Candidate model pipelines

Two model families are evaluated:

- Random Forest (RF), an ensemble of decision trees that is robust to nonlinearities and mixed feature types. Given that green roofs represent a minority class, training uses sample weights computed from class frequencies in the training split of each fold. In scikit-learn, compute_sample_weight(class_weight="balanced") assigns weights inversely proportional to class frequency

```
rf = RandomForestClassifier(
    n_estimators=500,
    min_samples_leaf=2,
    n_jobs=-1,
    random_state=42,
    class_weight="balanced_subsample",
)

pipe_rf = Pipeline(steps=[
    ("prep", preprocess_common),
    ("clf", rf),
])
```

- Multi-layer Perceptron (MLP), a feed-forward neural network trained with the Adam optimizer and configured with early stopping.

```
mlp = MLPClassifier(
    hidden_layer_sizes=(128, 64),
    solver="adam",
    alpha=1e-4,
    batch_size=512,
    learning_rate_init=1e-3,
    max_iter=80,
    early_stopping=True,
    n_iter_no_change=10,
    validation_fraction=0.15,
    random_state=42,
)

pipe_mlp = Pipeline(steps=[
    ("prep", preprocess_common),
    ("scale", StandardScaler(with_mean=False)),
    ("clf", mlp),
])
```

Both models are embedded into scikit-learn Pipelines to ensure preprocessing is fitted only on the training partition of each LOCO fold

5.5.5 Evaluation metrics and selection criteria

Model performance was evaluated under class imbalance using precision–recall based metrics. Precision–recall analysis is commonly preferred over ROC-based summaries when the positive class is rare, because it focuses directly on the retrieval quality of the minority class and is more informative for imbalanced classification (Saito & Rehmsmeier, 2015). The primary threshold-independent metric was PR-AUC, computed as Average Precision, which summarizes the precision–recall curve across all discrimination thresholds.

In addition to PR-AUC, an operating-point metric was reported to provide an interpretable threshold-based summary. For each LOCO test fold, the best F1-score was computed by evaluating the F1-score across all thresholds returned by the precision–recall curve and selecting the threshold that maximizes F1. The best-F1 threshold is used only as an evaluation summary per fold; final inventory thresholds for statewide inference are defined separately in the post-processing stage. The corresponding precision and recall at this threshold are reported as Precision@bestF1 and Recall@bestF1. This combination provides a compact summary of performance at an internally consistent operating point, while PR-AUC retains an overall ranking-based view independent of any single threshold.

```
precision, recall, thresholds = precision_recall_curve(y_true, y_prob)
p, r, t = precision[:-1], recall[:-1], thresholds
f1 = 2 * p * r / (p + r)
i = np.nanargmax(f1)
best_f1, prec_bestf1, rec_bestf1 = f1[i], p[i], r[i]
```

The LOCO loop trains on three cities and evaluates on the held-out city, reporting metrics per fold and aggregating mean and standard deviation across folds.

```
logo = LeaveOneGroupOut()
for fold_i, (tr_idx, te_idx) in enumerate(logo.split(X, y, groups=groups), 1):
    X_tr, X_te = X.iloc[tr_idx], X.iloc[te_idx]
    y_tr, y_te = y[tr_idx], y[te_idx]
    prob = pipe.predict_proba(X_te)[:, 1]
```

5.5.6 Model selection decision

Based on LOCO cross-validation across the four training cities, Random Forest (RF) was selected as the primary model for the statewide inference stage. RF showed more reliable high-precision operating behavior than the MLP in most held-out cities, which is aligned with the downstream requirement of producing an inventory product where false positives must be controlled. The MLP model was retained as a benchmark but was not used for final statewide prediction due to its weaker transfer performance and less stable operating-point behavior under LOCO evaluation (see Section 6.4.1).

5.6 Feature importance

To interpret the Random Forest (RF) classifier and to guide subsequent feature-set refinement, permutation feature importance was computed using Average Precision (PR-AUC) as the scoring function. Permutation importance quantifies the decrease in model performance after randomly shuffling one input feature while keeping the others unchanged; a larger performance drop indicates a larger contribution of that feature to predictive performance (Breiman, 2001). Because correlated predictors can share explanatory power and influence each other's importance values, the resulting ranking was treated as a screening signal rather than a definitive proof of causality; all feature dropping were therefore validated through cross-city evaluation (Strobl, et al., 2007).

```
pipe_full = fit_rf_all_data(B0_features, random_state=42)

perm = permutation_importance(
    pipe_full,
    X[B0_features], y,
    scoring="average_precision",
    n_repeats=10,
    random_state=42,
    n_jobs=-1,
)

perm_imp_raw = pd.Series(perm.importances_mean, index=B0_features).sort_values(ascending=False)
```

5.6.1 Permutation importance ranking

Table 4: Permutation Importance Ranking for RF Model

Rank	Feature	Importance
1	area_m2	0.102174
2	NDVI_avg_Winter	0.055276
3	height_m	0.049991
4	slope_deg	0.049436
5	roofType	0.038334
6	function	0.036510
7	NIR_avg_Summer	0.036069
8	NDVI_avg_Summer	0.035772
9	height_relief_m	0.029895
10	aspect_deg	0.023499
11	NDVI_std_Summer	0.018412
12	R_avg_Winter	0.018160
13	NDVI_std_Winter	0.017067
14	G_avg_Summer	0.016911
15	B_avg_Summer	0.016779
16	NIR_avg_Winter	0.016510
17	B_avg_Winter	0.014724
18	R_avg_Summer	0.013385
19	G_avg_Winter	0.012638

The permutation importance ranking shows that RF decisions are driven primarily by a combination of roof-plane geometry and vegetation-sensitive spectral information. The most influential variable is roof-plane area (`area_m2`, 0.1022), followed by winter vegetation intensity (`NDVI_avg_Winter`, 0.0553) and geometric descriptors (`height_m`, 0.0500; `slope_deg`, 0.0494). Building metadata (`roofType`, 0.0383; `function`, 0.0365) also provides notable contribution. Both summer and winter vegetation cues are represented among the upper ranks (`NIR_avg_Summer`, 0.0361; `NDVI_avg_Summer`, 0.0358), while several individual visible-band means show comparatively lower marginal contribution (e.g., `R_avg_Summer`, 0.0134; `G_avg_Winter`, 0.0126). This importance profile motivated a systematic check of whether the weakest-ranked variables can be dropped without harming LOCO generalization in Section 0, and it informed the ablation design in Section 5.7.

5.6.2 Targeted feature dropping

Starting from the full feature baseline (B0, 19 features), a first screening step dropped two lowest-importance variables (`G_avg_Winter`, `R_avg_Summer`) to define a 17-feature intermediate set (Experiment 1). In the next step, two targeted dropping were tested relative to Experiment 1: dropping `aspect_deg` (motivated by weak class separation in exploratory analysis) noted as Experiment 2 and dropping `height_relief_m` (testing whether the height variability proxy was necessary) noted as Experiment 3.

Results indicate that dropping aspect improved LOCO performance, while dropping `height_relief` reduced it. In particular, Experiment 1 (17 features) achieves PR-AUC = 0.6981, whereas dropping `aspect_deg` increases performance to PR-AUC = 0.7033. In contrast, dropping `height_relief_m` decreases performance to PR-AUC = 0.6920. Based on this evidence, aspect was treated as a candidate for exclusion, while `height_relief` was retained for later ablation design.

Table 5: Targeted Features Dropping Evaluated under LOCO (RF)

Configuration	Drop	Number of features	PR AUC mean	Best F1 mean	Precision@F1 mean	Recall@F1 mean
Baseline (B0)	-	19	0.698299	0.657291	0.624473	0.695961
Experiment 1	G_avg_Winter, R_avg_Summer	17	0.698119	0.656879	0.641492	0.673509
Experiment 2	G_avg_Winter, R_avg_Summer, Aspect	16	0.703343	0.661024	0.641555	0.683184
Experiment 3	G_avg_Winter, R_avg_Summer, height_relief_m	16	0.692001	0.658645	0.638012	0.681238

5.7 Ablation experiment

Ablation experiments were conducted to quantify the contribution of different feature subsets and to test whether predictive performance depends primarily on vegetation-sensitive spectral information, geometry, or contextual building attributes. In contrast to feature importance, which provides a model-dependent ranking, ablation directly evaluates the effect of dropping feature groups on model performance under a controlled evaluation protocol. All ablation variants were trained using the same Random Forest configuration and identical preprocessing (median imputation with missingness indicators for numeric variables, and one-hot encoding for categorical variables). Class imbalance was addressed via balanced sample weights during training.

Two complementary evaluation schemes were used:

- Leave-One-City-Out (LOCO) cross-validation, where each fold holds out one city as an unseen test region. This directly tests spatial transferability and is the primary evaluation protocol for this thesis. LOCO was implemented using a group-based splitter with city as the grouping variable.
- Mixed-sample cross-validation using Repeated Stratified K-Fold (RSKF 5×3), which ignores city grouping and evaluates performance under random stratified splits. This provides a robustness check under a less stringent, non-spatial generalization setting (Kohavi, 1995).

5.7.1 Experiment design and feature groups

All experiments start from the full feature universe (B0, 19 features). A compact baseline (B1, 16 features) drops three predictors identified as weak or unstable in the Section 5.6, i.e. `aspect_deg`, `G_avg_Winter`, and `R_avg_Summer`. Additional experiments were grouped into:

- Intuitive target configurations (C-series): designed to test redundancy and domain hypotheses (e.g., dropping raw Red/NIR while retaining NDVI).
- Group ablations (G-series): drop entire seasonal or spectral groups to quantify which information sources are essential.

The main feature groups were defined as follows:

- Winter spectra: Winter RGB + Winter vegetation features (NIR, NDVI, NDVI_std)
- Summer spectra: Summer RGB + Summer vegetation features (NIR, NDVI, NDVI_std)
- RGB (both seasons): all visible bands
- NIR/Vegetation (both seasons): NIR + NDVI + NDVI_std for both seasons
- Geometry-only: slope, aspect, area, height metrics.

```
BASE_FEATURES = FEATURE_COLS[:]  # full 19-feature universe (B0)

EXPERIMENTS = [
```

```
  dict(name="B0_baseline_all_features", drops=[]),
  dict(name="B1_baseline_feature_importance",
       drops=["aspect_deg","G_avg_Winter","R_avg_Summer"]),
  dict(name="G5_geometry_only", drops=ALL_SPECTRAL),
  # ... other experiments ...
]

def drop_feats(feats, to_drop):
    to_drop = set(to_drop)
    return [f for f in feats if f not in to_drop]
```

5.7.2 Ablation experiment results overview

Tables 6 and 7 summarize performance under LOCO and RSKF respectively. Each row reports mean ± standard deviation over folds, and delta_pr_auc_vs_baseline represents the difference relative to B0 under the same evaluation scheme. There appears to be some key pattern that can be observed from the ablation experiment.

From LOCO (spatial transferability):

- The compact baseline B1 performs slightly better than the full feature set B0, indicating that dropping weak predictors can marginally improve cross-city generalization.
- Dropping all vegetation-sensitive features (NIR/NDVI across both seasons) causes the largest reduction in LOCO PR-AUC (G4_no_nir_veg_both), confirming that vegetation signals are essential for green roof detection.
- The geometry-only model (G5_geometry_only) performs substantially worse than spectral-informed models, showing that geometry alone cannot reliably distinguish vegetated from non-vegetated roofs at the roof-plane level.
- Seasonal ablations show that both seasons contribute, but winter vs summer importance differs by experiment; dropping winter spectral features tends to reduce performance more than dropping summer features in this dataset.

Robustness check (RSKF mixed splits):

- Mixed-split PR-AUC values are higher than LOCO, as expected under a non-spatial split that allows training and testing on similar city distributions.
- Feature dropping pattern remains consistent with LOCO (e.g., dropping NIR/NDVI is strongly detrimental; geometry-only is worst), supporting the conclusion that vegetation cues drive discrimination.

Based on the ablation results, configuration C3 is selected as the final production feature set for state-scale inference because it preserves essentially the same predictive signal as the full feature set while reducing redundancy and stabilizing the model. Under LOCO, C3 achieves PR-AUC =

0.6938 ± 0.0955 (ΔPR-AUC = −0.0045 vs. B0), which is far closer to the full baseline than variants that drop key vegetation information (e.g., G4, ΔPR-AUC = −0.0857) or rely on geometry alone (G5, ΔPR-AUC = −0.1874). The mixed-sample evaluation confirms that C3 remains competitive (PR-AUC = 0.8440 ± 0.0081, ΔPR-AUC = +0.0012), indicating that the reduced feature set does not weaken discrimination under non-spatial splits. Conceptually, C3 drops raw Red and NIR band means while retaining NDVI/NDVI_std, which is a deliberate redundancy reduction because NDVI is computed from red and near-infrared reflectance (Rouse et al., 1974). This choice aligns with the overall goal of producing a stable 2025 Baden-Württemberg green roof inventory, where fewer, more meaningful predictors reduce failure modes and support consistent statewide deployment without sacrificing transferability.

Table 6: Ablation Experiment Summary (LOCO / city hold-out)

Experiment	Description	Type	Number of Features	PR-AUC (mean ± std)	Δ PR-AUC vs B0	Best F1 Score (mean ± std)	Precision @ F1	Recall @ F1
B0_baseline_all_features	All features retained (full feature set; no drops)	Baseline	19	0.6983 ± 0.0931	0.0000	0.6573 ± 0.0805	0.624	0.696
B1_baseline_feature_importance	Filter from feature importance (drops: aspect, G_avg_Winter, R_avg_Summer)	Baseline	16	0.7033 ± 0.0954	+0.0050	0.6610 ± 0.0819	0.642	0.683
C1_no_aspect_no_R_no_G	Remove aspect and all Red & Green bands (both seasons)	Intuitive target	14	0.6978 ± 0.0968	−0.0005	0.6611 ± 0.0801	0.646	0.678
C2_no_aspect_no_rgb_no_nir	Remove aspect and all RGB & NIR bands (both seasons)	Intuitive target	10	0.6732 ± 0.0872	−0.0251	0.6513 ± 0.0836	0.644	0.663
C3_no aspect_no_R_no_NIR	Remove aspect and Red & NIR (both seasons)	Intuitive target	14	0.6938 ± 0.0955	−0.0045	0.6532 ± 0.0838	0.629	0.680
C3S_C3_plus_no_summer_spectra	C3 but remove all the summer spectral	Intuitive target	10	0.6505 ± 0.0918	−0.0478	0.6374 ± 0.0915	0.599	0.684
C3W_C3_plus_no_winter_spectra	C3 but remove all the winter spectral	Intuitive target	10	0.6723 ± 0.0942	−0.0260	0.6389 ± 0.0732	0.600	0.684
G1_no_winter_spectra	Remove all winter spectral features (RGB + NIR + NDVI)	Group	13	0.6835 ± 0.0952	−0.0148	0.6429 ± 0.0763	0.604	0.688
G2_no_summer_spectra	Remove all summer spectral features (RGB + NIR + NDVI)	Group	13	0.6683 ± 0.0998	−0.0300	0.6412 ± 0.0866	0.616	0.672
G3_no_rgb_both	Remove all RGB bands (both seasons)	Group	13	0.6886 ± 0.0991	−0.0097	0.6576 ± 0.0846	0.620	0.700
G4_no_nir_veg_both	Remove NIR and NDVI (both seasons)	Group	13	0.6126 ± 0.0927	−0.0857	0.6111 ± 0.0799	0.557	0.679
G5_geometry_only	Geometry-only features (remove all spectral features)	Group	7	0.5109 ± 0.1071	−0.1874	0.5572 ± 0.0928	0.477	0.674

Table 7: Ablation Experiment Summary (Mixed samples / RSKF 5×3)

Experiment	Description	Type	Number of Features	PR-AUC (mean ± std)	Δ PR-AUC vs B0	Best F1 Score (mean ± std)	Precision @ F1	Recall @ F1
B0_baseline_all_features	All features retained (full feature set; no drops)	Baseline	19	0.8428 ± 0.0087	0.0000	0.7604 ± 0.0114	0.762	0.762
B1_baseline_feature_importance	Filter from feature importance (drops: aspect, G_avg_Winter, R_avg_Summer)	Baseline	16	0.8458 ± 0.0079	+0.0030	0.7653 ± 0.0096	0.772	0.760
C1_no_aspect_no_R_no_G	Remove aspect and all Red & Green bands (both seasons)	Intuitive target	14	0.8444 ± 0.0077	+0.0016	0.7659 ± 0.0098	0.764	0.769
C2_no_aspect_no_rgb_no_nir	Remove aspect and all RGB & NIR bands (both seasons)	Intuitive target	10	0.8296 ± 0.0098	−0.0132	0.7593 ± 0.0101	0.758	0.762
C3_no aspect_no_R_no_NIR	Remove aspect and Red & NIR (both seasons)	Intuitive target	14	0.8440 ± 0.0081	+0.0012	0.7651 ± 0.0085	0.767	0.764
C3S_C3_plus_no_summer_spectra	C3 but remove all the summer spectral	Intuitive target	10	0.8244 ± 0.0100	−0.0184	0.7535 ± 0.0066	0.745	0.766
C3W_C3_plus_no_winter_spectra	C3 but remove all the winter spectral	Intuitive target	10	0.8338 ± 0.0093	−0.0090	0.7572 ± 0.0089	0.763	0.754
G1_no_winter_spectra	Remove all winter spectral features (RGB + NIR + NDVI)	Group	13	0.8320 ± 0.0095	−0.0108	0.7532 ± 0.0118	0.742	0.766
G2_no_summer_spectra	Remove all summer spectral features (RGB + NIR + NDVI)	Group	13	0.8241 ± 0.0109	−0.0187	0.7459 ± 0.0080	0.742	0.754
G3_no_rgb_both	Remove all RGB bands (both seasons)	Group	13	0.8365 ± 0.0106	−0.0063	0.7596 ± 0.0104	0.761	0.760
G4_no_nir_veg_both	Remove NIR and NDVI (both seasons)	Group	13	0.7937 ± 0.0104	−0.0491	0.7223 ± 0.0107	0.717	0.729
G5_geometry_only	Geometry-only features (remove all spectral features)	Group	7	0.7354 ± 0.0141	−0.1074	0.6789 ± 0.0093	0.659	0.703

5.8 Final Model Configuration

The final classifier selected for this study is a Random Forest (RF), implemented with scikit-learn's RandomForestClassifier. The choice of RF over a multilayer perceptron (MLP) was based on the Leave-One-City-Out (LOCO) experiments (Section 5.5), where RF consistently achieved higher PR-AUC with more stable recall at high precision across the four held-out cities, while the MLP showed markedly weaker behavior in some folds (e.g. Freiburg im Breisgau, Tübingen) despite similar or higher complexity. In the context of a state-wide inventory with pronounced class imbalance and heterogeneous urban fabric, RF offered a more robust and interpretable baseline than the neural network alternative.

The final model operates on the C3 feature configuration, which removes raw Red and NIR band means from both seasons while retaining Blue and Green means and NDVI-based vegetation indices. This choice is supported by the ablation experiments result (Section 5.7.2), where C3 achieved PR-AUC values very close to the full 19-feature baseline under both LOCO and mixed-sample validation, while configurations that removed NDVI/NIR information showed substantial performance degradation. Given that NDVI is computed directly from red and near-infrared reflectance, dropping raw Red and NIR means reduces redundancy without discarding the principal vegetation signal.

The final feature set contains 14 predictors, consisting of 12 numeric features (geometric + seasonal vegetation cues) and 2 categorical attributes (`roofType`, `function`).

```
FEATURES_FINAL = [
    "slope_deg", "area_m2", "height_m", "height_relief_m",
    "G_avg_Winter", "B_avg_Winter", "NDVI_avg_Winter", "NDVI_std_Winter",
    "G_avg_Summer", "B_avg_Summer", "NDVI_avg_Summer", "NDVI_std_Summer",
    "roofType", "function",
]
```

Numerical features are preprocessed using median imputation with missingness indicators, and categorical features are handled via most-frequent imputation and one-hot encoding.

```
def make_preprocessor(num_cols, cat_cols):
    numeric = SimpleImputer(strategy="median", add_indicator=True)
    categorical = Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("ohe", OneHotEncoder(handle_unknown="ignore", sparse_output=True)),
    ])
    return ColumnTransformer(
        [("num", numeric, num_cols), ("cat", categorical, cat_cols)],
        remainder="drop",
        sparse_threshold=0.3,
    )
```

The RF hyperparameters were tuned empirically (starting from defaults and increasing tree count) and fixed as:

```python
def make_rf_pipeline(random_state=42):
    pre = make_preprocessor(NUM_COLS, CAT_COLS)
    rf = RandomForestClassifier(
        n_estimators=600,
        min_samples_leaf=2,
        n_jobs=-1,
        random_state=random_state,
        class_weight="balanced_subsample",
    )
    return Pipeline([("prep", pre), ("clf", rf)])
```

Key design decisions are:

- Many trees (n_estimators = 600) to stabilise predictions and feature usage.
- Leaf size ≥ 2 (min_samples_leaf = 2) to avoid overly specific leaves on noisy or rare patterns.
- Balanced subsampling (class_weight="balanced_subsample") to compensate for the strong class imbalance (≈16 % green roofs in the training set).

5.9 Thresholding strategies

The RF model outputs a continuous probability `p(green roof)` for each roof plane. Because an operational inventory requires binary layers, probability outputs are transformed into multiple thresholded products reflecting different use cases: a balanced best-F1 threshold from calculated from validation sample, high-confidence layers using fixed probability cutoffs (`p≥0.85`, `p≥0.90`), and ranking-based layers selecting the top 1–5% of roof planes per city with minimum-count guardrails. Selecting the best-F1 threshold on the validation set avoids tuning decision thresholds on the test set, maintaining an unbiased test evaluation.

Table 8: Different Thresholding Strategies for Slicing Green Roof Probability

Layer	Definition	Intended use
pred_f1	($p \geq T_{F1=0.61}$) (from validation)	Balanced inventory
pred_ge_085	($p \geq 0.85$)	High precision
pred_ge_090	($p \geq 0.90$)	Very high precision
pred_top1pct	Top 1% per city (min 50)	Candidate shortlist
pred_top2pct	Top 2% per city (min 100)	Candidate shortlist
pred_top5pct	Top 5% per city (min 250)	Broader shortlist

5.10 Performance Evaluation Matrices

Model performance is evaluated along two complementary axes:

- Model evidence, showing how well the RF separates green vs non-green roofs under spatial and random validation.
- Product slicing performance, showing how different thresholding and ranking strategies trade off precision, recall, and selection rate on the test set.

All primary metrics are based on the precision–recall (PR) curve, which is more informative than ROC curves in imbalanced settings where the positive class is rare (Saito & Rehmsmeier, 2015). The average precision (AP) is used as the scalar summary of the PR curve, often referred to as PR-AUC, and is equivalent to a weighted mean of precisions at different recall levels (Davis & Goadrich, 2006). In addition, the F1-score (harmonic mean of precision and recall) is used to identify an operating point on the PR curve. For that operating point, the balanced accuracy is reported, defined as the mean of the true positive rate for green roofs and the true negative rate for non-green roofs. This metric is relevant in the present setting because it prevents overall accuracy from being dominated by the majority class and explicitly shows whether the classifier is performing reasonably on both vegetated and non-vegetated roofs in an imbalanced dataset.

First, a Leave-One-City-Out (LOCO) experiment is conducted with the four cities Stuttgart, Karlsruhe, Freiburg im Breisgau, and Tübingen as groups. In each fold, three cities are used for training and the remaining city is held out entirely for testing. For every fold, the PR-AUC, the best F1-score on the PR curve, and the corresponding precision, recall, and balanced accuracy are computed from the predicted probabilities:

```
logo = LeaveOneGroupOut()
rows = []
for fold_i, (tr_idx, te_idx) in enumerate(logo.split(X_all, y_all, groups=groups_city), 1):
    X_tr, X_te = X_all.iloc[tr_idx], X_all.iloc[te_idx]
    y_tr, y_te = y_all[tr_idx], y_all[te_idx]

    pipe = make_rf_pipeline()
    sw   = compute_sample_weight("balanced", y_tr)
    pipe.fit(X_tr, y_tr, clf__sample_weight=sw)

    prob = pipe.predict_proba(X_te)[:, 1]
    ap   = pr_auc(y_te, prob)
    bf   = best_f1_from_pr(y_te, prob)
    bacc = bal_acc_at_threshold(y_te, prob, bf["thr"])
    ...
```

To complement the spatial evaluation, a stratified random split (60 % train, 20 % validation, 20 % test) is used. Generally, the train is used for fitting model, validation is used for making choices such as finding threshold for slicing, and test used for final unbiased evaluation.

The Random Forest is trained on the training subset, thresholds (including the best-F1 threshold) are selected on the validation subset, and all reported "product" metrics (e.g. threshold menus and ranking slices) are computed only on the held-out test subset.

```
# Stratified 60/20/20 split
X_tr, X_tmp, y_tr, y_tmp = train_test_split(
    X_all, y_all,
    test_size=0.4,
    stratify=y_all,
    random_state=42,
)
X_val, X_te, y_val, y_te = train_test_split(
    X_tmp, y_tmp,
    test_size=0.5,
    stratify=y_tmp,
    random_state=42,
)

# Fit on TRAIN, choose threshold on VAL
pipe_split = make_rf_pipeline()
sw = compute_sample_weight("balanced", y_tr)
pipe_split.fit(X_tr, y_tr, clf__sample_weight=sw)

p_val = pipe_split.predict_proba(X_val)[:, 1]
p_te  = pipe_split.predict_proba(X_te)[:, 1]

bf_val = best_f1_from_pr(y_val, p_val)
thr_f1 = bf_val["thr"]
```

The random-split protocol yields higher PR-AUC and F1-scores than LOCO because it only requires generalization to new roofs sampled from the same four cities, rather than to entire unseen cities. The gap between LOCO and mixed-sample performance is therefore expected and can be interpreted as the difference between spatial extrapolation (LOCO) and within-sample generalization (random split, mixed-sample CV).

To support practical interpretation for different inventory use cases, a product slicing matrix is computed on the test subset using the final RF model and several thresholding strategies (Table 8):

- F1-based threshold from validation (thr_f1)
- Fixed thresholds $p \geq 0.85$ and $p \geq 0.90$
- Top 1 %, 2 %, and 5 % slices with minimum-k guardrails

For each strategy, precision, recall, balanced accuracy, selection rate, and selected count are reported:

5.11 Deploying state-scale production

After model selection and evaluation, the final RF model is retrained on all available labeled roofs, using the full C3 feature set and the same preprocessing and class weighting scheme. This maximises the effective training data before state-wide inference:

```
pipe_final = make_rf_pipeline()
sw_full = compute_sample_weight("balanced", y_all)

pipe_final.fit(X_all, y_all, clf__sample_weight=sw_full)
```

Here, y_all comprises 18,216 labeled roof planes from the four training cities (≈16 % positives), filtered to retain only roof planes with valid zonal statistics from methods 1 and 2.

State-scale inference is performed by reading the roof-plane-level GeoParquet for Baden-Württemberg, splitting by city, and predicting in manageable chunks. For each city, the following steps are taken in sequence:

- Read only that city's roof planes from the BW GeoParquet.
- Apply the same strict method filter (method_winter ∈ {1,2} and method_summer ∈ {1,2}).
- Compute proba_greenroof using the final RF pipeline and the 14-feature C3 set.
- Derive multiple binary prediction layers (pred_f1, pred_ge_085, pred_ge_090, pred_top1pct, pred_top2pct, pred_top5pct).
- Save the resulting GeoDataFrame as a city-specific GeoParquet file under a structured output directory.

```
for city in cities:
    city_clean = safe_name(city)
    out_dir = os.path.join(OUT_DIR, f"city={city_clean}")
    os.makedirs(out_dir, exist_ok=True)
    out_path = os.path.join(out_dir, f"{city_clean}.geoparquet")

    gdf_city = gpd.read_parquet(BW_PATH, filters=[("city", "==", city)], columns=cols_needed)

    # strict method filter
    mask = (
        gdf_city["method_winter"].isin(ALLOWED_METHODS) &
        gdf_city["method_summer"].isin(ALLOWED_METHODS)
    )
    gdf_city = gdf_city.loc[mask].copy()

    proba = pipe_final.predict_proba(gdf_city[FEATURES_FINAL])[:, 1].astype(np.float32)
    gdf_city["proba_greenroof"] = proba

    # threshold-based layers
    gdf_city["pred_f1"]     = (proba >= thr_f1).astype("int8")
    gdf_city["pred_ge_085"] = (proba >= 0.85).astype("int8")
    gdf_city["pred_ge_090"] = (proba >= 0.90).astype("int8")

    # top-% layers with minimum-k guardrails
    pred1, thr1, k1 = top_pct_with_min(proba, 0.01, MIN_K["top1%"])
```

```
    pred2, thr2, k2 = top_pct_with_min(proba, 0.02, MIN_K["top2%"])
    pred5, thr5, k5 = top_pct_with_min(proba, 0.05, MIN_K["top5%"])

    gdf_city["pred_top1pct"] = pred1
    gdf_city["pred_top2pct"] = pred2
    gdf_city["pred_top5pct"] = pred5

    gdf_city.to_parquet(out_path)
```

6 Result and Discussion

This chapter presents and interprets the empirical results of the proposed green roof detection workflow. Building on the methods described in Chapter 5, it first clarifies the effective analysis domain by examining geometric and spectral coverage of LoD2 roof planes and PlanetScope mosaics. It then characterizes the labelled sample used for model training, including class balance and basic feature behavior, before turning to feature importance, ablation experiments, and final model performance.

6.1 Data coverage and missingness

The first step towards a state-scale green roof inventory is defining the *effective* population of roof planes that can be analysed with the available LoD2 and PlanetScope data. Although the official LoD2 dataset for Baden-Württemberg contains **14,695,948** roof planes in total, not all of these can be used for modelling due to geometry issues and missing spectral information.

On the geometry side, the LoD2 CityGML data contains a small but non-negligible share of problematic roof surfaces, including incomplete polygons, highly fragmented surfaces, very small technical structures (e.g. antenna masts, lift housings, cell towers) and artefacts produced by the original photogrammetric reconstruction. These objects are technically encoded as `bldg:RoofSurface` but do not represent meaningful roof planes in the sense of this study. The GML parsing workflow described in Chapter 5.2 extracts all roof polygons, computes their slope, aspect, area and height proxies, and discards polygons that cannot be converted into a valid 2D polygon or have degenerate 3D geometry. This already removes a fraction of extremely small or broken roof planes from further analysis but might still include self-intersecting or invalid geometry.

On the spectral side, the behavior of the zonal-statistics methods is strongly constrained by the moderate spatial resolution of PlanetScope. Each basemap pixel at 4.77 m resolution covers roughly 23 m² and is therefore comparable in size to many small roof planes. Narrow, highly articulated or very small LoD2 roof surfaces are often only one pixel wide or even smaller than a single pixel footprint. After reprojection and resampling, the exact alignment between LoD2 roof polygons and the Planet grid is also imperfect. As a result, whether a roof receives any valid pixels – and which pixels – is extremely sensitive to how rasterization and buffering are defined.

This explains the observed pattern in the pass-method statistics. Under method 0 (no buffer, all_touched = False) many small or narrow roofs do not intersect any pixel centres and therefore receive no spectral values at all. Method 1 adds a half-pixel buffer (≈ 2.39 m) around each roof while still using all_touched = False. This is a conservative way to compensate for pixel-grid misalignment: roof polygons are slightly enlarged so that at least one pixel centre falls inside most well-behaved roof surfaces, but pixels are still only counted if their centre lies within the buffered roof. Consequently, method 1 captures the vast majority of roofs that are large enough to be meaningful at Planet resolution and whose geometry is reasonably aligned with the grid.

Table 9: Number of Roof Planes Extracted by Each Passing Method

Method	Season	Number of roofs	Share of all roofs	Decision
1	Winter	13,118,400	0.8927	Used
2	Winter	234,689	0.0160	Used
3	Winter	112,222	0.0076	Rejected
4	Winter	69,400	0.0047	Rejected
(none)	Winter	1,161,237	0.0790	Rejected
1	Summer	13,118,400	0.8927	Used
2	Summer	1,082,481	0.0737	Used
3	Summer	254,940	0.0173	Rejected
4	Summer	110,836	0.0075	Rejected
(none)	Summer	129,291	0.0088	Rejected

Roofs that remain without valid pixels after method 1 are mostly at the resolution limit of PlanetScope or are geometrically unusual: tiny technical structures, thin slivers along ridges, or polygons whose planimetric extent is dominated by façade and edge effects. For these residual cases, method 2 switches to `all_touched = True` without buffering. This is an explicitly more permissive rule: any pixel that is touched by the roof polygon contributes to the statistics, even if its center lies outside the roof. At 4.77 m resolution, this often means using pixels that are partially mixed with adjacent streets, vegetation or fades. In this workflow, method 2 is therefore restricted to a fallback role and only applied to roofs that still have no valid alpha after method 1.

Methods 3 and 4 further combine `all_touched = True` with half-pixel and full-pixel buffers, respectively. In theory, these variants maximize coverage by aggressively pulling in any intersecting Planet pixels. In practice, visual inspection shows that at PlanetScope resolution this behavior is dominated by artefacts: tiny LoD2 roof planes and vertical structures suddenly receive statistics from pixels whose footprint is mostly off-roof, and mixed pixels at building edges are systematically assigned to roofs that are too small to be resolved. Figure 8 illustrates this effect: many of the surfaces that only become populated under methods 3–4 are precisely the very small or odd roofs that are least suited to 5 m imagery. Including them would nominally reduce the share of roofs with missing statistics but would also inject a long tail of physically implausible samples into the training and prediction domain.

Figure 12: Collection of Rejected Roof Plane (Red) shown in Planetscope (left) and Google Image (Left)

For this reason, the final workflow restricts the analytical domain to roofs for which both winter and summer zonal statistics are successfully computed under methods 1–2. The remaining group with method = NULL consists of roof planes that are either outside Planet coverage, fall entirely into masked or nodata regions, or are simply too small or geometrically unstable to be matched to a reliable Planet pixel at 4.77 m. Forcing spectral values onto these roofs would amount to inventing information below the sensor's effective resolution, which is incompatible with the "inventory-grade" goal of this thesis.

From the perspective of the thesis objectives, this section closes O1 (roof-plane dataset construction) and constrains O5 (state-scale inference): the 2025 inventory produced in later sections should be interpreted as an inventory of LoD2 roof planes that (i) pass basic geometric sanity checks and (ii) are sufficiently large and stable to receive valid winter and summer PlanetScope zonal statistics under the Pass Method 1 (strict) and Pass Method 2 (fallback) strategy. It is not a complete census of every physical roof object in Baden-Württemberg, but a resolution-aware inventory tailored to what can be reliably observed at ~5 m pixel size.

6.2 Sample characteristics and label behavior

Manual labelling was carried out for four cities, i.e. Stuttgart, Karlsruhe, Freiburg im Breisgau and Tübingen using high-resolution Google and Bing imagery in QGIS as visual reference (see Section 5.4). Roof planes were labelled as green (1) if approximately ≥60 % of the roof surface appeared vegetated in the high-resolution imagery; otherwise they were labelled as non-green (0). Ambiguous cases (e.g. partial vegetation, dense gravel, strong shadows) were generally not labelled to avoid contaminating the training set with borderline examples. In rare ambiguous cases where vegetation clearly exceeded ≈60 %, the roof plane was still labelled as 1; otherwise it was treated as 0. After filtering to roofs with valid winter and summer zonal statistics under methods {1, 2}, the labelled dataset contains 18,216 roof planes:

- Negative (non-green) roofs (label = 0): 15,263
- Positive (green) roofs (label = 1): 2,953

This corresponds to a positive class proportion of about 16.2 %, i.e. roughly one in six labelled roof planes is vegetated. The label distribution is therefore clearly imbalanced, which motivates the focus on precision–recall based metrics rather than accuracy or ROC curves in the evaluation. The labelled roofs are distributed across the four cities as follows (after all method filters):

- Stuttgart: 5,475 labelled roofs, of which 1,234 green
- Karlsruhe: 4,583 labelled roofs, of which 752 green

- Freiburg im Breisgau: 4,304 labelled roofs, of which 550 green
- Tübingen: 3,854 labelled roofs, of which 417 green

Stuttgart contributes the largest number of positive examples, reflecting both its size and a relatively higher presence of green roofs. Freiburg im Breisgau and Tübingen have fewer positives in absolute terms, which partly explains the higher variance observed for these cities under LOCO evaluation in Section 6.4.1.

The labelling protocol introduces two systematic biases that are relevant for interpreting the results:

- Emphasis on clear cases
  By deliberately skipping most ambiguous roofs, the training set is enriched in "clear" examples of green and non-green roofs. The model is therefore optimized for confidently vegetated and confidently non-vegetated roofs and may be more uncertain around borderline cases in the full inventory.
- City-specific sampling
  All labels come from four cities with comparatively good PlanetScope coverage and a certain level of green-roof adoption. The class balance and feature distributions observed here may not fully reflect conditions in smaller municipalities or industrial areas elsewhere in Baden-Württemberg.

Beyond class counts, the labelled dataset shows systematic differences in feature behavior between green and non-green roofs.

6.2.1 Geometric features behavior

For `slope`, green roofs are strongly concentrated on flat or almost flat roof planes in all four cities. The orange distributions (label = 1) show a pronounced peak at very small slopes (≈0–5°) and only a short tail towards higher inclinations, whereas non-green roofs (label = 0) occupy the full range up to 70–80° with a secondary mode around moderately pitched roofs. This confirms the intuitive expectation that vegetated roofs in the training data are predominantly flat or low-pitch, while steeper roofs are almost exclusively non-vegetated.

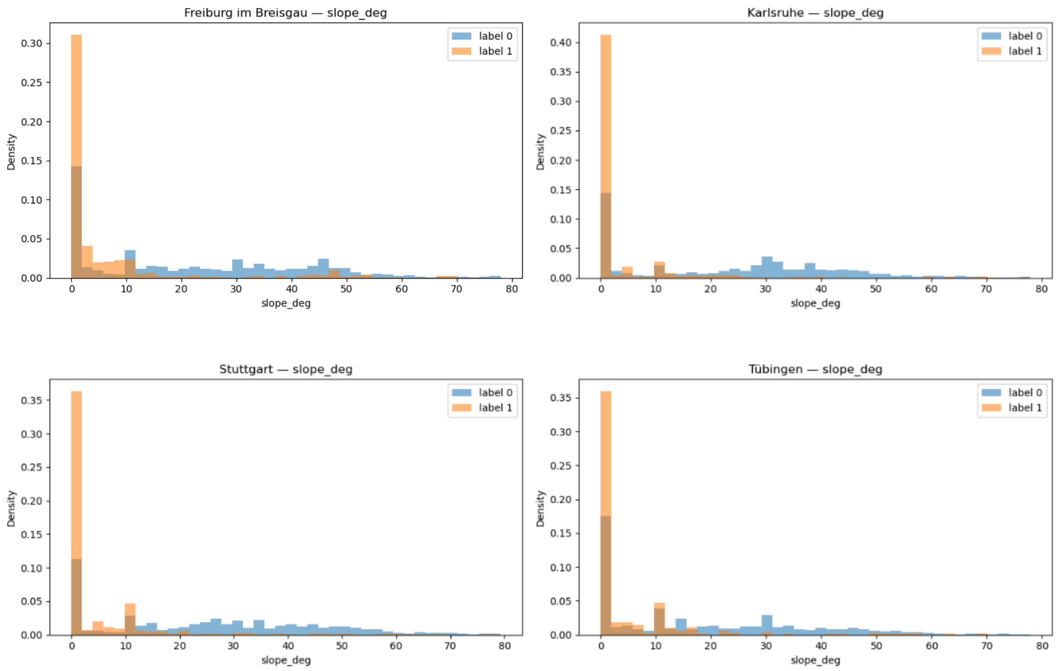


Figure 13: Sample Behavior in Slope

For `area`, both classes show the expected right-skewed distribution, with many small roofs and fewer large ones. However, green roofs are clearly shifted towards larger areas: the orange curves extend further into the several-hundred-square-metre range and are relatively under-represented among the very smallest roofs. In other words, small roof planes are mostly non-green, while larger continuous roof surfaces (e.g. blocks, halls, institutional buildings) are more likely to host vegetation.

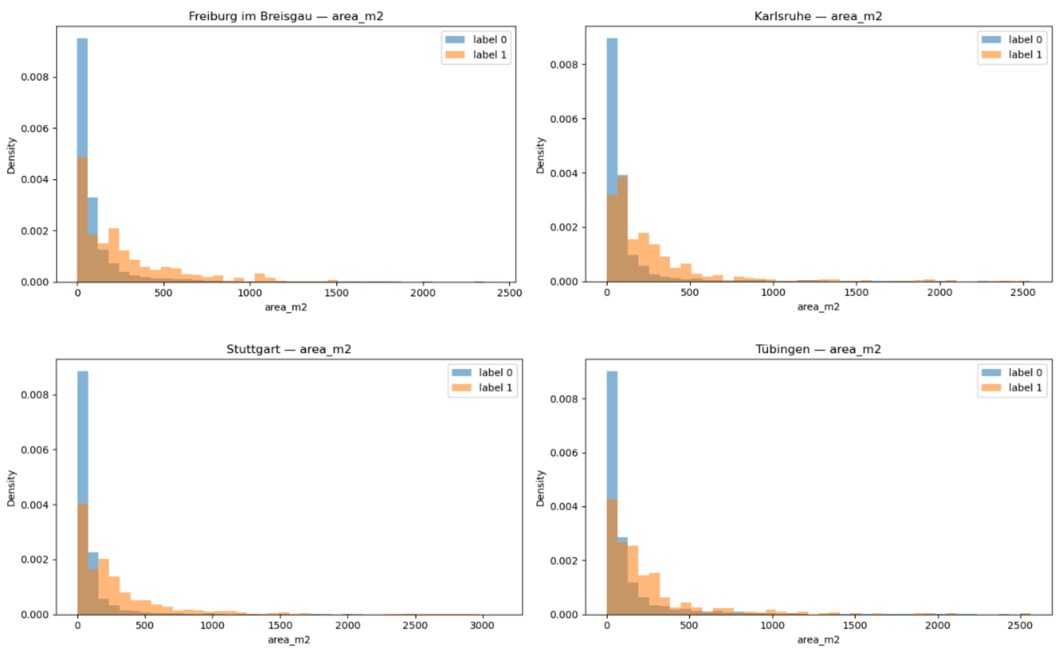


Figure 14: Sample Behavior in Area

In contrast, aspect shows almost complete overlap between the two classes. The blue and orange histograms are broadly similar within each city and do not exhibit a clear directional preference for green roofs beyond the general orientation patterns of the building stock. This visual impression matches the later finding that aspect has low permutation importance and can be dropped without harming cross-city performance, and it explains why aspect was excluded from the final feature set.

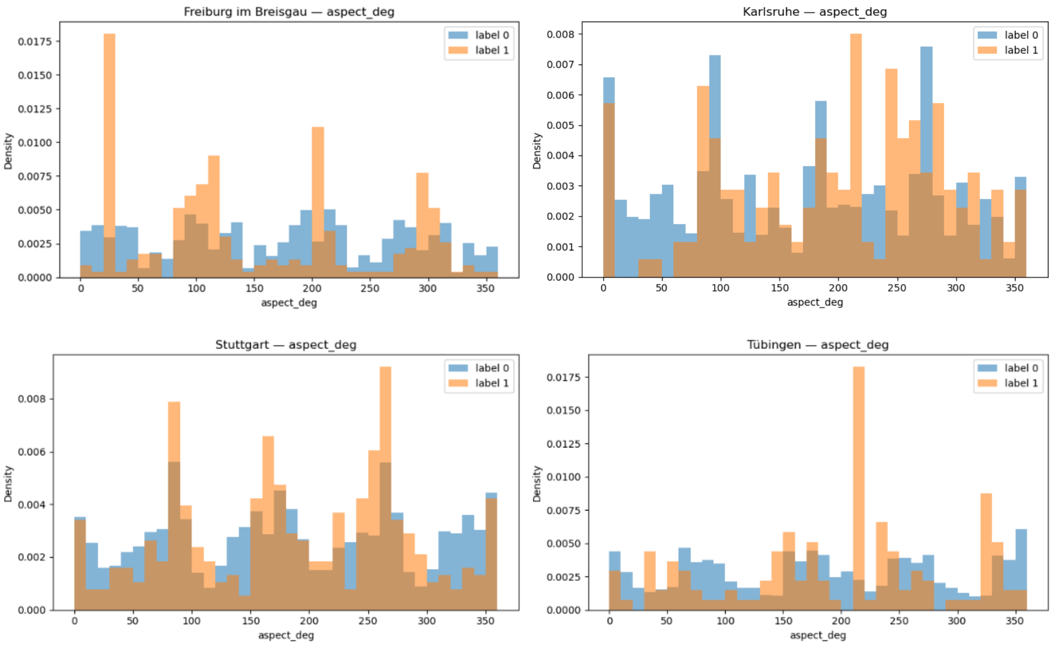


Figure 15: Sample Behavior in Aspect

6.2.2 Spectral Features Behavior

The vegetation indices show much clearer separation. For `NDVI_avg_Winter`, green roofs consistently occupy higher values than non-green roofs, with distributions shifted towards ≈0.30–0.45, while non-green roofs cluster at lower NDVI values with a mode around ≈0.15–0.30. Although there is still overlap (especially in Stuttgart where some non-green roofs reach intermediate NDVI levels), the winter NDVI plots show that vegetated roofs maintain a distinct spectral signal even under reduced canopy activity.

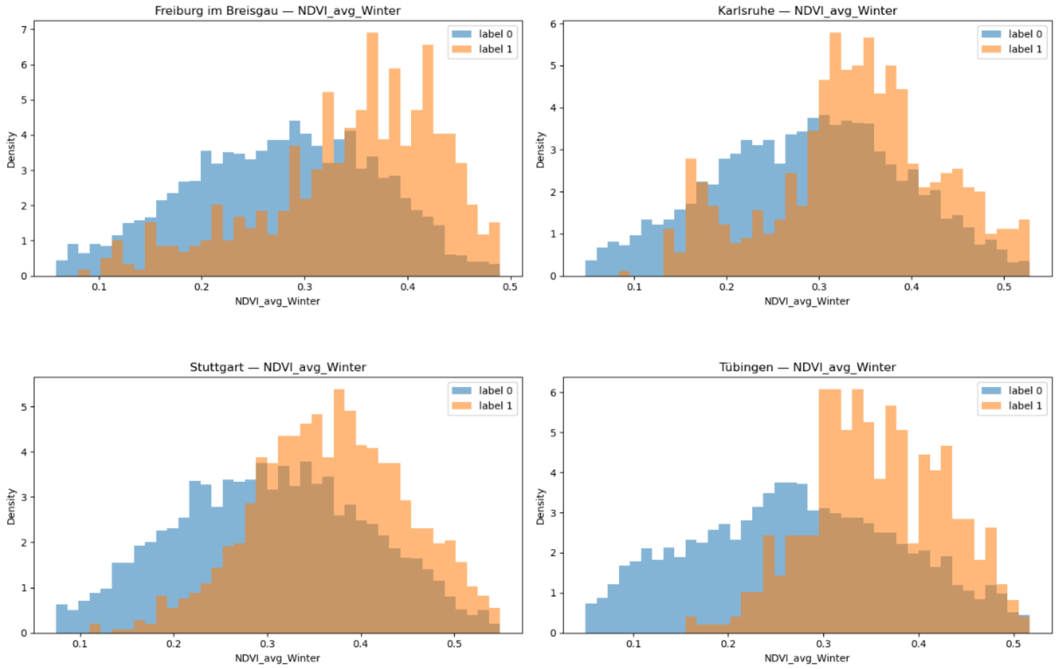
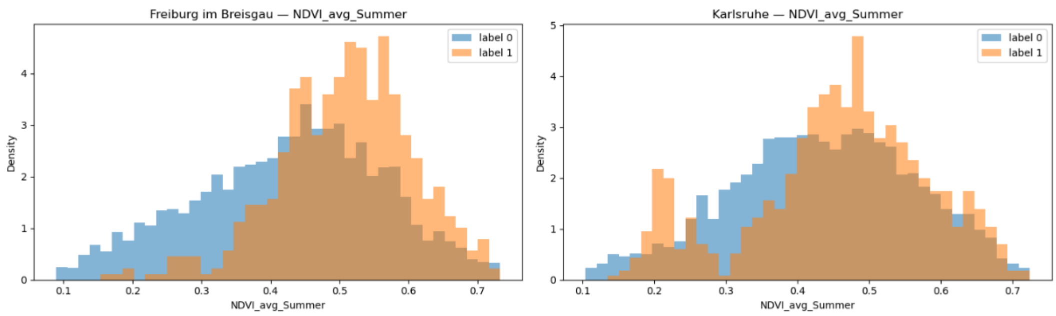


Figure 16: Sample Behavior in NDVI Winter

For `NDVI_avg_Summer`, the contrast between classes is similarly clear. In all four cities, green roofs are concentrated at higher NDVI (roughly 0.40–0.60), whereas non-green roofs peak at lower values and decline more quickly towards the upper tail. This indicates that summer imagery captures the full greening effect and provides strong discrimination between vegetated and non-vegetated roofs at PlanetScope resolution. Together, the winter and summer NDVI distributions support the design choice of using seasonal vegetation statistics rather than a single-date index.



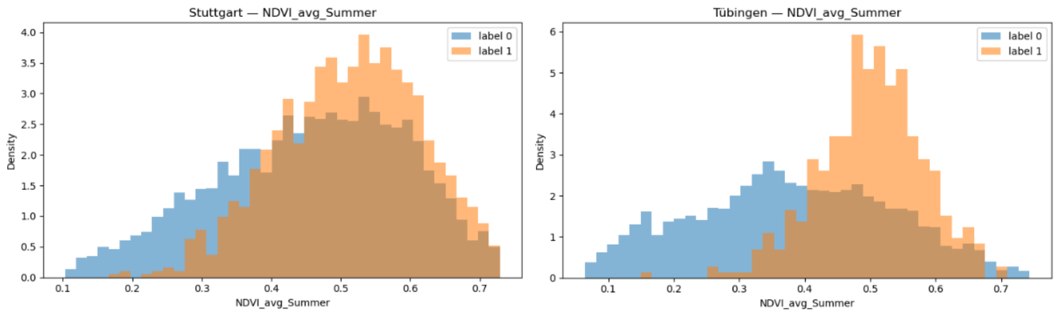
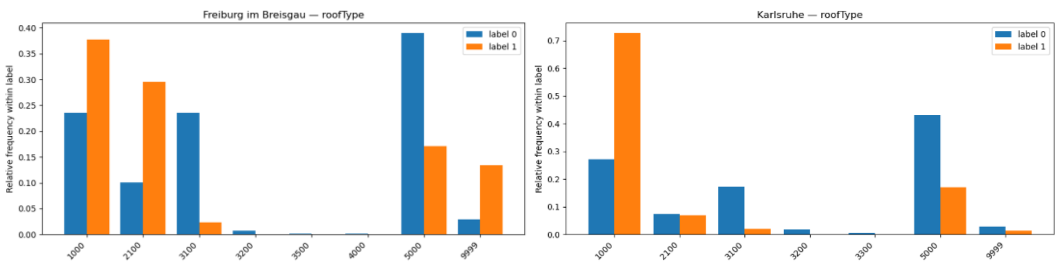


Figure 17: Sample Behavior in NDVI Summer

Although one might expect summer NDVI to be the strongest vegetation predictor, permutation importance ranks `NDVI_avg_Winter` slightly higher than `NDVI_avg_Summer`. A plausible explanation is that in winter, surrounding trees and ground vegetation are less vigorous, so PlanetScope pixels are less affected by off-roof vegetation and mixed-pixel contamination; vegetated roofs retain a comparatively stronger and more distinct NDVI signal than non-vegetated roofs. In contrast, summer NDVI is more strongly influenced by nearby trees, courtyards, and other leaf-on surfaces, and part of its signal is shared with other summer bands, which lowers its marginal contribution in the Random Forest.

6.2.3 Semantic features behavior

The `roof-type` distributions show a clear physical signal. In all cities, green roofs occur disproportionately on flat (1000) and mono-pitch (2100) roofs, which together account for the majority of vegetated roofs in the sample. Non-green roofs are much more evenly spread across gable (3100), hipped/mansard (3200–3400) and mixed/complex roof forms (5000), with these steeper or more articulated forms strongly over-represented among label 0. There is still overlap. Many flat roofs are not green, and some green roofs sit on gable or mixed types, indicating that roofType is a supporting contextual feature rather than a stand-alone discriminator, but it aligns well with the slope patterns and helps the model distinguish physically plausible green-roof candidates.



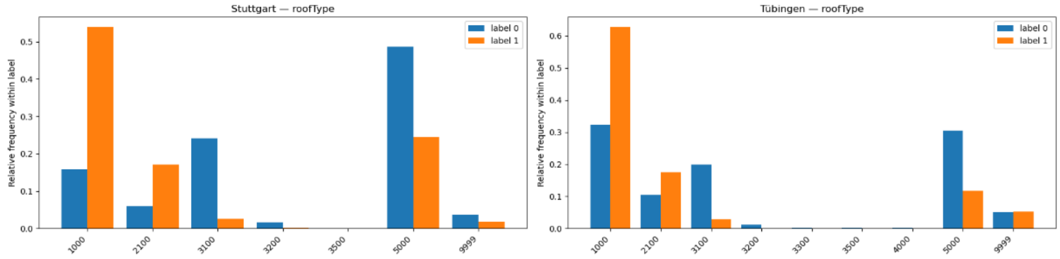


Figure 18: Sample Behavior in Roof Type

For `function` across all four cities, the labelled roofs are dominated by residential and mixed residential–commercial functions (codes starting with 31001), with smaller contributions from public, commercial and special-purpose buildings. Within this common baseline, green roofs are slightly over-represented on multi-storey residential and mixed-use buildings (e.g. 31001_1010, 31001_1123, 31001_2050/2070) and on some public or institutional buildings, while industrial/storage and utility buildings are more often non-green. The large "Other" bucket mainly collects rare or highly specific codes and should not be over-interpreted. Overall, building function acts as a soft prior—certain uses are more likely to host green roofs—but it is far from deterministic and must be combined with geometric and spectral information.

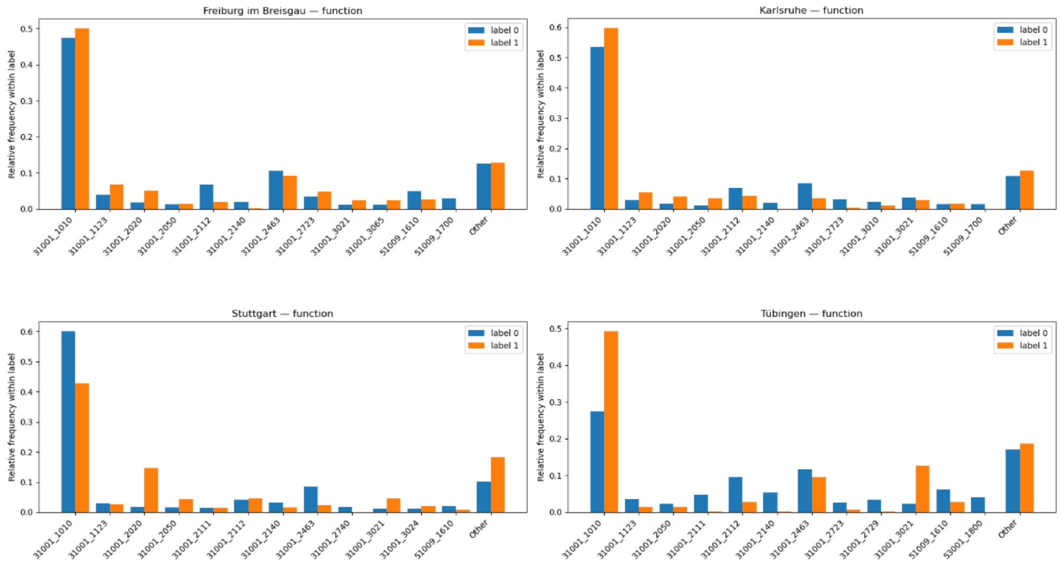


Figure 19: Sample Behavior in Building Function

Overall, these distributions confirm that:

- Geometry (slope and area) carries meaningful but not sufficient signal: green roofs are mostly flat, mid-rise, and somewhat larger.

- Aspect behaves essentially as noise for this task, with heavily overlapping class distributions.
- NDVI-based spectral features, especially in summer but also in winter, are the main discriminators between green and non-green roofs.
- Semantic attributes (`roofType` and `function`) provide useful priors that are consistent with the geometric patterns but cannot replace spectral information.

These empirical patterns are consistent across Stuttgart, Karlsruhe, Freiburg im Breisgau and Tübingen, despite their different urban fabrics, and they underpin the later model interpretation in Sections 6.3 and 6.4 They also provide a qualitative explanation for the results of the permutation-importance and ablation experiments: features that visibly separate the orange and blue histograms (NDVI, area, slope, height) emerge as important, while those with overlapping distributions (aspect, some RGB means) contribute little and can be removed. A full overview of all input features, including semantic categories, is given in Appendix.

To summarize, the labelled dataset fulfils its purpose with respect to the thesis objectives: it enables a realistic supervised learning setup for O3 (model training and comparison) and provides enough diversity in building types and spectral contexts to study RQ3 (feature contribution and interpretability) and RQ4 (inventory-grade behavior) under the cross-city evaluation described in subsequent sections.

6.3 Input Feature Tuning

6.3.1 Permutation feature importance (Random Forest)

Permutation importance was computed on the final Random Forest configuration using Average Precision (PR-AUC) as the scoring metric (Table 4). The resulting ranking shows that model decisions are driven by a combination of roof-plane geometry, building semantics, and vegetation-sensitive spectral information.

The most influential predictor is roof-plane area (`area_m2`), followed by winter vegetation intensity (`NDVI_avg_Winter`) and height-related descriptors (`height_m`, `height_relief_m`) and `slope_deg`. This confirms the patterns already visible in the sample distributions: green roofs in the labelled data tend to be larger, are more often mid-rise, and are almost always flat or low-pitch. Building semantics (`roofType` and `function`) also appear in the upper half of the importance ranking, indicating that typical usage and roof form carry additional information beyond geometry alone.

Vegetation-sensitive spectral features are strongly represented as well. Both winter and summer vegetation cues (`NIR_avg_Summer`, `NDVI_avg_Summer`, `NDVI_std_Winter/Summer`) show substantial importance values, whereas individual visible-band means, especially `R_avg_Summer` and `G_avg_Winter`, have comparatively low marginal importance. This pattern is consistent with the underlying physics: NDVI and NIR explicitly encode the red–NIR contrast that differentiates photosynthetically active vegetation from non-vegetated surfaces, while raw RGB averages are more easily confounded by roof material, illumination and shadows.

In contrast, `aspect_deg` appears in the lower part of the ranking. This matches the near-complete overlap between the aspect distributions of green and non-green roofs discussed in 6.2.1 and suggests that aspect contributes little to the discrimination task once slope and `roofType/function` are available.

Overall, the permutation results already hint at a compact, interpretable feature set: keep area, height, slope, vegetation indices and building semantics; treat aspect and some RGB means as secondary or redundant.

6.3.2 Targeted feature dropping

To test whether the weaker-ranked predictors can be safely removed, a first set of targeted dropping experiments was run (Table 5). Starting from the full 19-feature baseline (B0), three configurations were compared:

B0 (full): all 19 features.

- Experiment 1 (17 features): drops `G_avg_Winter` and `R_avg_Summer` (lowest importance).
- Experiment 2 (16 features): additionally drops `aspect_deg` (intuitively based on sample behavior)
- Experiment 3 (16 features): instead drops `height_relief_m` while keeping `aspect_deg`.

Under LOCO evaluation, the compact configurations behave differently. Dropping only `G_avg_Winter` and `R_avg_Summer` (Experiment 1) leaves PR-AUC essentially unchanged compared to B0, confirming that these RGB means are largely redundant once NDVI and other bands are included. Removing `aspect_deg` on top of this (Experiment 2) slightly improves LOCO PR-AUC (from ≈0.6983 to ≈0.7033) and marginally increases the mean best F1-score. This suggests that aspect acts more as noise than as a helpful discriminator under cross-city generalization.

By contrast, removing `height_relief_m` instead of aspect (Experiment 3) reduces PR-AUC to ≈0.6920 and slightly degrades F1. `Height_relief_m` was therefore retained as a useful descriptor of within-roof vertical structure, whereas `aspect_deg` was flagged as a candidate for full removal.

These targeted experiments already support two design decisions:

- Some low-importance RGB bands can be dropped without hurting performance.
- Aspect can be removed outright, while `height_relief_m` should be kept.

6.3.3 Ablation experiments

Building on the targeted dropping, a broader ablation experiment was carried out to quantify the contribution of entire feature groups (Section 5.7). All variants use the same Random Forest configuration and preprocessing; only the input feature set differs. The experiments fall into three categories:

- Baselines (B0, B1): full feature universe vs. a compact feature-importance baseline.
- Intuitive configurations (C-series): test redundancy hypotheses (e.g. drop raw R/NIR but keep NDVI).
- Group ablations (G-series): remove whole spectral or seasonal groups, or keep geometry only.

Table 6 (LOCO) and Table 7 (mixed-sample RSKF) summarize the results. Three main patterns emerge.

(i) Geometry-only is not sufficient

The geometry-only configuration (G5), which drops all spectral features and uses only slope, aspect, area and height metrics (plus semantics), performs substantially worse than any spectral-informed model. Under LOCO it reduces PR-AUC by roughly 0.19 compared to B0 (ΔPR-AUC ≈ −0.1874), and under mixed-sample splits it also lags behind the baselines.

This confirms that while geometry and building function relate to where green roofs are plausible, they cannot reliably distinguish vegetated from non-vegetated roofs at roof-plane level without vegetation-sensitive spectral information. In other words, geometry alone can at best define a prior; the actual detection depends on the PlanetScope signal.

(ii) NIR/NDVI groups are essential, RGB is secondary

Configurations that remove all NIR and NDVI information but keep RGB (G4) show the largest drop in LOCO PR-AUC among the spectral ablations (ΔPR-AUC ≈ −0.0857 vs. B0). The opposite configuration, which keeps NIR/NDVI but drops RGB (G3), performs much closer to the baseline. This asymmetry is consistent across LOCO and mixed-sample evaluation: losing vegetation-sensitive channels hurts the model far more than losing raw visible bands.

This behavior is fully aligned with the feature distributions from Section 6.2.2 and the physical interpretation of NDVI. At ~5 m resolution, many roofs are represented by only a few mixed pixels; in this regime, a robust red–NIR contrast (NDVI) provides a much cleaner vegetation signal than RGB averages, which are heavily influenced by roof material, shadows and facade contributions.

(iii) A compact "C3" configuration preserves almost all predictive power

Among the intuitive C-series configurations, C3 was designed to deliberately leverage this redundancy structure: it keeps Blue and Green means and NDVI/NDVI_std for both seasons, while dropping raw Red and NIR band means and aspect_deg. Conceptually, NDVI already encodes the essential red–NIR contrast, so retaining raw R/NIR means in addition is mostly redundant.

Empirically, C3 performs almost as well as the full 19-feature baseline:

- Under LOCO, C3 achieves PR-AUC ≈ 0.6938 ± 0.0955, which is only about 0.0045 below B0.
- Under mixed-sample RSKF, C3 slightly improves PR-AUC to ≈ 0.8440 ± 0.0081 (ΔPR-AUC ≈ +0.0012 vs. B0).

Other C- and G-configurations that remove NDVI or entire seasonal vegetation groups show clearly larger performance losses, particularly in the LOCO setting. C3 therefore emerges as a near-lossless reduction of the feature space: it removes variables that are either noisy (aspect) or strongly redundant (R/NIR means) while preserving the core vegetation cues and the most informative geometry and semantic attributes.

For these reasons, C3 was adopted as the final production feature set for the state-scale inventory (Section 5.8). The final Random Forest thus operates on 14 predictors: four geometric descriptors (`slope_deg`, `area_m2`, `height_m`, `height_relief_m`), eight seasonal vegetation and RGB summaries (G/B and NDVI/NDVI_std for winter and summer), and two semantic attributes (`roofType`, `function`).

6.3.4 Implications for RQ3 and feature design (O2)

Taken together, the feature importance and ablation results provide a coherent answer to RQ3 and validate the feature-design choices under O2:

- Most influential features
  Roof-plane area, slope, height, NDVI_avg (winter and summer), NDVI variability, and building semantics (`roofType`, `function`) are consistently important across cities and evaluation schemes. These variables jointly encode both physical feasibility (flat, larger roofs) and actual vegetation presence.
- Secondary or redundant features
  Aspect and several RGB band means show weak marginal contribution and heavily overlapping class distributions. Dropping them either preserves or slightly improves cross-city generalization.
- Critical feature families
  NIR/NDVI-based vegetation features are indispensable. Removing them causes the largest PR-AUC losses, whereas removing RGB while keeping NDVI/NIR has comparatively mild effects.
- Robust compression
  The C3 configuration demonstrates that a compact, interpretable feature set can achieve almost identical performance to the full 19-feature universe. This is important for inventory-grade deployment: fewer, well-understood predictors reduce the risk of overfitting spurious correlations and make the model easier to maintain and explain.

In summary, the evidence from importance ranking and ablation experiments shows that the model relies on physically meaningful and spatially consistent signals rather than exploiting incidental artefacts of the training cities. This supports the interpretability claims of the workflow and provides a solid foundation for the performance analysis and inventory behavior discussed in Sections 6.4 and 6.5.

6.4 Model performance

This section summarizes the predictive performance of the candidate models and evaluation schemes introduced in Chapter 5. It addresses two questions: (i) how well the workflow can separate green from non-green roofs under realistic spatial generalization (RQ1), and (ii) how Random Forest (RF) compares to the Multilayer Perceptron (MLP) under the same feature set and evaluation protocol (RQ2).

As described in Section 5.5.5, all primary metrics are based on the precision–recall (PR) curve, using Average Precision (AP, referred to as PR-AUC) as the threshold-independent summary and the best F1-score (with corresponding precision, recall and balanced accuracy) as a threshold-dependent operating point.

6.4.1 Spatial generalization under LOCO

The main evaluation for this thesis is the Leave-One-City-Out (LOCO) protocol, where each fold holds out one city (Stuttgart, Karlsruhe, Freiburg im Breisgau, Tübingen) as an unseen test domain while training on the other three. This directly reflects the intended use case of transferring a model trained in some cities to a different city that was not part of model fitting.

Figure 20 compares RF and MLP performance for each held-out city. Across all four folds, RF attains a higher PR-AUC in every city (mean PR-AUC ≈ 0.70 vs. ≈ 0.56 for MLP) and shows much more stable behavior, particularly in Freiburg im Breisgau and Tübingen, where the MLP's PR-AUC falls below 0.50. At the F1-optimal operating point, both models achieve comparable recall, but RF consistently provides higher precision – crucial in an inventory setting where false positives directly degrade the practical utility of the map. This already favors RF as the main model for statewide inference.

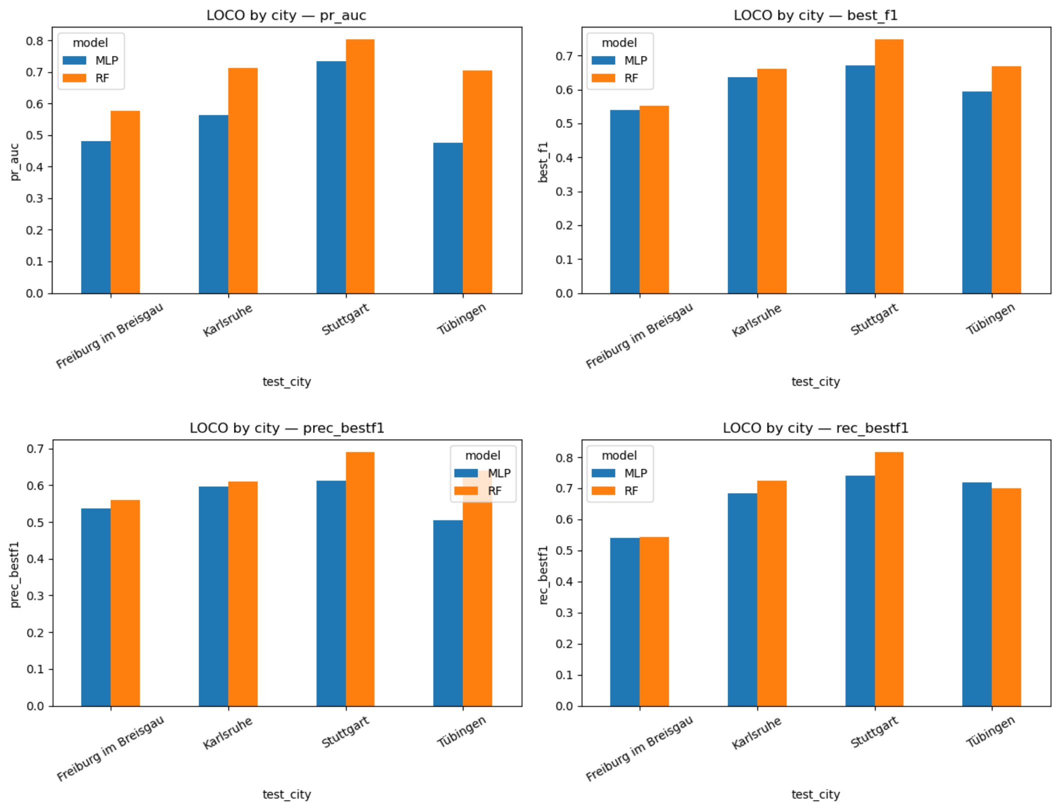


Figure 20: RF vs MLP Performance Graph under LOCO (city hold-out)

Using the final RF configuration and the C3 feature set, the LOCO experiment yields the summary shown in Table 10

Table 10: Final Random Forest Model Performance under LOCO in Fold

Fold	Test city	PR-AUC	Best F1	Precision	Recall	Balanced Accuracy	Fit time (s)
1	Freiburg im Breisgau	0.5668	0.5439	0.508	0.585	0.751	2.17
2	Karlsruhe	0.7086	0.6539	0.627	0.684	0.802	2.18
3	Stuttgart	0.7980	0.7481	0.705	0.797	0.850	2.00
4	Tübingen	0.7032	0.6698	0.652	0.688	0.822	2.69

Under strict city hold-out, the RF therefore reaches a moderate but clearly non-trivial level of separability between vegetated and non-vegetated roofs. A mean PR-AUC around 0.69 is far from perfect, but it is well above what would be expected from a naïve classifier in a strongly imbalanced setting (≈16 % positives). At the best-F1 operating point, the model recovers roughly two-thirds of all green roofs while misclassifying only a minority of non-green roofs as green; balanced accuracies above 0.80 indicate that both classes are handled reasonably despite the imbalance.

Fold-wise differences are evident. As already suggested by the sample characteristics in Section 6.2, folds with more positive examples and comparatively homogeneous building stock (notably Stuttgart) show higher PR-AUC and F1, whereas Freiburg im Breisgau and Tübingen with fewer positives and more heterogeneous urban fabric yield lower scores. This pattern is consistent with the idea that cross-city transfer is hardest where green roofs are rare, functional composition differs from the training cities, or PlanetScope mosaics suffer from more challenging illumination and seasonal conditions.

Compared to MLP under LOCO, RF consistently exhibits higher PR-AUC and more stable high-precision behavior across folds. The neural network occasionally matches recall at moderate thresholds but degrades more quickly when moving into high-precision regions, and in some folds its PR-AUC drops substantially below the RF baseline. In practice, the MLP is more sensitive to limited sample size, class imbalance and feature heterogeneity, while RF remains comparatively robust.

Across both evaluation schemes, the superior performance of the Random Forest compared to the Multilayer Perceptron is consistent with the structure of the problem. The labelled dataset is medium-sized (≈18,000 roofs) with a strongly imbalanced positive class and a set of hand-crafted, heterogeneous tabular features (geometric, spectral, semantic). In this regime, tree-based ensembles are typically more data-efficient and robust than feed-forward neural networks, which require substantially larger training sets and more extensive hyperparameter tuning to realize their potential. The Random Forest can naturally model non-linear interactions in tabular data, handles noisy or weakly informative features well through feature subsampling, and is comparatively insensitive to moderate label noise and class imbalance. By contrast, the MLP represents a single parametric function that is more sensitive to limited sample size, covariate shift between cities and the precise choice of optimization settings. Given that most of the representation learning has already been externalized into explicit features (e.g. NDVI, slope, height, building function), the MLP has little structural advantage and, in practice, delivered lower PR-AUC and less stable high-precision behavior under LOCO. Consequently, the Random Forest emerges as the more suitable model for this tabular, engineered-feature setting and for the operational constraints of a state-scale inventory.

From the perspective of the thesis objectives, the LOCO analysis leads to two interim conclusions:

- Under realistic spatial generalization, green-roof detection from LoD2 geometry and PlanetScope features is feasible but challenging. Performance is clearly above random, yet below what could be achieved in a single, city-specific setup.
- Among the candidate models, RF provides the more reliable basis for statewide inference, especially when high precision is required for inventory products.

6.4.2 Mixed-sample performance (random stratified split)

To put the LOCO results into context, a complementary evaluation was performed using a stratified 60/20/20 train/validation/test split that ignores city grouping (Section 5.10). In this mixed-sample setting, the model only needs to generalize to new roofs drawn from the same underlying pool of four cities, rather than to an entirely unseen city.

Using the same RF configuration and C3 feature set, the mixed-sample evaluation yields higher scores (Table 11):

Table 11: Summary of Final Random Forest Model Performance under LOCO and Mixed-sample

Metric	LOCO (mean ± std)	Mixed-sample (mean ± std)
PR-AUC	0.6941 ± 0.0954	0.8442 ± 0.0078
Best F1-score	0.6539 ± 0.0841	0.7651 ± 0.0085
Precision @ Best F1	0.6230 ± 0.0834	0.7608 ± 0.0212

Recall @ Best F1	0.6885 ± 0.0863	0.7703 ± 0.0194
Balanced Accuracy @ Best F1	0.8062 ± 0.0416	0.8616 ± 0.0076

These values correspond to a substantially easier task. When training and testing on randomly mixed samples from the same four cities, the RF can rely on very similar building types, roof materials and local conditions on both sides of the split. A PR-AUC above 0.84 and F1 around 0.77 indicate that, within the sampled cities, the model separates green and non-green roofs quite effectively.

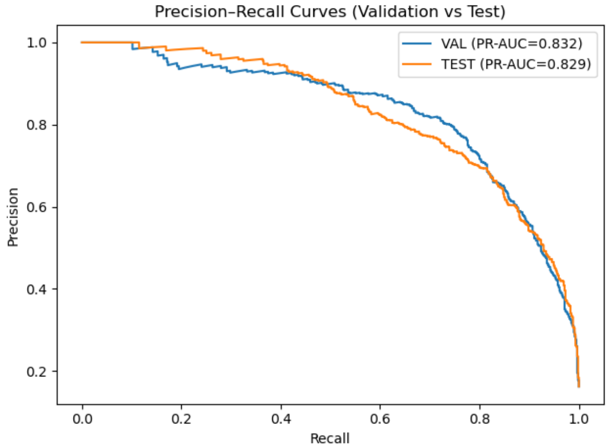


Figure 21: Precision-Recall Curve under Mixed Sample (Val vs Test)

Figure 21 overlays the PR curves for the validation and test splits in this mixed-sample setting. The two curves almost coincide (PR-AUC ≈ 0.832 for validation vs. ≈ 0.829 for test), which suggests that the model does not severely overfit the validation set and that its ranking of roofs by predicted probability is stable across unseen data drawn from the same distribution.

The gap between LOCO (mean PR-AUC ≈ 0.69) and mixed-sample (PR-AUC ≈ 0.84) therefore quantifies the domain-shift penalty incurred when moving from within-sample generalization to true cross-city transfer. In other words, a significant part of the difficulty stems not from the classifier itself but from differences in building stock, green-roof prevalence and imagery between cities.

As in the LOCO setting, RF remains competitive or superior to MLP under mixed-sample splits. The MLP does not provide a clear improvement over RF in this tabular, engineered-feature scenario, while being more delicate to optimize and more prone to unstable behavior on some splits.

6.4.3 Performance Summary

Taken together, the LOCO and mixed-sample experiments, the global PR curves and the calibration analysis provide a clear answer to the first two research questions.

RQ1 – Feasibility and accuracy.

An RF model that combines LoD2 geometric features, PlanetScope-derived spectral indices and semantic attributes can detect green roofs at roof-plane level with:

- Moderate cross-city accuracy: PR-AUC ≈ 0.69 and best F1 ≈ 0.65 under LOCO.
- High within-sample accuracy: PR-AUC ≈ 0.84 and best F1 ≈ 0.77 under random stratified splits.
- Reasonable probability reliability, especially for high-confidence predictions.

These levels of performance are sufficient to support high-precision inventory slices and analytical use, once probability-based thresholding is applied, but they also make clear that perfect separation is not realistic given the moderate spatial resolution of PlanetScope, the strong class imbalance and the heterogeneity of urban roofs.

RQ2 – Model comparison (RF vs. MLP).

- RF systematically outperforms or matches the MLP, which indicated by higher PR-AUC and more stable high-precision behavior under LOCO.
- at least comparable, often better, performance under mixed sample splits.
- easier training and more straightforward interpretation via permutation importance and ablation.

In the context of a state-scale inventory with imbalanced data and a limited number of labelled cities, RF is therefore the more appropriate choice for operational deployment. These findings directly support the model-selection decision in Section 5.5.6 and motivate using the RF-C3 configuration as the basis for the statewide 2025 green-roof inventory.

6.5 Thresholding behavior

The Random Forest model outputs is probabilistic, ranging from 0 to 1. To turn these continuous scores into a usable inventory, explicit thresholding decisions are required. There is no single “correct” probability cut-off: the optimal threshold depends on the use case, e.g. high-precision official statistics versus broad candidate lists for follow-up inspection. Building on the performance analysis in Section 6.4, this section describes how the RF probabilities are converted into several inventory products that support different planning and reporting needs (Table 12).

- Validation-optimal F1 threshold
  The first strategy uses the probability threshold that maximizes the F1-score on the validation split (`thr_f1_val = 0.6182`). This operating point is comparatively balanced: the model recovers around three quarters of all green roofs while keeping the false-positive rate moderate. In practical terms, this threshold is well suited for analytical tasks (e.g. estimating the overall prevalence of green roofs or exploring spatial patterns), where a reasonable compromise between coverage and reliability is needed and some manual cleaning is acceptable.
- High-confidence probability cut-offs
  The second group of strategies applies more conservative, fixed probability cut-offs, i.e. `p ≥ 0.85` and `p ≥ 0.90`. These thresholds effectively define a high confidence "core inventory". Roof planes flagged as green under $p \geq 0.85$ or $p \geq 0.90$ are, on average, very likely to be truly vegetated, which makes these strategies attractive for applications where false positives are costly (e.g. reporting numbers in policy documents, or selecting roofs for targeted field visits). The downside is that a majority of true green roofs fall below these cut-offs and are therefore not included.
- Rank-based top-k% selection
  The last group of strategies ignores absolute probability values and instead selects the top-k% highest-scoring roofs, with a minimum absolute count per city, i.e. `top1%` (min 50 roofs), `top2%` (min 100 roofs), `top5%` (min 250 roofs). By construction, these strategies return very small, extremely pure candidate sets. The top1% slice in particular contains only a handful of very obvious green roofs and can be interpreted as a "flagship" layer: if a roof appears here, it is almost certainly vegetated, but most green roofs do not reach this layer. The larger top5% slice roughly matches the precision of $p \geq 0.85$, but guarantees at least a fixed share of candidates in each city even if local probability distributions shift. These rank-based strategies are especially useful when the goal is to prioritise manual review or on-site inspection. For instance, municipalities could focus initial verification efforts on the top-1 % or top-2 % roofs, confident that almost all of them are genuine green roofs, before moving to broader thresholds.

Table 12: Product Slicing Performance Matrix

Strategy	Threshold used	Selected count	Selection rate	Precision	Recall	Balanced Accuracy
thr_f1_val	0.6182	559	0.1534	0.7621	0.7208	0.8386
p ≥ 0.85	0.8500	260	0.0714	0.9385	0.4129	0.7038
p ≥ 0.90	0.9000	170	0.0467	0.9706	0.2792	0.6388
top1% (min 50)	≈0.9624	50	0.0137	1.0000	0.0846	0.5423
top2% (min 100)	≈0.9396	100	0.0274	0.9900	0.1675	0.5836
top5% (min 250)	≈0.8572	250	0.0686	0.9440	0.3993	0.6974

Lets take a look at various snippets of different threshold type behavior in actual cases

Figure 22: Comparison of Predicted Green Roofs (green) and Non-green Roofs (red) for Three Thresholding Strategies: F1-based Threshold (left), Fixed Threshold p ≥ 0.90 (middle), and Top1% per city (right) over the same area

Across all examples, the F1-threshold behaves as the most permissive setting. In dense inner-city blocks, almost all visually clear green roofs are retained, and several borderline roofs with patchy or dark surfaces are still classified as green. This matches the numeric behaviour: the F1 operating point is tuned to balance precision and recall, so it deliberately accepts a small number of false positives in exchange for recovering many true greens.

The p90 threshold is visibly more conservative. Many marginal or weakly vegetated roofs that appear green under the F1 setting revert to red, especially narrow strips, partly vegetated courtyards and roofs with mixed gravel and vegetation. In several scenes (e.g. the institutional complex and the large housing estate), p90 keeps only the brightest, most homogeneous green roofs and drops a number of visually plausible but spectrally less clear candidates. This reflects the precision-oriented design of the p90 slice: only roofs with very high predicted probability survive, which reduces false positives but also sacrifices recall.

The top-1% slice is the most selective and produces a very "exclusive" inventory. In each example, only a small subset of roofs remain green – typically the largest, most uniformly vegetated blocks. Thin roof segments, shaded parts and roofs with mixed materials are pruned away, even when they are clearly green in the imagery. At the same time, some neighbourhoods where many roofs have extremely high probabilities can still contain more green polygons under top-1% than under p90, even though top-1% selects a smaller fraction overall. This happens because top-1% uses a global ranking: it keeps the highest-scoring roofs statewide, which can cluster in certain areas, whereas p90 removes any roof whose probability falls just below the fixed 0.90 cut-off, regardless of how it compares to other roofs.

In the context of RQ4, this suggests structuring the 2025 Baden-Württemberg green roof inventory as a multi-layer product rather than a single binary map.

6.6 Spatial patterns of the green roof 2025 inventory

At state scale, the inventory confirms that vegetated roofs remain a minority phenomenon. Only a relatively small share of all LoD2 roof planes with valid PlanetScope statistics is classified as green, which is consistent with the ≈16 % positive rate observed in the labelled training set (Section 6.2). When mapped using the balanced pred_f1 layer (probability ≥ thr_f1), green roofs appear as a sparse but clearly structured network of points and patches concentrated in urbanized corridors and larger settlements, with isolated occurrences in smaller towns and villages. In other words, the model does not predict a diffuse, everywhere-present green roof layer, but a geographically selective pattern that closely follows the underlying urban hierarchy.

At city scale (Figure 20), green roofs form a clear non-random pattern. They cluster in dense inner-urban areas, recent large-scale developments and major commercial or institutional zones, while remaining sparse in low-density residential suburbs and almost absent in small villages and industrial fringe areas. This city-wide structure is consistent across all four examples, but the absolute density of predicted green roofs differs: larger, economically stronger cities such as Stuttgart show visibly denser and more extensive clusters than smaller cities such as Tübingen. The inventory therefore captures both a general "urban core vs. periphery" contrast and inter-city differences in green-roof adoption.

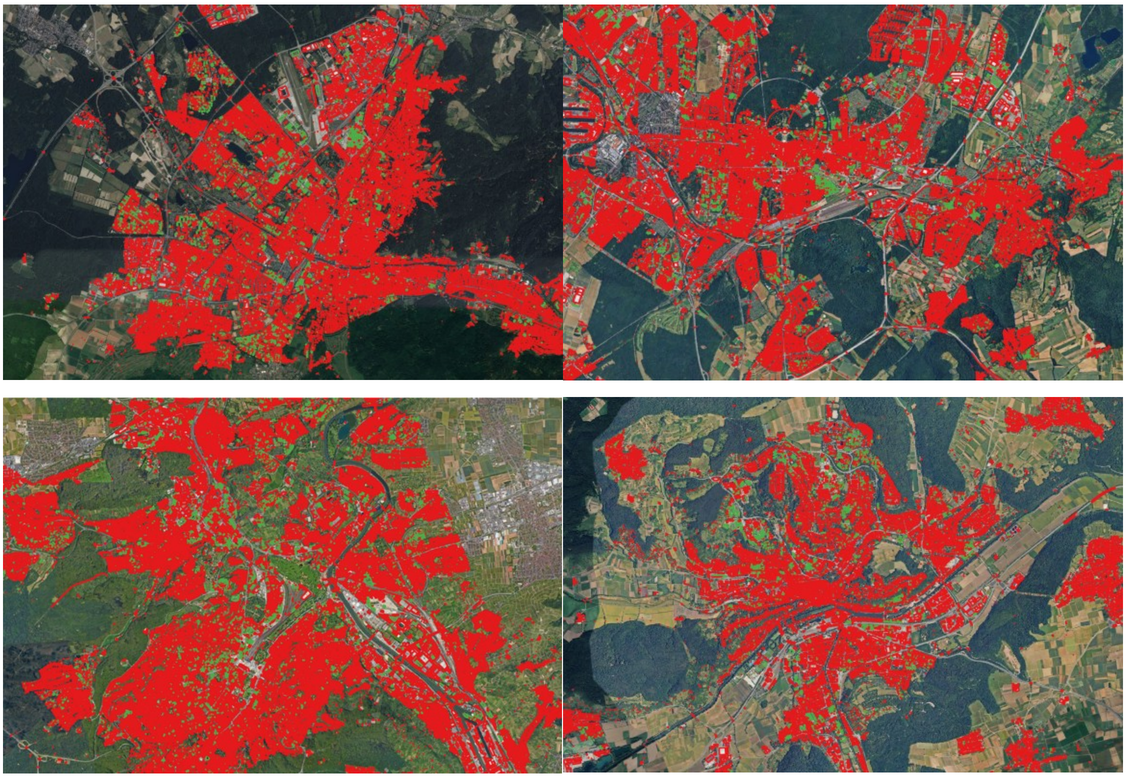


Figure 23: City-wide spatial distribution of the of the green roof (green) and non-green roof (red) by F1-threshold for the four labelled cities (Freiburg im Breisgau, Karlsruhe, Stuttgart, and Tübingen)

The neighborhood-scale views (left panels in each row of Figure 24) highlight how the model tends to light up whole blocks or estates where flat, mid-rise buildings dominate. In recently built quarters and post-war housing estates, continuous sequences of green roofs appear along entire street fronts or courtyards, often matching visually homogeneous building typologies. At the same time, the maps reveal pockets where green roofs are nearly absent despite similar densities and building heights, suggesting local differences in planning regulations or investment rather than purely physical constraints.

At building scale (right panels in Figure 24), the expected strengths and weaknesses of the workflow become visible. Large, homogeneously vegetated roofs are almost always detected, with clean green polygons tightly matching the footprints. In contrast, small roofs with patchy or low vegetation cover, narrow extensions and complex roof geometries are more frequently missed or only partially classified as green. In mixed blocks, some roofs with visually similar greenness are classified differently, reflecting both the moderate PlanetScope resolution and the influence of surrounding roof planes and façades on the zonal NDVI statistics.

Figure 24: Neighborhood cluster (left) and building scale (right) examples of the green roof (green) and non-green roof (red) by F1-threshold in the four labelled cities (Freiburg im Breisgau, Karlsruhe, Stuttgart, and Tübingen).

Taken together, these examples show that the 2025 inventory behaves plausibly at multiple scales: it reproduces known macro-patterns of green-roof adoption, picks out major green-roof clusters that are visible in high-resolution imagery, and still provides meaningful information at the level of individual building complexes. At the same time, the maps already hint at systematic errors and biases, particularly with respect to small, heterogeneous roofs and spectrally "messy" environments which are discussed in more detail in Section 6.7 on error modes and limitations.

6.7 Error modes, biases and limitations

Although the RF–C3 model reaches reasonable PR-AUC scores and produces visually plausible inventory maps (Sections 6.4–6.6), a closer inspection at building scale reveals systematic error modes and data-driven biases. Figure 25 shows six representative examples (first three are where model relatively well behaved, and the last three are where model relatively poorly behaved), each rendered over PlanetScope, Google and Bing imagery for qualitative cross-checking.

Figure 25: Examples of local classification behavior (F1-threshold) of the green roof (green) and non-green roof (red) over the same area overlaid by PlanetScope basemaps (left), Google (middle), and Bing Imagery (right)

6.7.1 Resolution and mixed-pixel effects

The most prominent limitation is the mismatch between LoD2 roof geometry and PlanetScope's moderate spatial resolution (4.77 m). Many roof planes are only a few Planet pixels wide, so the zonal statistics used as input features often integrate:

- vegetation on the roof surface,
- adjacent roof material (gravel, asphalt, metal), and
- nearby context such as tree crowns, courtyards or streets.

In the "well-behaved" examples, the model performs best where roofs are large and homogeneously vegetated. Here, most pixels within the roof polygon are genuinely green, NDVI values are high and stable, and spectral leakage from the surroundings is limited. In contrast, the "moderate" and "poor" examples often involve:

- small, fragmented roofs surrounded by trees or narrow courtyards, where a single off-roof pixel can pull NDVI down;
- mixed roofs with partial vegetation, technical installations and gravel, producing intermediate NDVI that is hard to separate from non-green roofs;
- narrow extensions or inner courtyards where the LoD2 footprint includes shaded vertical surfaces or neighbouring facades.

In these situations, the model tends to (i) miss true green roofs whose averaged NDVI is diluted by non-vegetated pixels, or (ii) hallucinate green roofs where trees or adjacent green spaces dominate the PlanetScope signal inside the polygon.

6.7.2 Temporal inconsistencies between imagery sources

A second source of error is the temporal mismatch between data used for labelling and for prediction. Labels were derived from high-resolution Google and Bing imagery, whereas features are extracted from PlanetScope monthly mosaics. In many locations, the three image sources clearly correspond to different acquisition dates and sometimes even different seasons. This leads to several characteristic failure cases visible in Figure 25, such as

- Ephemeral or seasonal vegetation (e.g. moss, dry grass, or recently installed roofs) that is present in one imagery source but not in the others. In such cases, the label may be "green" based on high-resolution imagery, while the PlanetScope composite still reflects a pre-greening or post-maintenance state, resulting in a low NDVI and a false negative prediction.
- Renovations or new constructions where a roof has been converted between green and non-green between acquisition dates. Depending on which imagery is more recent, this produces apparent false positives or false negatives that are in fact temporal discrepancies rather than pure model error.
- Seasonal illumination differences, especially on north-facing or shadowed roofs, where NDVI is systematically depressed in winter mosaics and amplified in bright summer scenes.

These temporal inconsistencies mean that some disagreements between prediction and "ground truth" are partly unresolvable without a time-stamped reference inventory.

6.7.3 Label and sampling biases

The manual labelling protocol (Section 6.2) also introduces specific biases that translate into inventory behavior:

- Preference for clear cases
  Roofs with visually unambiguous vegetation cover were more likely to be labelled, while borderline cases (sparse moss, small planters, mixed gravel–sedum roofs) were often skipped. The model therefore learns a relatively strict notion of "green roof" and tends to under-represent marginal or low-intensity greening.

- City bias
  All labels originate from four cities with comparatively good PlanetScope coverage and relatively high green-roof adoption. Industrial areas, small towns and rural municipalities elsewhere in Baden-Württemberg may exhibit different relationships between NDVI, roof materials and building function. Some seemingly systematic over- or under-prediction patterns outside the labelled cities likely reflect this domain shift.
- Building-type bias
  As shown in Section 6.2, green roofs in the training data are concentrated on mid-rise residential blocks and public or office buildings with flat roofs. The model consequently tends to assign higher probabilities to similar building types across the state, while being more conservative on rare or structurally different roofs (e.g. small detached houses, complex mixed forms).

6.7.4 Implications for use

Taken together, these error modes imply that the 2025 inventory is best interpreted as a probabilistic, inventory-grade layer, not as a wall-to-wall cadastral truth. At F1-threshold, the map captures most clearly vegetated roofs but still misses a share of partial or small green roofs, and includes some false positives where surrounding vegetation contaminates the PlanetScope signal. Users should therefore:

- treat individual roof labels as plausible candidates rather than definitive classifications, especially outside the four labelled cities;
- be cautious in very heterogeneous blocks with many small roofs and trees, where mixed-pixel artefacts are strongest; and
- rely on aggregated statistics (e.g. at city or district level) and/or supplementary visual checks when using the inventory for planning or reporting.

These limitations are not specific to the chosen RF model; they arise mainly from the combination of moderate-resolution satellite imagery with detailed 3D building geometry and sparse, city-biased labels. They highlight where future work could most effectively improve the inventory: higher-resolution spectral data, temporally aligned imagery for labelling, and a broader, more balanced training sample across different municipality types.

7 Conclusion

7.1 Main Findings

This thesis set out to build and evaluate a state-wide 2025 green roof inventory for Baden-Württemberg at roof-plane level, using only data that are available consistently across the state: LoD2 building geometry from the Landesamt für Geoinformation und Landentwicklung and PlanetScope multispectral imagery. The core idea was to combine geometric, spectral and semantic information into a supervised machine learning workflow that can be trained on a few labelled cities and then transferred to the rest of the state.

The work addressed four research questions:

RQ1 – Feasibility and accuracy

Under strict city hold-out, the final RF model achieves a mean PR-AUC of about 0.69 and a best F1-score around 0.65 across Freiburg im Breisgau, Karlsruhe, Stuttgart and Tübingen. Within each fold, the best-F1 operating point typically recovers roughly two-thirds of all green roofs while keeping false positives at a manageable level (balanced accuracy > 0.80).

When city boundaries are ignored and data from all four cities are randomly split into train, validation and test sets, performance rises to around 0.84 PR-AUC and 0.77 F1. This gap between LOCO and mixed-sample evaluation is an explicit measure of the domain-shift penalty incurred when transferring a model from a small set of labelled cities to previously unseen municipalities.

Taken together, these results show that:

- Green roof detection from LoD2 and PlanetScope is feasible at roof-plane level.
- Cross-city performance is clearly above what would be expected from blind baselines in a 16 % positive-class setting, but far from perfect, especially in cities with few green roofs or heterogeneous urban fabrics.
- Within the four sampled cities, the model can separate green and non-green roofs quite effectively, which justifies using its probability outputs as the basis for an inventory.

In other words, the produced inventory is good enough for planning and analytical use under appropriate thresholding and visual checking, but it must not be interpreted as a complete and error-free census.

RQ2 – Model comparison: RF vs MLP

Across all LOCO folds and in the mixed-sample evaluation, the Random Forest consistently outperforms or matches the Multilayer Perceptron:

- RF attains higher PR-AUC in each held-out city and shows much more stable behaviour, especially in Freiburg im Breisgau and Tübingen, where the MLP's PR-AUC drops below 0.50.
- At the F1-optimal thresholds, recall is similar between both models, but RF delivers higher precision, which is crucial for inventory products where each false positive roof might be followed up manually.
- RF is easier to train with the available sample size and class imbalance, and its decisions can be interrogated with permutation importance, ablation experiments and feature distributions. The MLP, in contrast, is more sensitive to hyperparameters and split variance while offering limited additional insight.

Given these results, and considering the practical requirement for a robust, interpretable and computationally lightweight model for state-wide prediction, the thesis concludes that RF is the more appropriate choice for operational deployment. The final inventory is therefore based on the RF–C3 configuration.

RQ3 – Feature contribution and interpretability

A combination of distribution plots, permutation importance and ablation experiments reveals a coherent picture of how different feature groups contribute:

Geometric features

- Slope: Green roofs in the labelled dataset are overwhelmingly flat or nearly flat (0–5°), while non-green roofs span the full slope range up to steep pitched roofs. Slope emerges as an important discriminator and aligns with building-physics expectations.
- Area and height: Green roofs tend to be somewhat larger and on mid-rise structures, often corresponding to blocks, halls or institutional buildings. Small and highly fragmented roofs are mostly non-green. These features carry useful signal but are not sufficient on their own.

Spectral features

- Seasonal NDVI statistics are the main drivers of class separation. Both winter and summer NDVI show clear differences between green and non-green roofs, and permutation importance indicates that winter NDVI contributes slightly more to the RF's predictive performance. This suggests that using both seasons together provides a more stable vegetation signal than relying on a single NDVI snapshot.
- R and NIR band means contribute secondary information, but their class distributions redundant with NDVI and they can be dropped with little loss in LOCO performance.
- The experiments confirm that using both winter and summer mosaics is beneficial, as it stabilizes the vegetation signal across phenological stages.

Semantic features

- Roof type acts as a physically meaningful contextual feature: vegetated roofs are disproportionately found on flat and mono-pitch forms, whereas gable, hipped and complex roofs are mostly non-green.
- Building function provides a soft prior: green roofs are more common on multistory residential, mixed-use and public/institutional buildings than on industrial or storage structures.
- Neither semantic feature is deterministic, but together they help the RF focus on plausible host buildings and improve cross-city stability.
- Overall, the feature analysis confirms that the model behavior is well aligned with urban form, construction practice and vegetation physics, which increases confidence in applying the resulting inventory for spatial planning.

RQ4 – Inventory-grade thresholding behaviour

Because the RF outputs probabilities rather than binary labels, several thresholding strategies were tested to support different use cases:

- A data-driven F1 threshold (≈0.62 on the validation set) yields a balanced, coverage-oriented inventory. It recovers most true green roofs while keeping a moderate false-positive rate and selecting around 15 % of all roof planes.
- Fixed probability cut-offs (e.g. $p \geq 0.85$, 0.90) provide high-confidence candidate layers. Precision increases towards 0.95–0.97, but at the cost of sharply reduced recall and coverage.

- Top-k% layers (e.g. top 1 %, 2 %, 5 % per city) offer a rank-based view of the “greenest” roofs. They are particularly useful when the absolute calibration of probabilities is uncertain but relative ranking is still meaningful.

Visual inspection over multiple example areas shows that:

- All three strategies agree on the most obvious, homogeneous green roofs.
- Fixed high thresholds tend to miss small or spectrally mixed roofs, especially where PlanetScope pixels leak vegetation from courtyards, trees or neighboring roofs.
- Top-k% layers sometimes re-introduce moderate-probability roofs in neighborhoods with few green roofs, which is useful for exploratory analysis but less ideal for strict monitoring.

The thesis therefore recommends:

- Using the F1-based threshold as the default “balanced” inventory layer for analytical work.
- Supplementing it with high-precision slices (e.g. $p \geq 0.90$ or top 1–2 %) for targeted interventions, site scouting or communication, always accompanied by imagery-based verification.

7.2 Limitations

Several limitations constrain the interpretation and potential reuse of the 2025 green roof inventory:

- Moderate spatial resolution of PlanetScope
  At 4.77 m ground sampling distance, many roof planes are represented by only a handful of pixels. Mixed roofs, narrow terraces and clustered small roofs frequently suffer from spectral leakage to walls, streets or surrounding vegetation. This limits the model’s ability to detect fine-scale or partially vegetated roofs and explains many of the observed error modes.
- Label bias and city coverage
  Manual labelling focused on clear cases and four comparatively well-documented cities. Ambiguous roofs and marginal greening were mostly skipped. As a result, the model is tuned to confidently vegetated and non-vegetated roofs and may behave unpredictably around borderline treatments or in municipalities with very different building stocks.

- Temporal inconsistency between data sources
  PlanetScope mosaics, Google imagery and Bing imagery are not temporally aligned, and their exact capture dates are partly unknown. Some apparent "errors" in the inventory may actually reflect real temporal change (e.g. new green roofs), or misalignment between the year of the inventory and the year of the reference imagery.
- Simplified treatment of uncertainty
  While probability outputs and multiple thresholds are used, the inventory is still delivered as discrete layers. More formal uncertainty quantification, e.g. confidence intervals, ensemble variance or explicit spatial error models—was beyond the scope of this thesis.
- Model family and feature space
  Only two model families (RF and MLP) and a specific engineered feature set were explored. Other approaches such as gradient boosting, convolutional models over image patches, or joint building-and-context representations might yield better cross-city performance, but would also require more complex implementation and more extensive training data.

7.3 Outlook

Despite these limitations, the thesis demonstrates that state-level, roof-plane green roof inventories are technically achievable using widely available geospatial data and open-source tools. Building on this work, several extensions are promising:

- Extended labelling and additional cities
  Expanding the labelled dataset to more cities, including smaller towns and industrial areas, would reduce domain shift and allow more nuanced analysis of regional patterns in green roof adoption.
- Higher-resolution and multi-sensor imagery
  Integrating very high-resolution aerial orthophotos or commercial satellite data, as well as additional spectral bands (e.g. red edge), could improve detection of small or partially vegetated roofs and help disentangle roofs from surrounding vegetation.
- Temporal analysis and change detection
  Repeating the workflow for multiple years would enable time-series monitoring of green roof development, which is relevant for tracking the impact of policy instruments and climate adaptation strategies.

- Integration with planning indicators
  The produced inventory can be linked with socio-demographic, climatic or land-use data to build indicators of green roof potential, equity and effectiveness in different neighborhoods, feeding into 15-minute city concepts and urban climate adaptation planning.
- Open data and reproducible workflows
  Publishing the code and, where licensing allows, derived features for selected cities would support further research, benchmarking and adaptation of the approach to other regions.

This thesis delivers a first 2025 green roof inventory for Baden-Württemberg based on a transparent, reproducible and interpretable machine learning workflow. It shows that combining LoD2 building models, PlanetScope imagery and relatively simple models such as Random Forests can already provide actionable information at state scale, while also highlighting the challenges of working at the edge of what moderate-resolution imagery can resolve.

The results are not a definitive map of all green roofs, but a probability-based, evidence-driven starting point that can support planning, monitoring and further research. With more labelled data, improved imagery and continued methodological development, similar approaches can contribute to a broader ecosystem of spatial indicators for climate-resilient and environmentally just cities.

Overview of tools used

Activity	**Description**	**Used AI tools** (if applicable)[1]	**Description of the use of AI tools** (Type of use, affected sections of the work, etc.)
Idea generation & conceptualization	– Development of ideas, research goals, objectives and research questions – Identification and definition of relevant concepts	-	-
Literature research and analysis	– Search for relevant literature – Review of potentially relevant literature – Summaries of relevant literature	ChatGPT	Search for relevant literature and summarizing workflow used
Methodology	– Search for a suitable methodology – Development and adaptation of the methodology to the research question(s)	-	-
Coding	– Creating and documenting code, algorithms, and software – Testing and debugging existing code, algorithms, and software – Understanding of existing code, algorithms, software	ChatGPT	Writing code for specific tasks based on my need
Data collection and analysis	– Collection of primary or secondary data – Qualitative data analysis (including summaries and coding) – Quantitative data analysis (including statistical evaluation) – Mathematical, computer-aided, or other formal techniques for modeling, simulation, and analysis	-	-

[1] List the names of the AI tools used, e.g., ChatGPT, DeepL Write, Microsoft 365 Copilot, Neuroflash, Grammarly, DeepL Translator, Google Translate, Perplexity, Elicit, Explainpaper, ResearchRabbit, GitHub Copilot, DeepSeek.

Interpretation and validation	– Interpretation of the results – Derivation of implications for research and practice	-	-
Text generation	– Generation of text on various topics in different sections of the work (including titles and summaries)	-	-
Generation of tables and figures	– Generation of tables, diagrams, graphs, etc. on various topics or presentation of results in different sections of the work.	-	-
Text review & editing	– Critical review, feedback, or revision of the content, organization, or grammar of the work – Proofreading – Reformulation or paraphrasing of text	ChatGPT	Proofreading and refining some of the paragraph narration
Source management	– Creating the reference list – Formatting of references	ChatGPT	Formatting my compiled list of resources into XML readable by Ms. Word
Other activities		-	-

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Appendix

Appendix 1: Building Function Dictionary List

Code	DE	English (EN)
31001_1000	Wohngebäude	Residential building
31001_1010	Wohnhaus	Dwelling house
31001_1020	Wohnheim	Residential home
31001_1021	Kinderheim	Children's home
31001_1022	Seniorenheim	Retirement home
31001_1023	Schwesternwohnheim	Nurses' residence
31001_1024	Studenten-, Schülerwohnheim	Student or pupil dormitory
31001_1025	Schullandheim	School camp
31001_1100	Gemischt genutztes Gebäude mit Wohnen	Mixed-use building with housing
31001_1110	Wohngebäude mit Gemeinbedarf	Residential building with community facilities
31001_1120	Wohngebäude mit Handel und Dienstleistungen	Residential building with trade and services
31001_1121	Wohn- und Verwaltungsgebäude	Residential and administration building
31001_1122	Wohn- und Bürogebäude	Residential and office building
31001_1123	Wohn- und Geschäftsgebäude	Residential and commercial building
31001_1130	Wohngebäude mit Gewerbe und Industrie	Residential building with trade and industry
31001_1131	Wohn- und Betriebsgebäude	Residential and operations building
31001_1210	Land- und forstwirtschaftliches Wohngebäude	Agricultural/forestry residential building
31001_1220	Land- und forstwirtschaftliches Wohn- und Betriebsgebäude	Agricultural/forestry residential and operations building
31001_1221	Bauernhaus	Farmhouse
31001_1222	Wohn- und Wirtschaftsgebäude	Residential and farm building
31001_1223	Forsthaus	Forestry house
31001_1310	Gebäude zur Freizeitgestaltung	Building for leisure activities
31001_1311	Ferienhaus	Holiday home
31001_1312	Wochenendhaus	Weekend house
31001_1313	Gartenhaus	Garden house
31001_2000	Gebäude für Wirtschaft oder Gewerbe	Building for economy or trade
31001_2010	Gebäude für Handel und Dienstleistungen	Building for trade and services
31001_2020	Bürogebäude	Office building
31001_2030	Kreditinstitut	Bank
31001_2040	Versicherung	Insurance office
31001_2050	Geschäftsgebäude	Commercial building
31001_2051	Kaufhaus	Department store
31001_2052	Einkaufszentrum	Shopping centre
31001_2053	Markthalle	Market hall
31001_2054	Laden	Shop
31001_2055	Kiosk	Kiosk
31001_2056	Apotheke	Pharmacy
31001_2060	Messehalle	Exhibition hall
31001_2070	Gebäude für Beherbergung	Building for accommodation
31001_2071	Hotel, Motel, Pension	Hotel, motel, guesthouse

31001_2072	Jugendherberge	Youth hostel
31001_2073	Hütte (mit Übernachtungsmöglichkeit)	Hut (with overnight accommodation)
31001_2074	Campingplatzgebäude	Campsite building
31001_2080	Gebäude für Bewirtung	Building for food service
31001_2081	Gaststätte, Restaurant	Restaurant
31001_2082	Hütte (ohne Übernachtungsmöglichkeit)	Hut (without overnight accommodation)
31001_2083	Kantine	Canteen
31001_2090	Freizeit- und Vergnügungsstätte	Leisure and amusement facility
31001_2091	Festsaal	Banquet hall
31001_2092	Kino	Cinema
31001_2093	Kegel-, Bowlinghalle	Bowling alley
31001_2094	Spielkasino	Casino
31001_2095	Spielhalle	Amusement arcade
31001_2100	Gebäude für Gewerbe und Industrie	Building for trade and industry
31001_2110	Produktionsgebäude	Production building
31001_2111	Fabrik	Factory
31001_2112	Betriebsgebäude	Operations building
31001_2113	Brauerei	Brewery
31001_2114	Brennerei	Distillery
31001_2120	Werkstatt	Workshop
31001_2121	Sägewerk	Sawmill
31001_2130	Tankstelle	Filling station
31001_2131	Waschstraße, Waschanlage, Waschhalle	Car wash, washing facility, wash hall
31001_2140	Gebäude für Vorratshaltung	Building for storage
31001_2141	Kühlhaus	Cold store
31001_2142	Speichergebäude	Storage building
31001_2143	Lagerhalle, Lagerschuppen, Lagerhaus	Warehouse, storage shed, storehouse
31001_2150	Speditionsgebäude	Freight forwarding building
31001_2160	Gebäude für Forschungszwecke	Building for research purposes
31001_2170	Gebäude für Grundstoffgewinnung	Building for raw material extraction
31001_2171	Bergwerk	Mine
31001_2172	Saline	Salt works
31001_2180	Gebäude für betriebliche Sozialeinrichtung	Building for company social facilities
31001_2200	Sonstiges Gebäude für Gewerbe und Industrie	Other building for trade and industry
31001_2210	Mühle	Mill
31001_2211	Windmühle	Windmill
31001_2212	Wassermühle	Watermill
31001_2213	Schöpfwerk	Pumping station
31001_2220	Wetterstation	Weather station
31001_2310	Gebäude für Handel und Dienstleistung mit Wohnen	Building for trade and services with housing
31001_2320	Gebäude für Gewerbe und Industrie mit Wohnen	Building for trade and industry with housing
31001_2400	Betriebsgebäude zu Verkehrsanlagen (allgemein)	Operations building for transport facilities (general)
31001_2410	Betriebsgebäude für Straßenverkehr	Operations building for road traffic
31001_2411	Straßenmeisterei	Road maintenance depot
31001_2412	Wartehalle	Waiting hall
31001_2420	Betriebsgebäude für Schienenverkehr	Operations building for rail traffic

31001_2421	Bahnwärterhaus	Gatekeeper's house
31001_2422	Lokschuppen, Wagenhalle	Engine shed, carriage hall
31001_2423	Stellwerk, Blockstelle	Signal box, block post
31001_2424	Betriebsgebäude des Güterbahnhofs	Operations building of the freight yard
31001_2430	Betriebsgebäude für Flugverkehr	Operations building for air traffic
31001_2431	Flugzeughalle	Aircraft hangar
31001_2440	Betriebsgebäude für Schiffsverkehr	Operations building for shipping
31001_2441	Werft (Halle)	Shipyard (hall)
31001_2442	Dock (Halle)	Dock (hall)
31001_2443	Betriebsgebäude zur Schleuse	Operations building for a lock
31001_2444	Bootshaus	Boathouse
31001_2450	Betriebsgebäude zur Seilbahn	Operations building for cableway
31001_2451	Spannwerk zur Drahtseilbahn	Tensioning works for cable railway
31001_2460	Gebäude zum Parken	Building for parking
31001_2461	Parkhaus	Multi-storey car park
31001_2462	Parkdeck	Parking deck
31001_2463	Garage	Garage
31001_2464	Fahrzeughalle	Vehicle hall
31001_2500	Gebäude zur Versorgung	Utility building
31001_2501	Gebäude zur Energieversorgung	Building for energy supply
31001_2510	Gebäude zur Wasserversorgung	Building for water supply
31001_2511	Wasserwerk	Waterworks
31001_2512	Pumpstation	Pumping station
31001_2513	Wasserbehälter	Water tank
31001_2520	Gebäude zur Elektrizitätsversorgung	Building for electricity supply
31001_2521	Elektrizitätswerk	Power station
31001_2522	Umspannwerk	Substation
31001_2523	Umformer	Converter station
31001_2527	Reaktorgebäude	Reactor building
31001_2528	Turbinenhaus	Turbine house
31001_2529	Kesselhaus	Boiler house
31001_2540	Gebäude für Fernmeldewesen	Telecommunications building
31001_2560	Gebäude an unterirdischen Leitungen	Building at underground lines
31001_2570	Gebäude zur Gasversorgung	Building for gas supply
31001_2571	Gaswerk	Gasworks
31001_2580	Heizwerk	Heating plant
31001_2590	Gebäude zur Versorgungsanlage	Building for utility plant
31001_2591	Pumpwerk (nicht für Wasserversorgung)	Pumping station (not for water supply)
31001_2600	Gebäude zur Entsorgung	Building for waste disposal
31001_2610	Gebäude zur Abwasserbeseitigung	Building for wastewater disposal
31001_2611	Gebäude der Kläranlage	Sewage treatment plant building
31001_2612	Toilette	Public toilet
31001_2620	Gebäude zur Abfallbehandlung	Building for waste treatment
31001_2621	Müllbunker	Waste bunker
31001_2622	Gebäude zur Müllverbrennung	Waste incineration building
31001_2623	Gebäude der Abfalldeponie	Landfill building

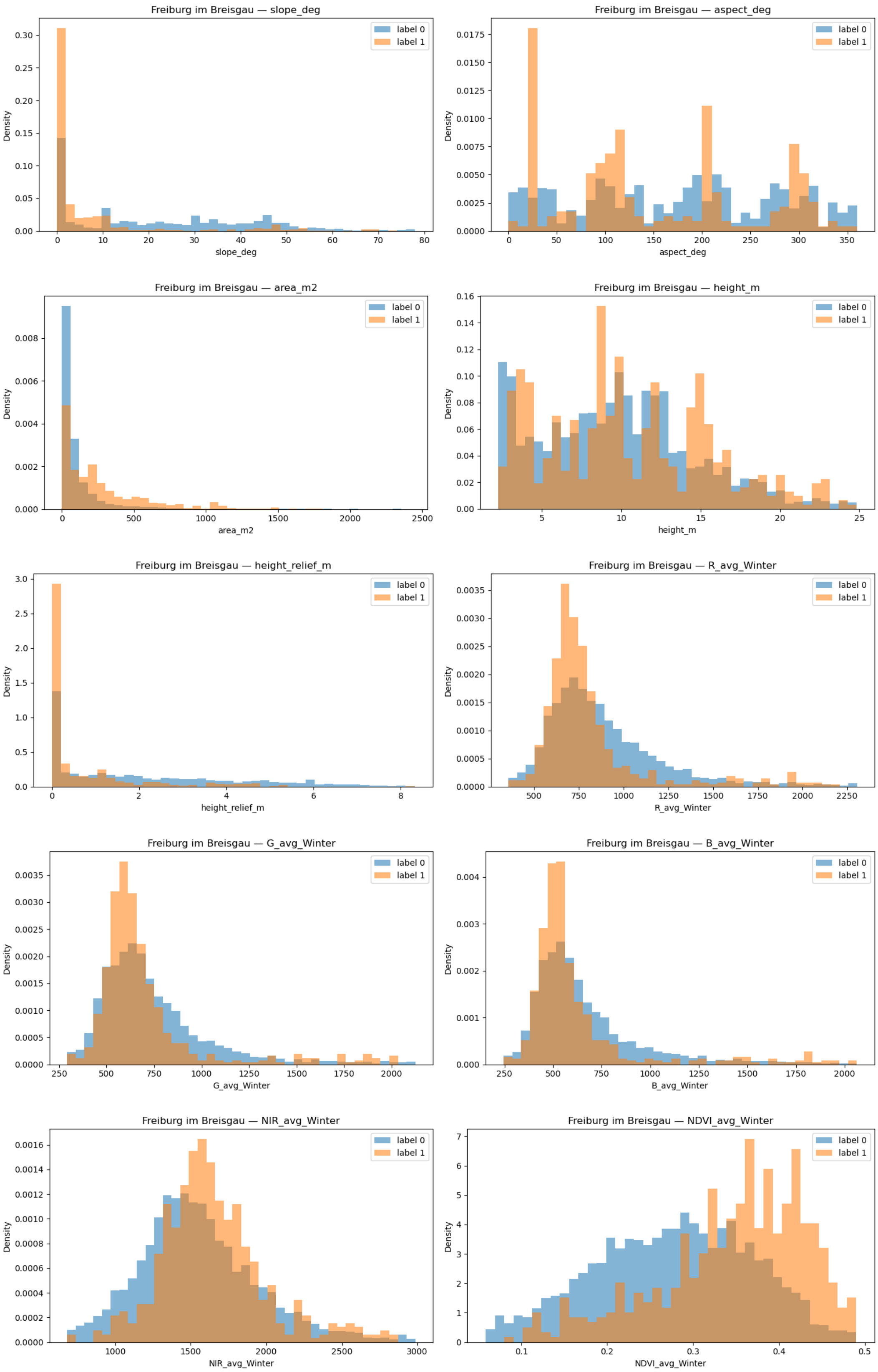
31001_2700	Gebäude für Land- und Forstwirtschaft	Building for agriculture and forestry
31001_2720	Land- und forstwirtschaftliches Betriebsgebäude	Agricultural/forestry operations building
31001_2721	Scheune	Barn
31001_2723	Schuppen	Shed
31001_2724	Stall	Stable
31001_2726	Scheune und Stall	Barn and stable
31001_2727	Stall für Tiergroßhaltung	Livestock barn
31001_2728	Reithalle	Riding hall
31001_2729	Wirtschaftsgebäude	Farm building
31001_2732	Almhütte	Alpine hut
31001_2735	Jagdhaus, Jagdhütte	Hunting lodge, hunting hut
31001_2740	Treibhaus, Gewächshaus	Greenhouse
31001_2741	Treibhaus	Greenhouse
31001_2742	Gewächshaus, verschiebbar	Movable greenhouse
31001_3000	Gebäude für öffentliche Zwecke	Building for public purposes
31001_3010	Verwaltungsgebäude	Administration building
31001_3011	Parlament	Parliament
31001_3012	Rathaus	Town hall
31001_3013	Post	Post office
31001_3014	Zollamt	Customs office
31001_3015	Gericht	Court
31001_3016	Botschaft, Konsulat	Embassy, consulate
31001_3017	Kreisverwaltung	District administration
31001_3018	Bezirksregierung	District government
31001_3019	Finanzamt	Tax office
31001_3020	Gebäude für Bildung und Forschung	Building for education and research
31001_3021	Allgemein bildende Schule	General education school
31001_3022	Berufsbildende Schule	Vocational school
31001_3023	Hochschulgebäude (Fachhochschule, Universität)	University/college building
31001_3024	Forschungsinstitut	Research institute
31001_3030	Gebäude für kulturelle Zwecke	Building for cultural purposes
31001_3031	Schloss	Palace
31001_3032	Theater, Oper	Theatre, opera
31001_3033	Konzertgebäude	Concert hall
31001_3034	Museum	Museum
31001_3035	Rundfunk, Fernsehen	Broadcasting, television
31001_3036	Veranstaltungsgebäude	Event building
31001_3037	Bibliothek, Bücherei	Library
31001_3038	Burg, Festung	Castle, fortress
31001_3040	Gebäude für religiöse Zwecke	Building for religious purposes
31001_3041	Kirche	Church
31001_3042	Synagoge	Synagogue
31001_3043	Kapelle	Chapel
31001_3044	Gemeindehaus	Parish/community centre
31001_3045	Gotteshaus	House of worship
31001_3046	Moschee	Mosque

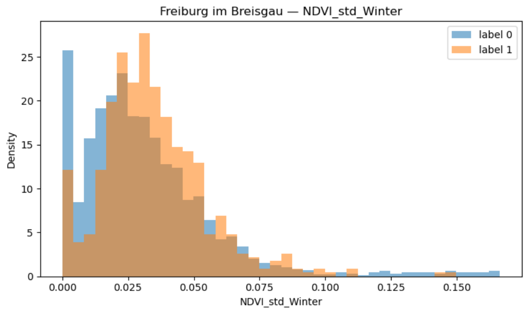
31001_3047	Tempel	Temple
31001_3048	Kloster	Monastery
31001_3050	Gebäude für Gesundheitswesen	Building for health care
31001_3051	Krankenhaus	Hospital
31001_3052	Heilanstalt, Pflegeanstalt, Pflegestation	Sanatorium, nursing home, care station
31001_3053	Ärztehaus, Poliklinik	Medical centre, polyclinic
31001_3054	Rettungswache	Rescue station
31001_3060	Gebäude für soziale Zwecke	Building for social purposes
31001_3061	Jugendfreizeitheim	Youth recreation centre
31001_3062	Freizeit-, Vereinsheim, Dorfgemeinschafts-, Bürgerhaus	Recreation/club house, village/community centre
31001_3063	Seniorenfreizeitstätte	Senior citizens' recreation centre
31001_3064	Obdachlosenheim	Homeless shelter
31001_3065	Kinderkrippe, Kindergarten, Kindertagesstätte	Crèche, kindergarten, day-care centre
31001_3066	Asylbewerberheim	Asylum seekers' home
31001_3070	Gebäude für Sicherheit und Ordnung	Building for safety and order
31001_3071	Polizei	Police station
31001_3072	Feuerwehr	Fire station
31001_3073	Kaserne	Barracks
31001_3074	Schutzbunker	Shelter bunker
31001_3075	Justizvollzugsanstalt	Prison
31001_3080	Friedhofsgebäude	Cemetery building
31001_3081	Trauerhalle	Funeral hall
31001_3082	Krematorium	Crematorium
31001_3090	Empfangsgebäude	Station building (reception)
31001_3091	Bahnhofsgebäude	Railway station building
31001_3092	Flughafengebäude	Airport terminal building
31001_3094	Gebäude zum U-Bahnhof	Building at underground station
31001_3095	Gebäude zum S-Bahnhof	Building at suburban rail station
31001_3097	Gebäude zum Busbahnhof	Building at bus station
31001_3098	Empfangsgebäude Schifffahrt	Harbour terminal building
31001_3100	Gebäude für öffentliche Zwecke mit Wohnen	Building for public purposes with housing
31001_3200	Gebäude für Erholungszwecke	Building for recreation
31001_3210	Gebäude für Sportzwecke	Building for sports
31001_3211	Sport-, Turnhalle	Sports hall, gymnasium
31001_3212	Gebäude zum Sportplatz	Building at sports ground
31001_3220	Badegebäude	Bathing facility building
31001_3221	Hallenbad	Indoor swimming pool
31001_3222	Gebäude im Freibad	Building at open-air pool
31001_3230	Gebäude im Stadion	Building in stadium
31001_3240	Gebäude für Kurbetrieb	Building for spa operations
31001_3241	Badegebäude für medizinische Zwecke	Bathing facility for medical purposes
31001_3242	Sanatorium	Sanatorium
31001_3260	Gebäude im Zoo	Building in zoo
31001_3261	Empfangsgebäude des Zoos	Zoo entrance building
31001_3262	Aquarium, Terrarium, Voliere	Aquarium, terrarium, aviary
31001_3263	Tierschauhaus	Animal exhibition house

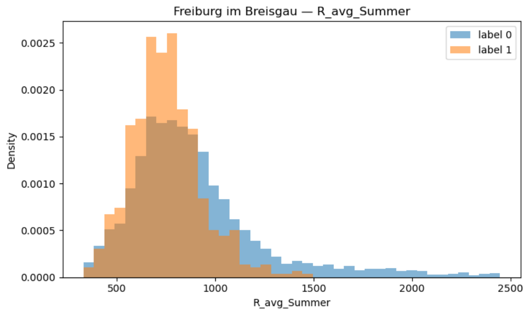
31001_3264	Stall im Zoo	Stable in zoo
31001_3270	Gebäude im botanischen Garten	Building in botanical garden
31001_3271	Empfangsgebäude des botanischen Gartens	Botanical garden entrance building
31001_3272	Gewächshaus (Botanik)	Greenhouse (botany)
31001_3273	Pflanzenschauhaus	Plant exhibition house
31001_3280	Gebäude für andere Erholungseinrichtung	Building for other recreation facility
31001_3281	Schutzhütte	Shelter hut
31001_3290	Touristisches Informationszentrum	Tourist information centre
31001_9998	Nach Quellenlage nicht zu spezifizieren	Not specified based on source
51001_1001	Wasserturm	Water tower
51001_1002	Kirchturm	Church tower
51001_1003	Aussichtsturm	Observation tower
51001_1004	Kontrollturm	Control tower
51001_1005	Kühlturm	Cooling tower
51001_1006	Leuchtturm	Lighthouse
51001_1007	Feuerwachturm	Fire lookout tower
51001_1008	Sende-,Funkturm	Broadcast/communication tower
51001_1009	Stadt-,Torturm	City/gate tower
51001_1010	Förderturm	Headframe (pithead)
51001_1011	Bohrturm	Drilling rig
51001_1012	Schloss-, Burgturm	Palace/castle tower
51001_9998	Nach Quellenlage nicht zu spezifizieren	Not specified based on source
51001_9999	Sonstiges	Other
51002_1215	Biogasanlage	Biogas plant
51002_1220	Windrad	Wind turbine
51002_1230	Solarzellen	Solar panels
51002_1250	Mast	Mast
51002_1251	Freileitungsmast	Overhead line mast
51002_1260	Funkmast	Radio mast
51002_1280	Radioteleskop	Radio telescope
51002_1290	Schornstein	Chimney
51002_1330	Kran	Crane
51002_1331	Drehkran	Slewing crane
51002_1332	Portalkran	Gantry crane
51002_1333	Laufkran	Bridge crane
51002_1350	Hochofen	Blast furnace
51002_1400	Umformer	Converter station
51002_9999	Sonstiges	Other
51003_1201	Silo	Silo
51003_1205	Tank	Tank
51003_1206	Gasometer	Gas holder
51003_9999	Sonstiges	Other
51006_1430	Zuschauertribüne	Grandstand
51006_1431	Zuschauertribüne, überdacht	Grandstand, covered
51006_1432	Zuschauertribüne, nicht überdacht	Grandstand, uncovered
51006_1440	Stadion	Stadium

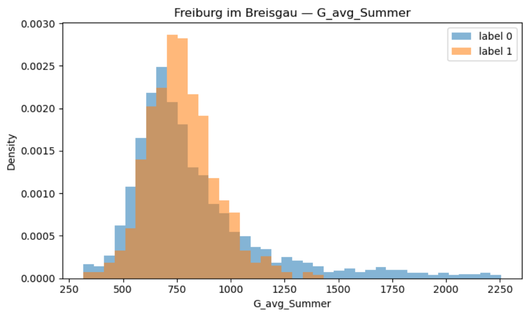
51006_1441	Stadion, überdacht	Stadium, covered
51006_1442	Stadion, nicht überdacht	Stadium, uncovered
51006_1470	Sprungschanze (Anlauf)	Ski jump (inrun)
51006_1490	Gradierwerk	Graduation tower
51006_9999	Sonstiges	Other
51007_1110	Aquädukt	Aqueduct
51007_1210	Wachturm	Watchtower
51007_1400	Befestigung (Burgruine)	Fortification (castle ruins)
51007_1500	Historische Mauer	Historic wall
51007_1510	Stadtmauer	City wall
51007_1520	Sonstige historische Mauer	Other historic wall
51007_9999	Sonstiges	Other
51009_1610	Überdachung	Roofing (cover)
51009_1611	Carport	Carport
51009_1700	Mauer	Wall
51009_1750	Denkmal	Monument
51009_9999	Sonstiges	Other
52003_1010	Schiffshebewerk	Boat lift
52003_1020	Kammerschleuse	Chamber lock
53001_1800	Brücke	Bridge
53001_1806	Drehbrücke	Swing bridge
53001_1807	Hebebrücke	Lift bridge
53001_1808	Zugbrücke	Drawbridge
53001_1830	Hochbahn, Hochstraße	Elevated railway/road
53001_1890	Schleusenkammer	Lock chamber
53009_2030	Staumauer	Dam
53009_2050	Wehr	Weir
53009_2060	Sicherheitstor	Safety gate
53009_2070	Siel	Sluice
53009_2080	Sperrwerk	Storm surge barrier
53009_2090	Schöpfwerk	Pumping station

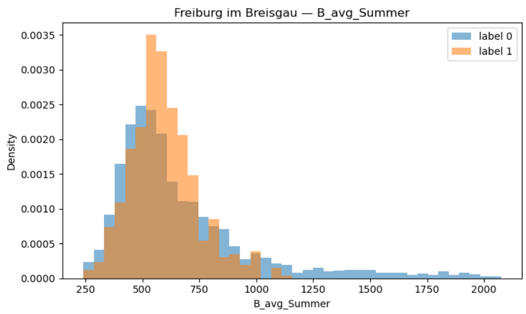
Appendix 2: Complete Sample Feature Behavior for Freiburg




Freiburg im Breisgau — NDVI_std_Winter
label 0
label 1
Density
NDVI_std_Winter


Freiburg im Breisgau — R_avg_Summer
label 0
label 1
Density
R_avg_Summer


Freiburg im Breisgau — G_avg_Summer
label 0
label 1
Density
G_avg_Summer


Freiburg im Breisgau — B_avg_Summer
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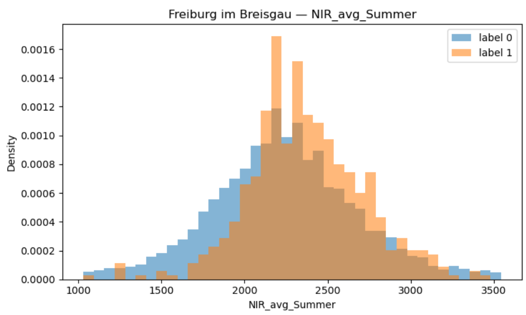
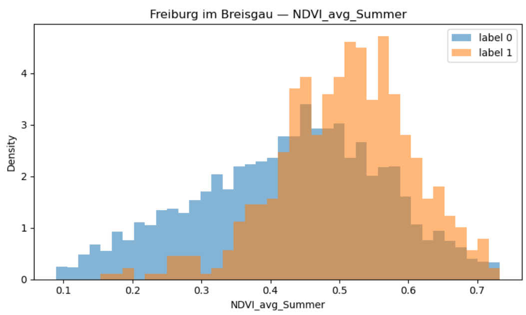
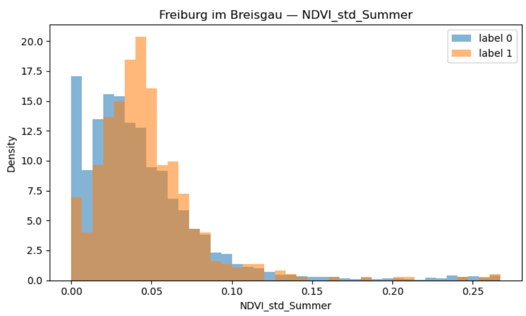
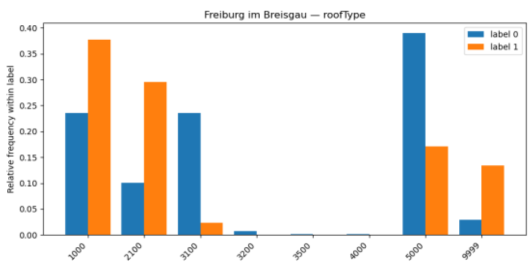
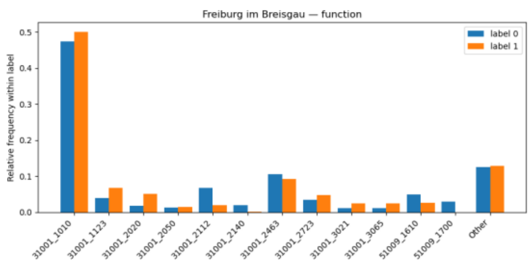
label 1
Density
B_avg_Summer


Freiburg im Breisgau — NIR_avg_Summer
label 0
label 1
Density
NIR_avg_Summer

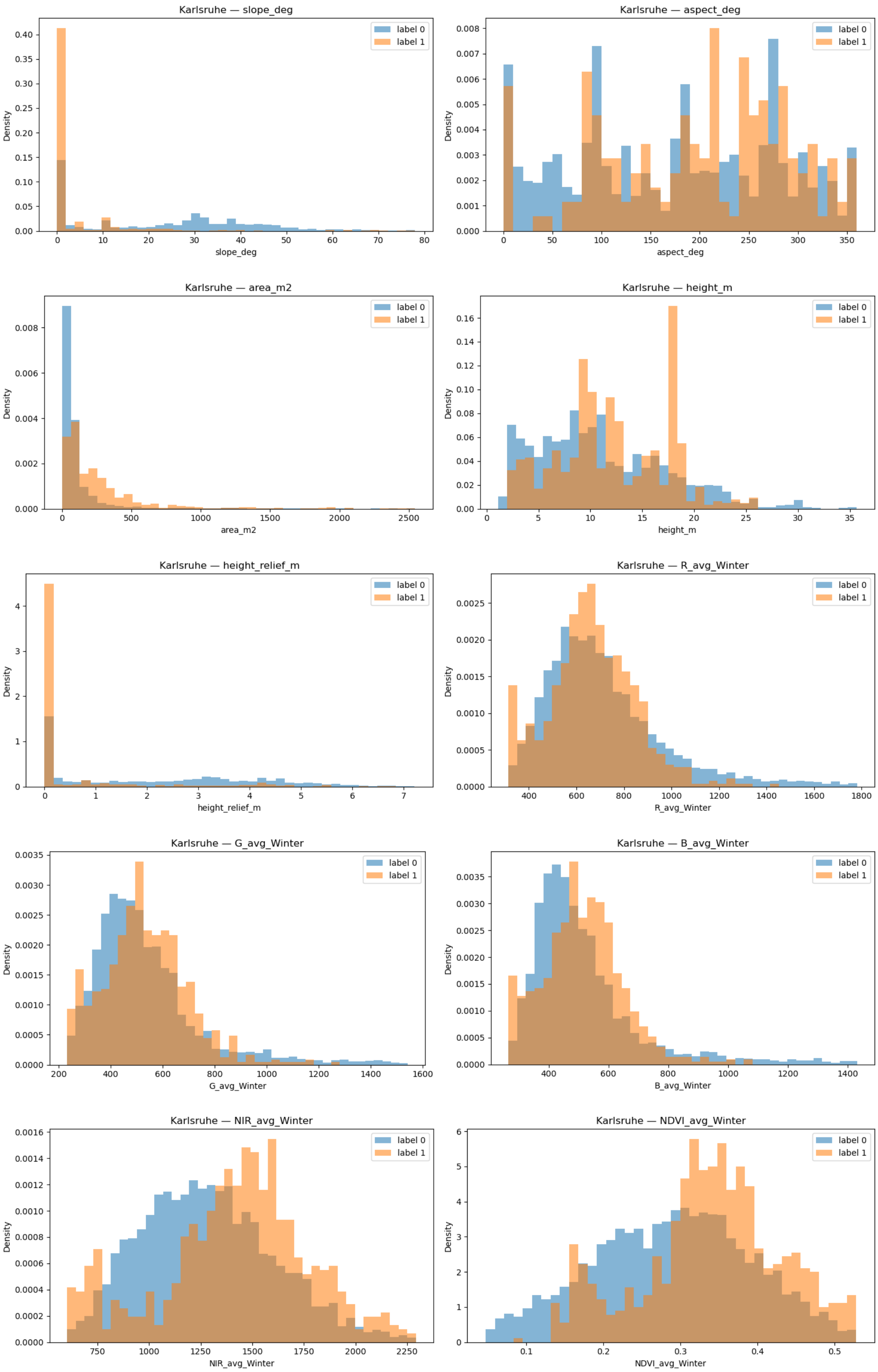

Freiburg im Breisgau — NDVI_avg_Summer
label 0
label 1
Density
NDVI_avg_Summer

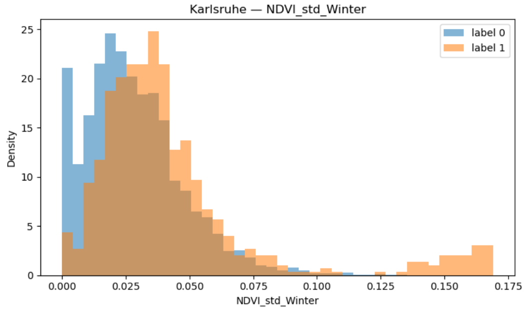

Freiburg im Breisgau — NDVI_std_Summer
label 0
label 1
Density
NDVI_std_Summer

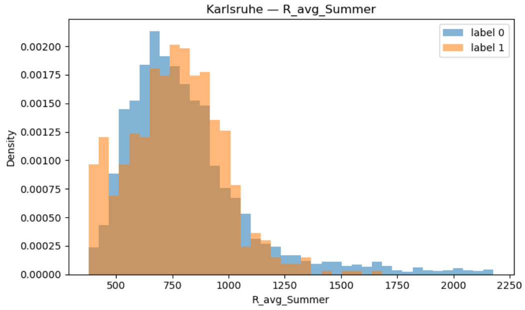

Freiburg im Breisgau — roofType
label 0
label 1
Relative frequency within label
1000
2100
3100
3200
3500
4000
5000
9999

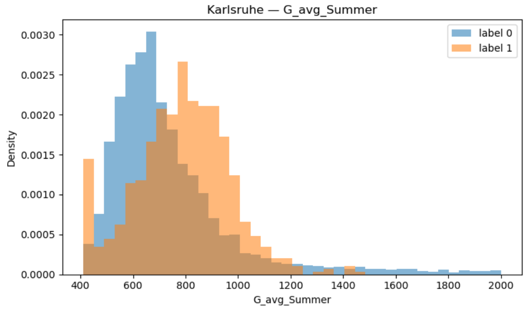

Freiburg im Breisgau — function
label 0
label 1
Relative frequency within label
31001_1010
31001_1123
31001_2020
31001_2050
31001_2112
31001_2140
31001_2463
31001_2723
31001_3021
31001_3065
51009_1610
51009_1700
Other

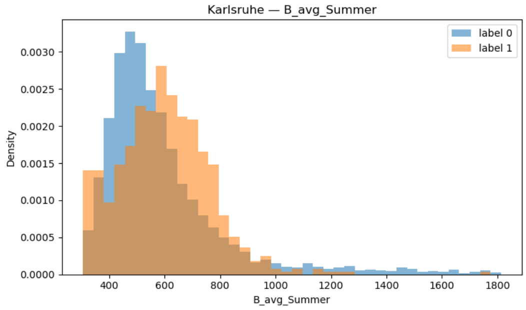
Appendix 3: Complete Sample Feature Behavior for Karlsruhe

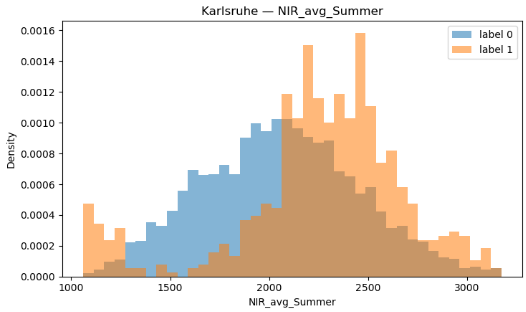


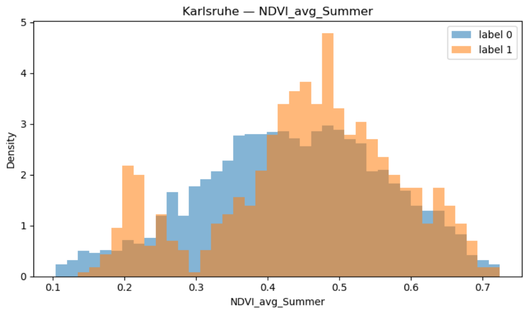

Karlsruhe — NDVI_std_Winter
label 0
label 1
Density
NDVI_std_Winter

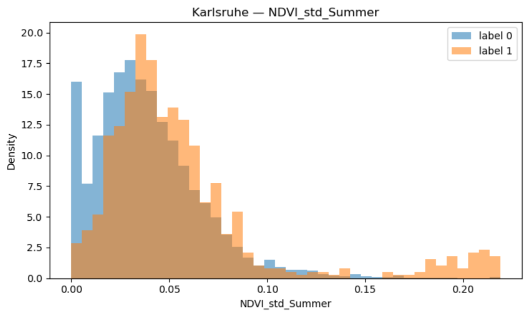

Karlsruhe — R_avg_Summer
label 0
label 1
Density
R_avg_Summer

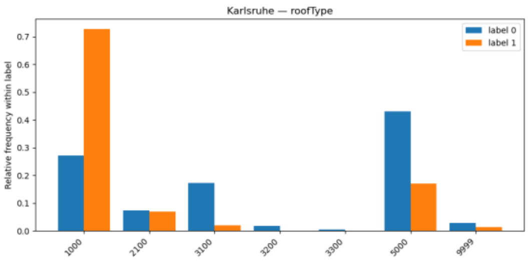

Karlsruhe — G_avg_Summer
label 0
label 1
Density
G_avg_Summer

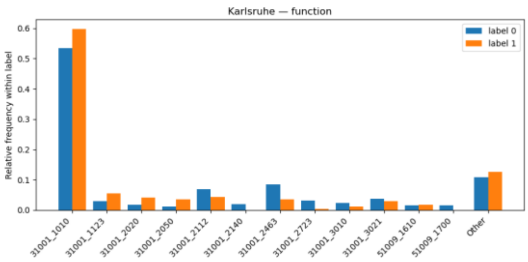

Karlsruhe — B_avg_Summer
label 0
label 1
Density
B_avg_Summer


Karlsruhe — NIR_avg_Summer
label 0
label 1
Density
NIR_avg_Summer

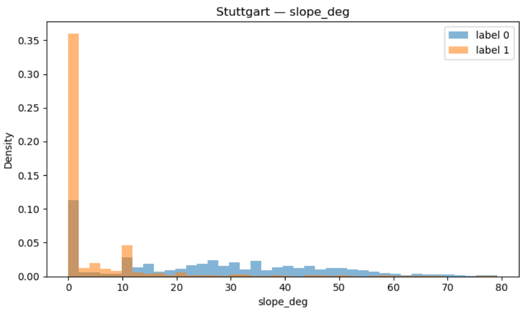

Karlsruhe — NDVI_avg_Summer
label 0
label 1
Density
NDVI_avg_Summer

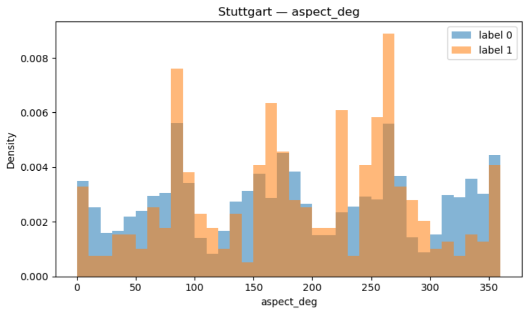

Karlsruhe — NDVI_std_Summer
label 0
label 1
Density
NDVI_std_Summer

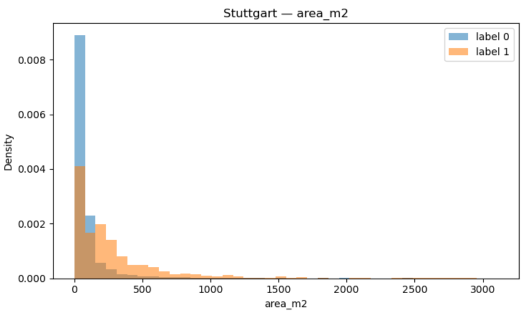

Karlsruhe — roofType
label 0
label 1
Relative frequency within label
1000
2100
3100
3200
3300
5000
9999

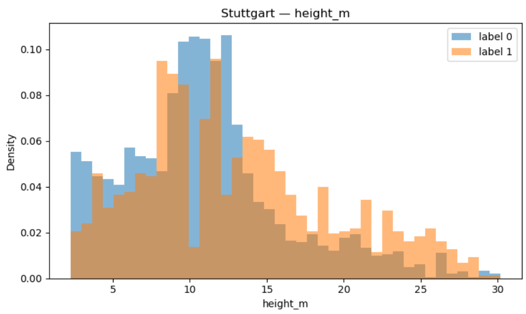

Karlsruhe — function
label 0
label 1
Relative frequency within label
31001_1010
31001_1123
31001_2020
31001_2050
31001_2112
31001_2140
31001_2463
31001_2723
31001_3010
31001_3021
51009_1610
51009_1700
Other

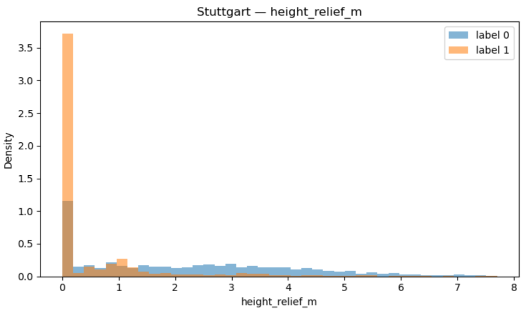
Appendix 4: Complete Sample Feature Behavior for Stuttgart

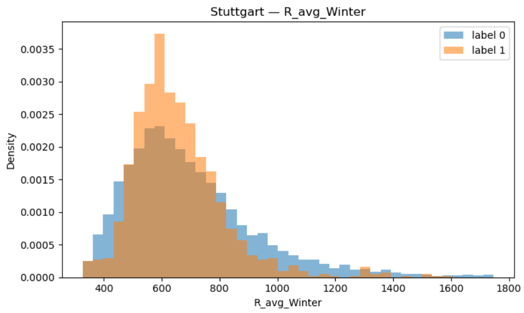


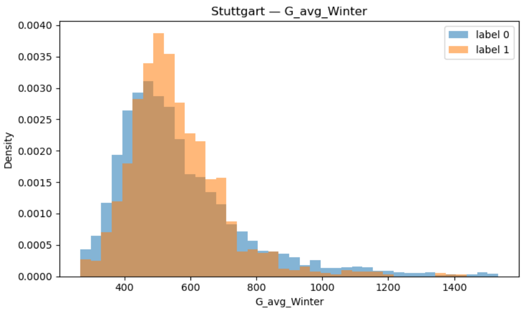


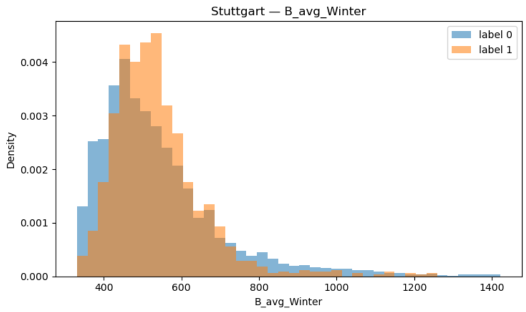


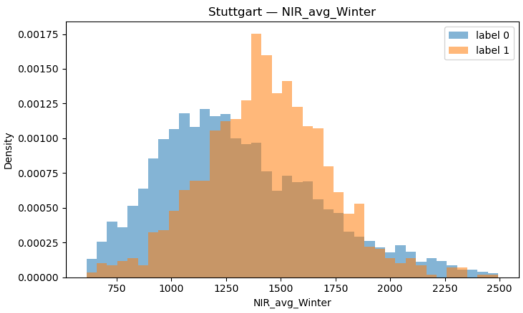


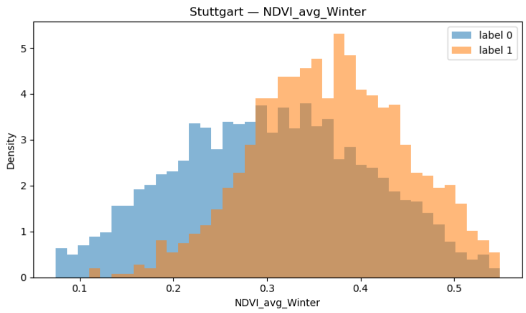


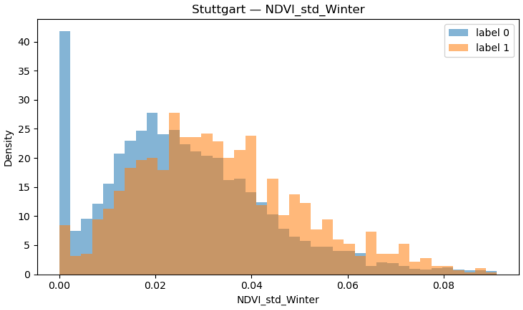


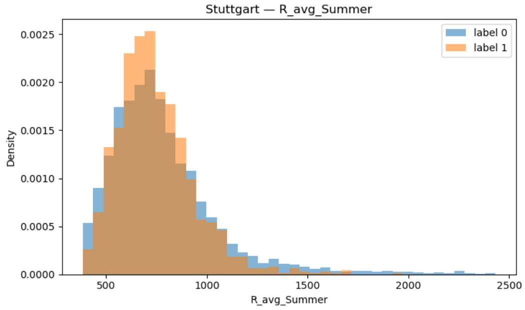


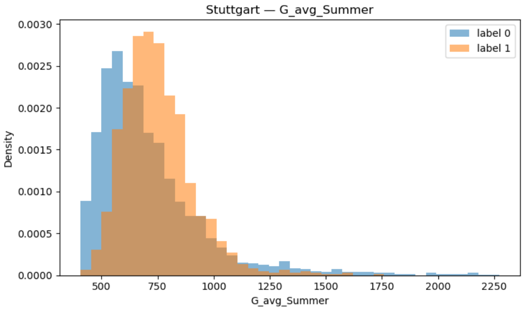


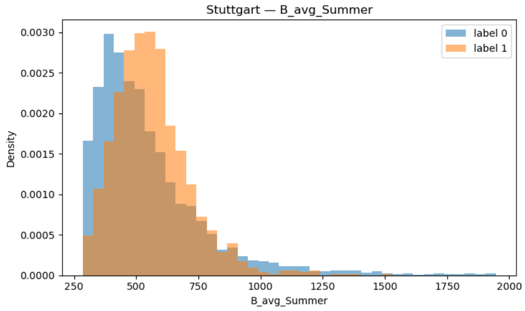


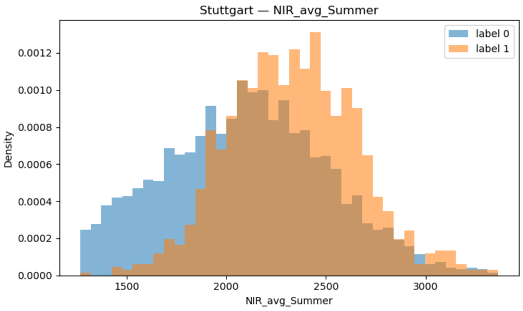


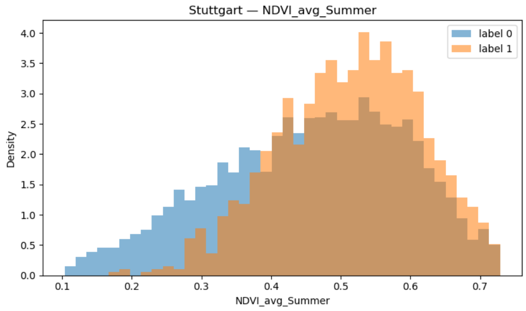

Stuttgart — NDVI_std_Winter
Density
NDVI_std_Winter
label 0
label 1

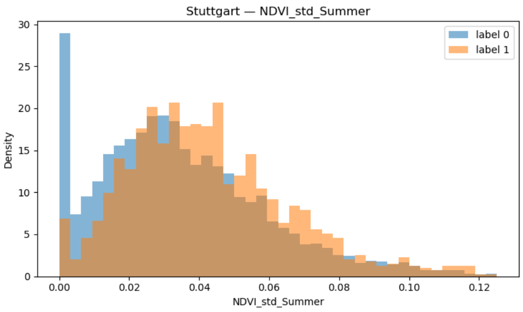

Stuttgart — R_avg_Summer
Density
R_avg_Summer
label 0
label 1

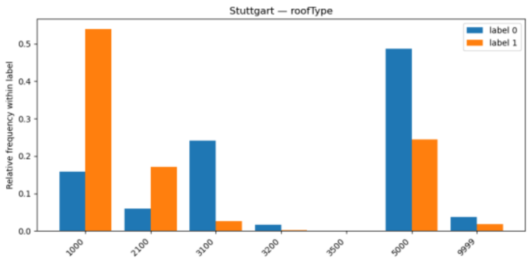

Stuttgart — G_avg_Summer
Density
G_avg_Summer
label 0
label 1

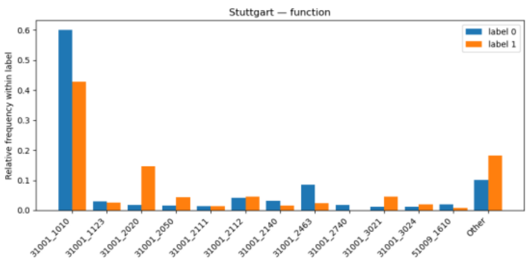

Stuttgart — B_avg_Summer
Density
B_avg_Summer
label 0
label 1


Stuttgart — NIR_avg_Summer
Density
NIR_avg_Summer
label 0
label 1

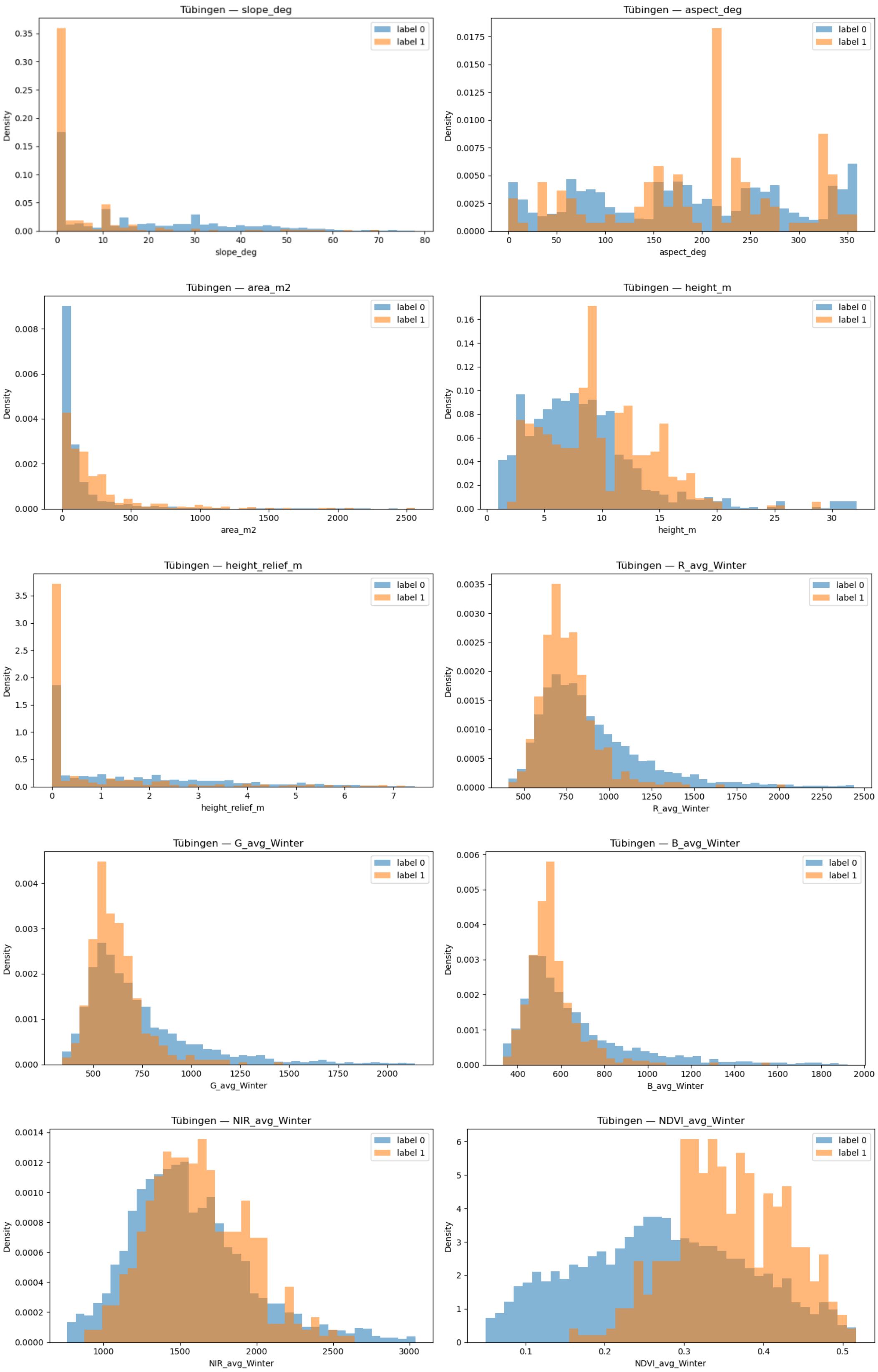

Stuttgart — NDVI_avg_Summer
Density
NDVI_avg_Summer
label 0
label 1

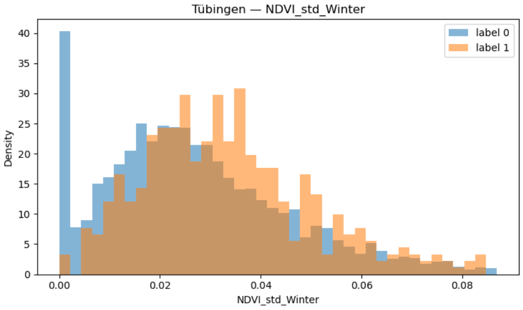

Stuttgart — NDVI_std_Summer
Density
NDVI_std_Summer
label 0
label 1

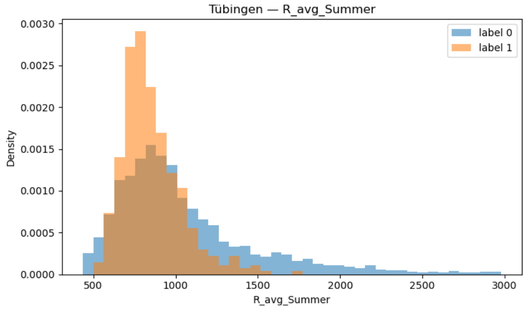

Stuttgart — roofType
Relative frequency within label
label 0
label 1
1000
2100
3100
3200
3500
5000
9999

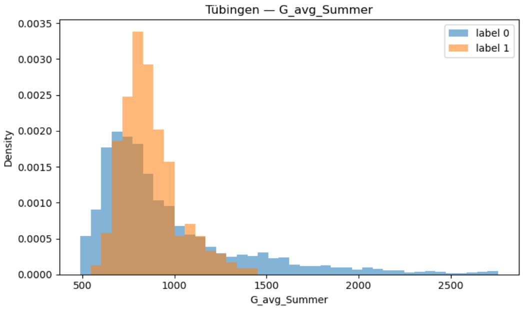

Stuttgart — function
Relative frequency within label
label 0
label 1
31001_1010
31001_1123
31001_2020
31001_2050
31001_2111
31001_2112
31001_2140
31001_2463
31001_2740
31001_3021
31001_3024
51009_1610
Other

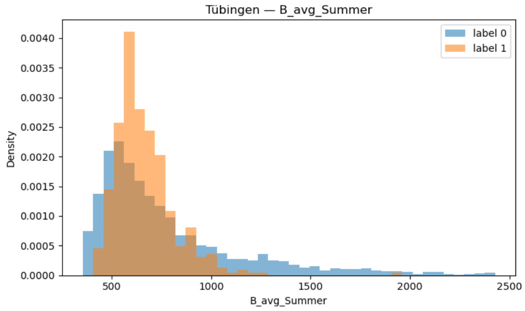
Appendix 5: Complete Sample Feature Behavior for Tübingen

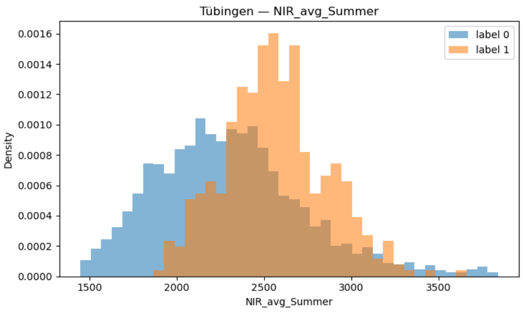


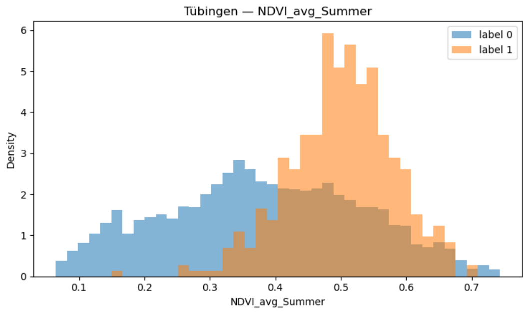

Tübingen — NDVI_std_Winter
label 0
label 1
Density
NDVI_std_Winter

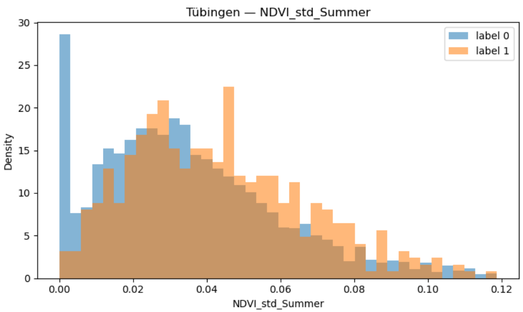

Tübingen — R_avg_Summer
label 0
label 1
Density
R_avg_Summer

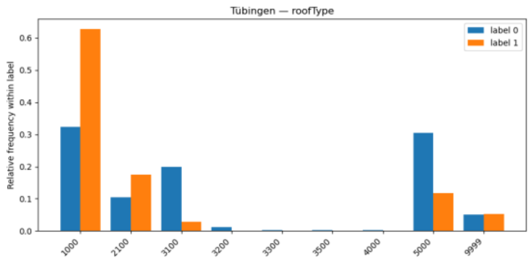

Tübingen — G_avg_Summer
label 0
label 1
Density
G_avg_Summer

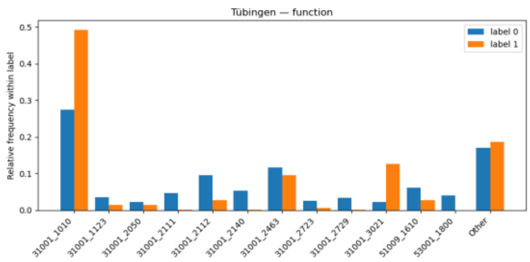

Tübingen — B_avg_Summer
label 0
label 1
Density
B_avg_Summer


Tübingen — NIR_avg_Summer
label 0
label 1
Density
NIR_avg_Summer

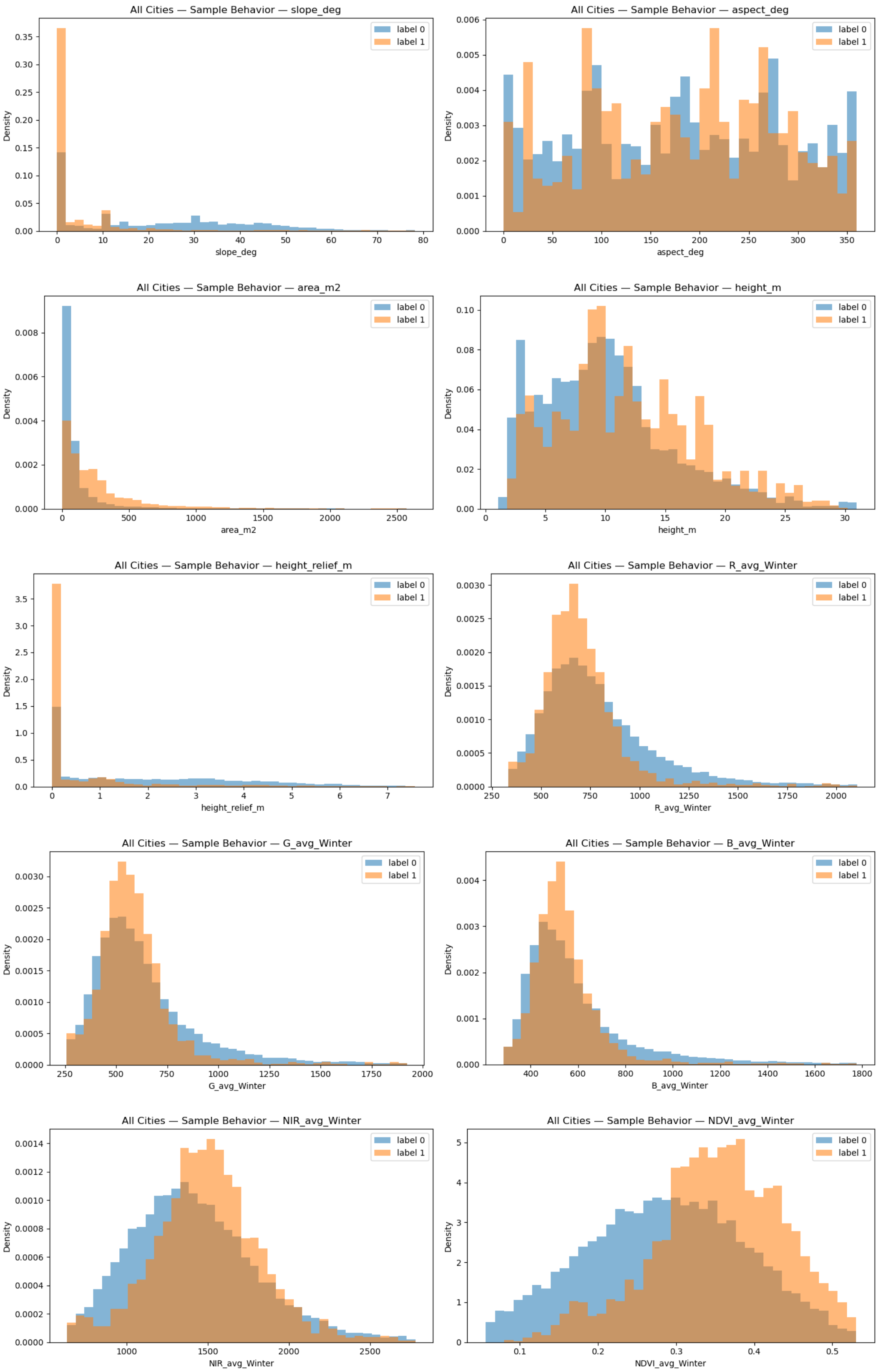

Tübingen — NDVI_avg_Summer
label 0
label 1
Density
NDVI_avg_Summer

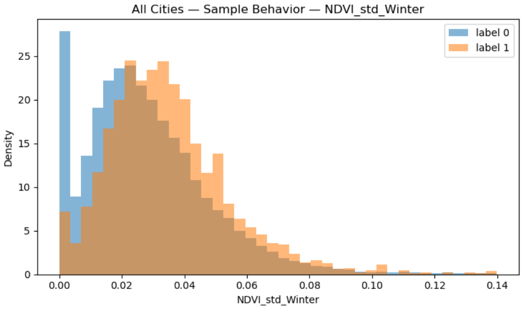

Tübingen — NDVI_std_Summer
label 0
label 1
Density
NDVI_std_Summer

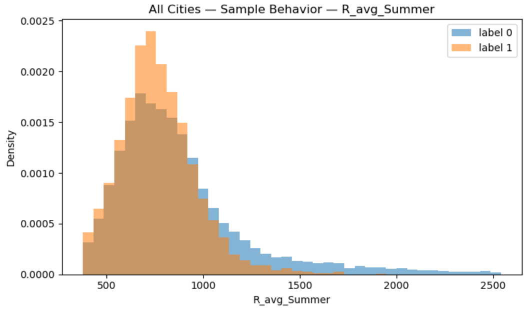

Tübingen — roofType
label 0
label 1
Relative frequency within label
1000
2100
3100
3200
3300
3500
4000
5000
9999

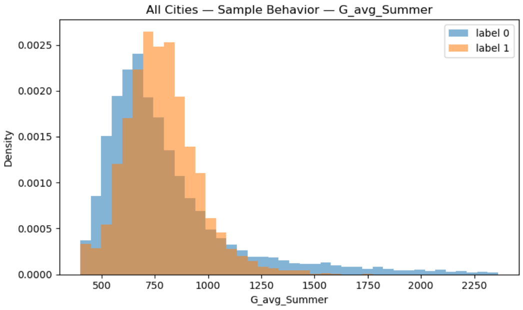

Tübingen — function
label 0
label 1
Relative frequency within label
31001_1010
31001_1123
31001_2050
31001_2111
31001_2112
31001_2140
31001_2463
31001_2723
31001_2729
31001_3021
51009_1610
53001_1800
Other

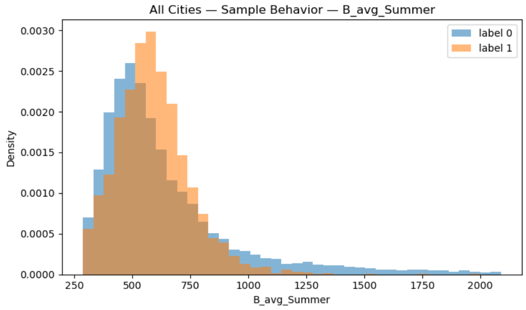
Appendix 6: Complete Sample Feature Behavior for All Cities

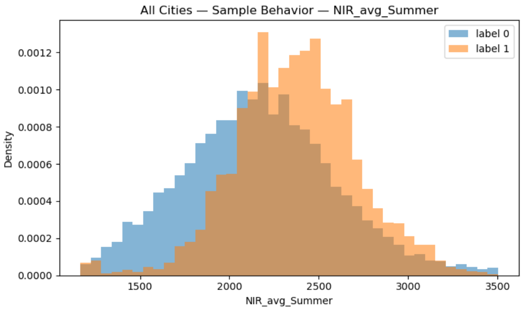


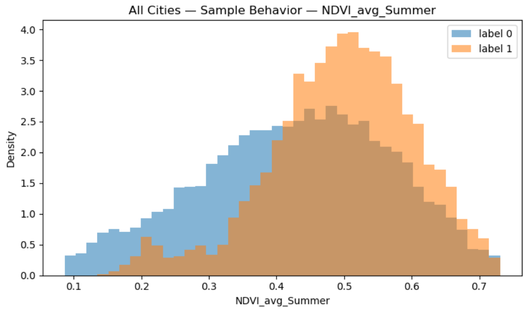

All Cities — Sample Behavior — NDVI_std_Winter
label 0
label 1
Density
NDVI_std_Winter
0
5
10
15
20
25
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14

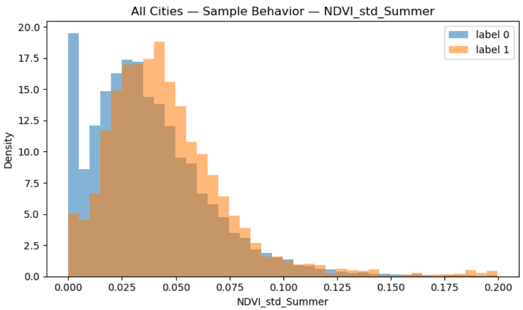

All Cities — Sample Behavior — R_avg_Summer
label 0
label 1
Density
R_avg_Summer
0.0000
0.0005
0.0010
0.0015
0.0020
0.0025
500
1000
1500
2000
2500

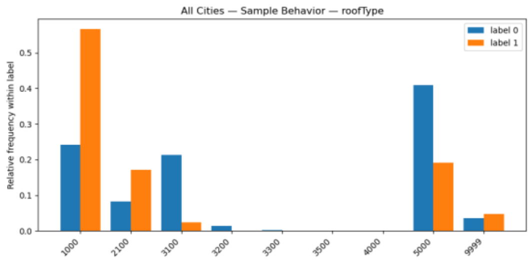

All Cities — Sample Behavior — G_avg_Summer
label 0
label 1
Density
G_avg_Summer
0.0000
0.0005
0.0010
0.0015
0.0020
0.0025
500
750
1000
1250
1500
1750
2000
2250

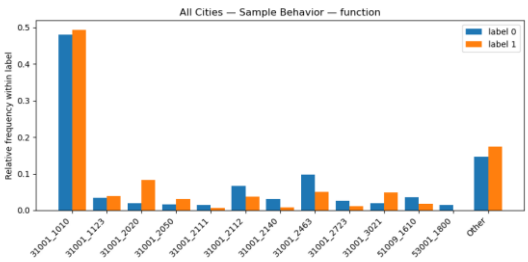

All Cities — Sample Behavior — B_avg_Summer
label 0
label 1
Density
B_avg_Summer
0.0000
0.0005
0.0010
0.0015
0.0020
0.0025
0.0030
250
500
750
1000
1250
1500
1750
2000


All Cities — Sample Behavior — NIR_avg_Summer
label 0
label 1
Density
NIR_avg_Summer
0.0000
0.0002
0.0004
0.0006
0.0008
0.0010
0.0012
1500
2000
2500
3000
3500


All Cities — Sample Behavior — NDVI_avg_Summer
label 0
label 1
Density
NDVI_avg_Summer
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7


All Cities — Sample Behavior — NDVI_std_Summer
label 0
label 1
Density
NDVI_std_Summer
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200


All Cities — Sample Behavior — roofType
label 0
label 1
Relative frequency within label
0.0
0.1
0.2
0.3
0.4
0.5
1000
2100
3100
3200
3300
3500
4000
5000
9999


All Cities — Sample Behavior — function
label 0
label 1
Relative frequency within label
0.0
0.1
0.2
0.3
0.4
0.5
31001_1010
31001_1123
31001_2020
31001_2050
31001_2111
31001_2112
31001_2140
31001_2463
31001_2723
31001_3021
51009_1610
53001_1800
Other